Advanced Financial and Macro-Econometrics

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Chapter 1

Bootstrap in Econometrics

1.1 Why Bootstrap

Bootstrap methodology comes into econometrics as a technique which enables us to generate "new artificial data" starting from our sample data. The reasons behind this methodology rely on:

- **Asymptotic Approximation can be unreliable in finite sample**
- **Test Statistics with unknown or non-standard limiting distribution**
- **Misspecified model**
- **Higher order accuracy**

Now, to understand why Bootstrap is important and where it can be applied, we proceed by presenting the following example:

Given an ARCH(q+1) model we can write:

$$\begin{cases} x_t = \sigma_t z_t, \quad z_t \sim iid(0,1) \\ \sigma^2 = \omega + \sum_{i=1}^{q} \alpha_i x_{t-i}^2 + \alpha_{q+1} x_{t-q-1} \end{cases}$$

We now want to test $H_0 : \alpha_{q+1} = 0 \quad vs \quad H_1 : \alpha_{q+1} > 0$. Using a classical test statistic, inference is problematic because the null hypothesis lies on the boundary of the parameter space. As a result, the test statistic does not converge to a standard normal distribution under H_0. So we rely on the Likelihood Ratio test which is given by:

$$LR = -2(\ell_U - \ell_R), \text{where} \quad LR = \inf_R Q(\lambda) - \inf_U Q(\lambda)$$

to get to this, we can proceed with a quadratic Taylor approximation:

$$\ell(\theta) \approx \ell(\theta_0) + (\text{score})'(\theta - \theta_0) - 1/2(\theta - \theta_0)' I(\theta_0)(\theta - \theta_0)$$

since we are minimising on a specific subset of the space $\mathbb{R}$, hence we have a constrained minimisation then:

$$LR = \inf_{\lambda \in \{0\} \times \Lambda} Q(\lambda) - \inf_{\lambda \in [0,\infty) \times \Lambda} Q(\lambda)$$

so LR is the distance between the best fit under the null constrained and the alternative unconstrained. Here $\Lambda = [0,\infty)^k \times (-\infty,\infty)^{q-k}$ where k is the number zero valued α_i since also other α's could be at the boundary. So the structure of LR becomes complicated and **Unknown**.

Now, as we have illustrated Bootstrap procedure can be used for approaching and tackling the problem. In fact, we can use a well-designed, consistent bootstrap to obtain $\{LR^*\}_{b=1}^B$ and compute the $1-\alpha$ quantiles to test whether to reject our H_0.

When we use Bootstrap we are moving and playing within two different universe, the true one related to the population and the data generating process and the one generated by the bootstrap schemes which generate new *artificial data*.

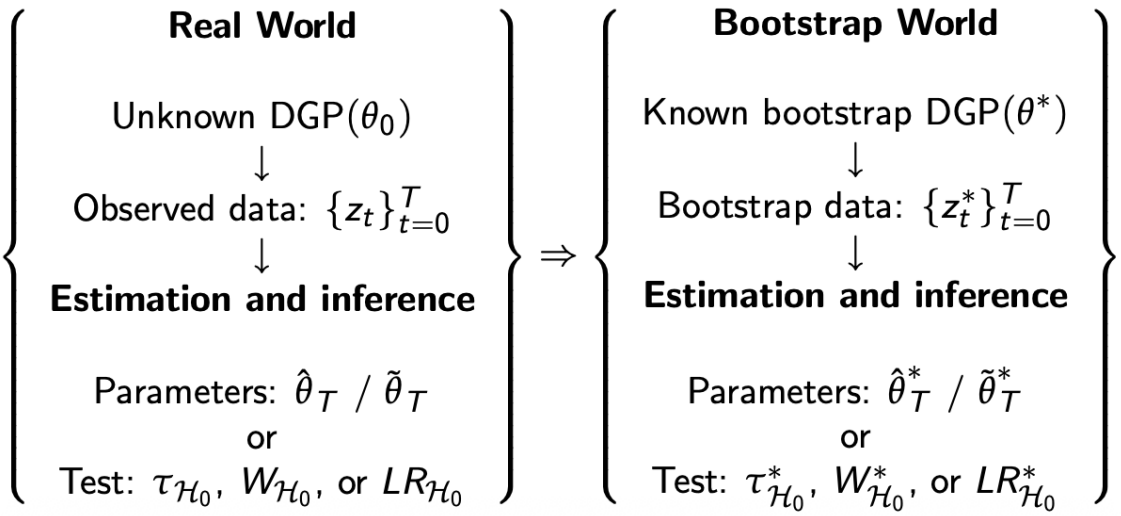


Figure 1.1

Figure 6.5 illustrates the logic of the bootstrap by contrasting the *real world* with the *bootstrap world*.

In the **real world**, data $\{z_t\}_{t=0}^T$ are generated by an unknown data-generating process (DGP) indexed by the true parameter θ_0. Since the DGP is unknown, statistical inference is based on estimators $\hat{\theta}_T$ (or restricted estimators $\tilde{\theta}_T$ under H_0) and on test statistics such as the Wald, Lagrange Multiplier, or Likelihood Ratio statistics, denoted generically by W_{H_0}, τ_{H_0}, or LR_{H_0}.

In the **bootstrap world**, the unknown DGP is replaced by a *known bootstrap DGP*, indexed by a bootstrap parameter θ^*, typically obtained from the sample (either unrestricted or imposed under H_0). Bootstrap samples $\{z_t^*\}_{t=0}^T$ are generated from this DGP, and the same estimation and testing procedures are applied, yielding bootstrap estimators $\hat{\theta}_T^*$ (or $\tilde{\theta}_T^*$) and bootstrap test statistics $W_{H_0}^*$, $\tau_{H_0}^*$, or $LR_{H_0}^*$.

Bootstrap validity requires that, under the null hypothesis H_0, the conditional distribution of the bootstrap statistics consistently approximates the sampling distribution of their real-world counterparts. In this way, the bootstrap world provides a feasible approximation to inference in the real world, despite the DGP being unknown.

Repeat the bootstrap world B times to obtain statistics

$$S_T^{*(b)}, \quad b = 1, 2, \ldots, B.$$

Inference is based on the empirical distribution

$$\{S_T^{*(1)}, \ldots, S_T^{*(B)}\}.$$

When applying the bootstrap procedure we need to ensure **Bootstrap validity** in particular we need to show that:

- **Parameter estimation**: as $\hat{\theta}_T \xrightarrow{d} D$ as $T \to \infty$ also the bootstrap parameter $\hat{\theta}^*_T \xrightarrow{d} D$
- **Hypothesis Testing**: meaning that was the H_0 is true then $S_T \xrightarrow{d} D$ therefore the bootstrap statistic should also replicate that. It is important that we need to account for the fact that, as H_0 is not true, then $S_T \to \pm\infty$; however, the bootstrap statistic **cannot do that**, otherwise also the critical values would diverge. So we need to have bounded probability hence $S^*_T \to \tilde{D}$

When considering Bootstrap schemes, it is important to take into account two key features: **robustness** and **validity**. The former ensures that the bootstrap universe mirrors the DGP process in all its features, whereas the latter ensures that we can make safe inferences.

1.2 Bootstrap Algorithms

Starting from the classic Regression model:

$$y_t = \beta x_t + \varepsilon_t$$

we can work out:

$$\hat{\beta} = \frac{\sum_i^n y_t x_t}{\sum_i^n x_t^2}, \quad \sqrt{n}(\hat{\beta} - \beta) = \frac{\frac{1}{\sqrt{n}} \sum_i^n \varepsilon_t x_t}{\frac{1}{n} \sum_i^n x_t^2} \xrightarrow{d} N(0, \Sigma)$$

where $\Sigma = \frac{\sigma^2_\varepsilon}{\sigma^2_x}$ and as we would suggest $\tau = (\hat{\beta} - \beta)^2 \sum_i^n x_i^2 / \hat{\sigma}^2_\varepsilon \xrightarrow{d} \chi^2_1$. Now we can present the different types of Bootstrap algortihm.

Parametric Bootstrap

Fixed regressor. Consider the linear regression model with fixed regressors. Bootstrap samples are generated by simulating new disturbances

$$\varepsilon^*_t \sim \text{i.i.d. } \mathcal{N}(0, \hat{\sigma}^2),$$

and constructing pseudo–data as

$$y^*_t = b^* x_t + \varepsilon^*_t.$$

The choice of b^* depends on the bootstrap scheme:

- *Restricted bootstrap*: $b^* = \beta$, the value imposed under H_0;
- *Unrestricted bootstrap*: $b^* = \hat{\beta}$, the estimator obtained from the original sample.

For each bootstrap replication $b = 1, \ldots, B$ (typically $B = 399$), the test statistic $\tau^{*(b)}$ is computed, yielding the empirical bootstrap distribution $\{\tau^*\}_{b=1}^B$.

Nonparametric Bootstrap

Residual (fixed regressor) bootstrap. With fixed regressors, bootstrap errors are obtained by resampling from the estimated residuals

$$\{\hat{\varepsilon}_1, \ldots, \hat{\varepsilon}_n\}, \qquad \varepsilon_t^* \sim \text{i.i.d. } \{\hat{\varepsilon}_1, \ldots, \hat{\varepsilon}_n\}.$$

Pseudo–samples are then generated as $y_t^* = \hat{\beta} x_t + \varepsilon_t^*$. Although the resulting distribution of the test statistic is not exactly χ_1^2, it often provides a good approximation in small samples and under mild model misspecification.

Wild bootstrap. In the wild bootstrap, bootstrap disturbances are constructed as

$$\varepsilon_t^* = \hat{\varepsilon}_t w_t,$$

where $\{w_t\}_{t=1}^n$ are i.i.d. random variables with zero mean and unit variance, such as

$$w_t \sim \mathcal{N}(0,1) \quad \text{or} \quad w_t \in \{-1, +1\} \text{ (Rademacher)}.$$

This procedure is particularly effective in the presence of heteroskedasticity, as it preserves the conditional variance structure of the errors.

Pairs bootstrap. The pairs bootstrap resamples directly from the joint empirical distribution of the data,

$$\{(y_1, x_1), \ldots, (y_n, x_n)\},$$

drawing observations with replacement. This approach can perform poorly in the presence of extreme observations or heavy tails, since resampling may generate bootstrap samples with repeated extreme values, leading to distorted inference.

1.3 Cross-Sectional Data

Whenever we deal with cross section data we are dealing with $\{y_t, x_t\}_{t=1}^T$ obtained by randomly sampling from fixed population. Assuming then a DGP given by a linear model of the type:

$$y_t = x_t'\beta_0 + \varepsilon_t, \quad \varepsilon_t \sim iid(0, \sigma_0^2)$$

We can then proceed with: *(i) Estimation of* β or *(ii) test parameters.* Now the first point can be addressed through the OLS or QLME estimator of β which is:

$$\hat{\beta} = \left(\frac{1}{T}\sum_{t=1}^{T} x_t x_t'\right)^{-1}\left(\frac{1}{T}\sum_{t=1}^{T} x_t y_t\right).$$

For Testing we can instead proceed by providing an hypothesis of the following type H_0 : $\beta_0 = a$ using the classic **t-statistic**:

$$\tau_{\beta_1=\alpha} = \frac{\hat{\beta}_1 - \alpha}{se(\hat{\beta}_1)}$$

When H_0 is true, then we have that $\tau_{\beta_1=\alpha} \xrightarrow{d} N(0,1)$ as $T \to \infty$, since on average we will be very close to the true value. In case H_0 is not true, then our t-statistic will diverge to $\pm\infty$.

If the population data-generating process (DGP) were known, we could generate bootstrap samples directly from it and obtain an estimator that is asymptotically identical to the one derived under the classical sampling scheme. However, knowledge of the true population DGP is typically infeasible. What is feasible instead is to **consider** $\{y_t, x_t\}_{t=1}^T$ as the population and draw bootstrap samples $\{y_t^*, x_t^*\}_{t=1}^T$, in order to obtain for each bootstrap sample a specific $\hat{\beta}^{*(b)}$ and therefore proceed with estimation. However, in a non-parametric setting where resampling is performed directly from the observed data (pair bootstrap), hypothesis testing requires special care. The plug-in principle that justifies bootstrap consistency for parameter estimation does not extend directly to hypothesis testing, since the resampling scheme does not impose the null hypothesis.

A naïve bootstrap procedure would construct the bootstrap test statistic as

$$\tau^*_{\beta_1=a} = \frac{\hat{\beta}_1^* - a}{\mathrm{se}(\hat{\beta}_1^*)},$$

where the standard error is computed from the bootstrap sample. If the null hypothesis holds, this statistic converges in distribution to a standard normal. However, when the null is false, the statistic diverges to $\pm\infty$, mirroring the behavior of the original test statistic. As a consequence, bootstrap critical values would also diverge, rendering the test invalid.

To overcome this issue, one instead considers a modified bootstrap statistic that recenters the bootstrap estimator around the original sample estimate,

$$\tau^*_{\beta_1=\hat{\beta}_1} = \frac{\hat{\beta}_1^* - \hat{\beta}_1}{\mathrm{se}(\hat{\beta}_1^*)}.$$

Under the null hypothesis, this statistic converges to a standard normal distribution. Crucially, even when the null hypothesis is false, both $\hat{\beta}_1^*$ and $\hat{\beta}_1$ converge to the true parameter value, implying that the bootstrap statistic remains asymptotically standard normal.

The empirical distribution of the bootstrap statistics

$$\left(\tau^{*(1)}_{\beta_1=\hat{\beta}_1}, \ldots, \tau^{*(B)}_{\beta_1=\hat{\beta}_1}\right)$$

can therefore be used to conduct inference. In particular, a bootstrap p-value is obtained by comparing the original test statistic to the simulated bootstrap distribution, or alternatively by computing bootstrap critical values. This can be seen by decomposing the naïve bootstrap statistic as

$$\tau^*_{\beta_1=a} = \frac{\hat{\beta}^*_1 - a}{\text{se}(\hat{\beta}^*_1)} = \frac{\hat{\beta}^*_1 - \hat{\beta}_1}{\text{se}(\hat{\beta}^*_1)} + \frac{\hat{\beta}_1 - a}{\text{se}(\hat{\beta}^*_1)}.$$

The first term converges in distribution to a standard normal, reflecting the stochastic variation of the bootstrap estimator around the sample estimate. The second term, however, depends on the distance between the true parameter and the value imposed under the null hypothesis. If the null hypothesis is false, this term diverges to $\pm\infty$, implying that the bootstrap statistic diverges as well. As a consequence, the corresponding bootstrap critical values would also diverge, invalidating the test.

By contrast, recentering the bootstrap statistic around $\hat{\beta}_1$ removes this diverging component and ensures that the bootstrap distribution remains bounded, even when the null hypothesis is false.

1.3.1 Parametric Bootstrap

In a parametric bootstrap, we can generally re-sample the data given that we allow for a specific parametric scheme. Therefore, we can specify a fully parametric model

$$y_t|x_t = N(x_t'\beta_0, \sigma_0^2)$$

and then bootstrap samples from that, however, we do not know the exact true paraemters therefore, we rely on the parametric draws given the following:

$$y^*_t|x_t = N(x_t'\hat{\beta}_0, \hat{\sigma}_0^2)$$

In this way, since we are able to control the parameters of our model, we can parametrise by allowing $\hat{\beta}_0 = (\alpha, \hat{\beta}_1, ...)'$ and use the **restricted** version of the test where we use the following statistic:

$$\tau^*_{\beta_1=a} = \frac{\hat{\beta}^*_1 - a}{\text{se}(\hat{\beta}^*_1)}$$

1.4 Time Series

In time series, the bootstrap procedure is a bit more complicated because we use data that are time-dependent; therefore, our models aim to capture this structure. Using a naive pairwise bootstrap would destroy such a structure, rendering the bootstrap procedure invalid. Specifying a classic AR(1) model have:

$$y_t = \delta + \theta y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim iid(0, \sigma_0^2)$$

and the $\Theta = \{(\delta, \theta, \sigma_0^2) \in \mathbb{R}^{\not\in} : |\theta| < 1\}$, as usual, we can derive the classic t-statistic or LR statistic for parameter testing. We generally also rely on the asymptotic approximation due to inference assuming $T \to \infty$, but how large should T be? In most cases, T is considered small; therefore, the asymptotic approximation does not work, so we can rely on the bootstrap.

For small sample sizes T, the asymptotic $\chi^2(1)$ distribution can be a poor approximation to the finite-sample distribution of the likelihood ratio statistic. This is illustrated by considering an AR(1) data generating process with $\delta_0 = 1$, $\theta_0 = 0.9$, $\sigma_0^2 = 1$ and $T = 25$, where artificial data are generated and the null hypothesis $H_0 : \theta_0 = 0.9$ is tested. Table 1.1 reports nine representative replications based on Gaussian innovations. The estimated autoregressive coefficient $\hat{\theta}_T$ varies substantially across replications, leading to likelihood ratio statistics ranging from values close to zero to values exceeding six. In particular, one replication produces $LR_T = 6.3217$ with an associated asymptotic p-value of 0.0119, implying rejection at the 5% level even though the null hypothesis is true. This illustrates the unreliability of asymptotic inference in small samples.

Table 1.1: Finite-sample replications for $T = 25$

Replication	$\hat{\theta}_T$	$LR_T(\theta_0 = 0.9)$	Ass. p-value	5% decision
1	0.55746	3.7060	0.054219	Accept
2	0.84060	0.26249	0.60841	Accept
3	0.67647	2.3594	0.12453	Accept
4	0.89195	0.0072239	0.93227	Accept
5	0.84330	0.27578	0.59948	Accept
6	0.74726	1.1603	0.28140	Accept
7	0.93019	0.17125	0.67900	Accept
8	0.40131	6.3217	0.011927	Reject
9	0.89567	0.0014226	0.96991	Accept

The size distortions of the asymptotic test become even clearer in a Monte Carlo experiment with $N = 10{,}000$ replications, where the empirical rejection frequency (ERF) is computed for different sample sizes. Table 1.2 reports the ERF obtained using asymptotic critical values. For $T = 15$, the ERF equals 0.1971, almost four times the nominal level of 0.05. Even for $T = 25$ and $T = 50$, the rejection frequencies remain very high at 0.1665 and 0.1180, respectively. Although the ERF decreases as T increases, convergence to the nominal size is slow, and even for $T = 500$ the ERF is still 0.0601. Only asymptotically, as $T \to \infty$, does the rejection frequency converge to the correct size of 0.05. This slow convergence is partly due to the autoregressive parameter being close to unity.

Finally, the discrepancy between the empirical rejection frequencies and the nominal level cannot be attributed to simulation uncertainty. At the 5% significance level, the rejection indicator is Bernoulli with parameter 0.05, implying that for $N = 10{,}000$ replications the standard error of the ERF is $\sqrt{0.05(1-0.05)/N} \approx 0.0022$. By a central limit theorem argument, the ERF should lie in the interval [0.0456, 0.0544] under correct size. Since the values reported in Table 1.2 lie well outside this interval for all finite T, the asymptotic likelihood ratio test is clearly oversized in finite samples.

Table 1.2: Empirical rejection frequency of the asymptotic test

T	ERF (Asymptotic)
15	0.1971
25	0.1665
50	0.1180
100	0.0902
250	0.0651
500	0.0601
∞	0.0500

1.4.1 Parametric Bootstrap

Also, in time series, the parametric bootstrap, is allowed, therefore we can resample y_t^* recursively from:

$$y_t^*|y_{t-1}^* = N(\hat{\delta} + \hat{\theta} y_{t-1}^*, \hat{\sigma}_0^2)$$

with $y_0^* = y_0$. Identical to such a scheme, we could also consider:

$$y_t^* = \hat{\delta} + \hat{\theta} y_{t-1}^* + \varepsilon_t^*, \quad \varepsilon_t^* \sim N(0, \hat{\sigma}^2)$$

By going parametric, we can proceed with a restricted bootstrap.

1.4.2 Residual Based Bootstrap

In residual based bootstrap we generate artificial data by allowing the model to use random sampled innovations taken from the sample estimation of the model, so:

$$y_t^* = \hat{\delta} + \hat{\theta} y_{t-1}^* + \varepsilon_t^*, \quad \varepsilon_t^* \sim \{\hat{\varepsilon}_1, ..., \hat{\varepsilon}_t\}$$

However, we need to ensure that the residuals are centred in order to get $E^*[\varepsilon_t] = \frac{1}{T}\sum \hat{\varepsilon}_t = 0$. In our AR(1) residuals are centered since we use the intercept however, we could recenter the data by:

$$\{\tilde{\varepsilon}_t^c\}_{t=1}^T, \qquad \text{where} \qquad \tilde{\varepsilon}_t^c = \tilde{\varepsilon}_t - \frac{1}{T}\sum_{t=1}^{T} \tilde{\varepsilon}_t.$$

With 10,000 replications, we record the empirical rejection frequency (ERF) and compare it with the nominal level 0.05.

T	Asymptotic	i.i.d. Bootstrap
15	0.1971	0.0558
25	0.1665	0.0529
50	0.1180	0.0513
100	0.0902	0.0498
250	0.0651	0.0485
500	0.0601	0.0527
∞	0.0500	0.0500

In terms of size, the bootstrap test is much more accurate than the asymptotic test.

1.4.3 Wild Bootstrap

As we have seen when considering bootstrap, we also need to account for robustness; therefore, given that our data exhibit heteroskedasticity, we would like a bootstrap method that preserves it. To account for this issue, we rely on the **Wild Bootstrap**, which generates bootstrap residuals according to

$$\varepsilon_t^* = \tilde{\varepsilon}_t \cdot \eta_t, \qquad \eta_t \overset{i.i.d.}{\sim} (0,1).$$

The distribution of η_t is chosen so that the bootstrap residuals $\{\varepsilon_t^*\}_{t=1}^T$ mimic the main features of the original residuals $\{\tilde{\varepsilon}_t\}_{t=1}^T$. Ideally, one would like to match all moments, that is,

$$\big(\mathbb{E}\eta_t,\ \mathbb{E}\eta_t^2,\ \mathbb{E}\eta_t^3,\ \mathbb{E}\eta_t^4,\ \mathbb{E}\eta_t^5, \ldots\big) = (0,1,1,1,1,1,\ldots),$$

so that the bootstrap errors reproduce the full moment structure of the true disturbances.

However, such a distribution does not exist. In particular, it holds that

$$\mathbb{E}\eta_t^4 \ \geq\ (\mathbb{E}\eta_t^3)^2 + 1,$$

implying that not all moments can be simultaneously matched. As a consequence, the choice of the distribution of η_t involves a trade–off. Equality in the above inequality is attained for two–point distributions, which motivates common wild bootstrap choices such as the Rademacher or Mammen distributions.

Common choices for the distribution of η_t in the wild bootstrap include the following.

Gaussian distribution. Let $\eta_t \sim \mathcal{N}(0,1)$. In this case,

$$\big(\mathbb{E}\eta_t,\ \mathbb{E}\eta_t^2,\ \ldots,\ \mathbb{E}\eta_t^6\big) = (0,1,0,3,0,15).$$

This choice imposes symmetry of the bootstrap errors but inflates higher even–order moments relative to the ideal case.

Rademacher distribution (sign–flip). Let η_t follow a two–point distribution on

$$\{1,-1\} \quad \text{with probabilities} \quad \left\{\tfrac{1}{2},\tfrac{1}{2}\right\}.$$

Then,

$$\big(\mathbb{E}\eta_t,\ \mathbb{E}\eta_t^2,\ \ldots,\ \mathbb{E}\eta_t^6\big) = (0,1,0,1,0,1).$$

This distribution assumes symmetry and exactly matches all even moments.

Mammen distribution. Let η_t follow a two–point distribution on

$$\left\{\frac{1-\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2}\right\} \approx \{-0.618,\ 1.618\},$$

with probabilities

$$\left\{\frac{\sqrt{5}+1}{2\sqrt{5}}, \frac{\sqrt{5}-1}{2\sqrt{5}}\right\} \approx \{0.724,\ 0.276\}.$$

For this choice,

$$\big(\mathbb{E}\eta_t,\ \mathbb{E}\eta_t^2,\ \ldots,\ \mathbb{E}\eta_t^6\big) = (0,1,1,2,3,5).$$

Hence, the Mammen distribution matches the skewness of the error distribution but inflates higher–order moments.

1.5 Bootstrap for AR(1)

In this section, we start introducing the main characteristics of the classical AR(1) model in order to present a full picture of all characteristics we will replicate through the Bootstrap.

1.5.1 Introduction to AR(1)

AR(1) Properties

We start discussing and assessing the AR(1) model, deriving its main components and properties. We start defining the following AR(1) model:

$$\begin{cases} x_t = \rho x_{t-1} + \varepsilon_t \\ x_0 \quad \text{fixed} \end{cases}$$

we assume now that $\varepsilon_t \sim iidN(0, \sigma^2)$, $|\rho| < 1$ and we want to test the following $H_0 : \rho = \rho_0$, so we can write our log-likelihood function as:

$$l_T(\rho|x_1, ..., x_T) = -\frac{1}{2}\left[T \log \sigma^2 - \sum_{t=1}^{T} \frac{(x_t - \rho x_{t-1})^2}{\sigma^2}\right]$$

Before proceeding, we aim to examine some key characteristics of our AR(1) model, specifically determining the conditions under which stationarity and ergodicity hold. So we start from the AR(1), which we can write as follows:

$$x_t = \rho x_{t-1} + \varepsilon_t = \underbrace{\rho^t x_0}_{\text{initial value}} + \underbrace{\sum_{t=0}^{t-1} \rho^i \varepsilon_{t-i}}_{\text{shock accumulation}}$$

In this setting this is not stationary as $E(\sum_{i=1}^{t-1} \rho^i \varepsilon_{t-i})^2 = \sigma^2 \frac{1-\rho^{2t}}{1-\rho^2}$, however we can give x_0 an initial distribution:

$$x_0 = \sum_{i=0}^{\infty} \rho^i \varepsilon_{-i}$$

So we can plug this in and obtain:

$$x_t^{stat} = \sum_{i=0}^{\infty} \rho_0^i \varepsilon_{t-i}^i$$

To prove that the process is also ergodic, we can rely on the drift criterion, so we assume the AR(1) process to be a **Markov Chain**, so $x_t|x_{t-1} = x \sim N(\rho x, \sigma^2)$ and we define a drift function $V(x) = 1 + x^2$. We then compute the conditional expectation so we get:

$$\begin{aligned} E[V(x)|x_{t-1} = x] = 1 + E[x_t^2|x] &= 1 + \rho^2 x^2 + \sigma^2 \\ &= \rho^2(1 + x^2) + (1 + \sigma^2 - \rho^2) \end{aligned}$$

Introducing the Foster–Lyapunov drift condition, we obtain

$$E[V(x_t) \mid x_{t-1} = x] \leq \lambda V(x) + b,$$

where we choose the Lyapunov function $V(x) = 1 + x^2$. For the AR(1) process

$$x_t = \rho x_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim iid(0, \sigma^2),$$

we have

$$E\big[x_t^2 \mid x\big] = \rho^2 x^2 + \sigma^2.$$

Therefore,

$$E[V(x_t) \mid x] = 1 + \rho^2 x^2 + \sigma^2 = \rho^2(1 + x^2) + (1 + \sigma^2 - \rho^2).$$

Hence we can identify

$$\lambda = \rho^2 < 1, \qquad b = 1 + \sigma^2 - \rho^2.$$

Since $\lambda < 1$, the conditional expectation of $V(x_t)$ is strictly smaller than $V(x)$ (up to the bounded constant b) whenever $|x|$ is sufficiently large. This establishes a contraction property outside a compact set, implying that the process is pulled back toward the origin. Consequently, the AR(1) process admits a unique invariant distribution and is geometrically ergodic. In particular, ergodicity ensures the validity of the Law of Large Numbers and the Central Limit Theorem, which justify the asymptotic properties of the maximum likelihood estimator.

Likelihood Ratio test

To propose a likelihood-type test we can start assessing a *restricted* and *unrestricted* version of the log likelihood, so:

$$l_T(\rho|x_1, ..., x_T) = -\frac{1}{2}\left[T \log \sigma^2 - \sum_{t=1}^{T} \frac{(x_t - \rho_0 x_{t-1})^2}{\sigma^2}\right] \quad \text{Restricted}$$

$$l_T(\rho|x_1, ..., x_T) = -\frac{1}{2}\left[T \log \sigma^2 - \sum_{t=1}^{T} \frac{(x_t - \hat{\rho} x_{t-1})^2}{\sigma^2}\right] \quad \text{Unrestricted}$$

We can then propose a Likelihood ratio statistic as a testing procedure, we use:

$$\begin{aligned} LR(\rho = \rho_0) &= 2[l(\hat{\rho}) - l(\rho_0)] \\ &= \underbrace{(\hat{\rho} - \rho_0)}_{\text{distance}} \underbrace{\frac{\sum_{t=1}^{T} x_{t-1}^2}{\sigma^2}}_{\text{scaling factor}} \xrightarrow{d} \chi_1^2 \end{aligned}$$

But why this statistic converges to a χ_1^2?. We start assuming stationarity and ergodicity of our AR(1) process, then we can proceed by writing:

$$\hat{\rho} - \rho_0 = \frac{\sum_{t=1}^{T} \varepsilon_t x_{t-1}}{\sum_{t=1}^{T} x_{t-1}^2}$$

We can then rewrite:

$$\sqrt{(T)}(\hat{\rho} - \rho_0) = \frac{\frac{1}{\sqrt{T}} \sum_{t=1}^{T} \varepsilon_t x_{t-1} \xrightarrow[CLT]{d} N(0, \sigma^4/(1-\rho^2))}{\frac{1}{T} \sum_{t=1}^{T} x_{t-1}^2 \xrightarrow[LLN]{p} E[x_{t-1}^2] = \frac{\sigma^2}{1-\rho^2}} \xrightarrow[\text{slutsky}]{d} N(0, 1 - \rho^2)$$

So, as we can see, since we have strict stationarity and ergodicity, we can use the *Law of Large Numbers* for the denominator. The numerator instead is a martingale; therefore, we need a specific central limit theorem for martingales.

Brown 1971 Assume that $\{m_t, \mathcal{F}_t\}$ is a martingale difference sequence (MDS), i.e.

$$E(m_t \mid \mathcal{F}_{t-1}) = 0.$$

Assume furthermore that, as $n \to \infty$:

(i)
$$\frac{1}{n}\sum_{t=1}^{n} E\big(m_t^2 \mid \mathcal{F}_{t-1}\big) \xrightarrow{P} \sigma_m^2 > 0,$$

(ii)
$$\frac{1}{n}\sum_{t=1}^{n} E\big(m_t^2 \mathbf{1}\{|m_t| > \sqrt{n}\delta\}\big) \longrightarrow 0, \qquad \text{for every } \delta > 0.$$

Then
$$\frac{1}{\sqrt{n}}\sum_{t=1}^{n} m_t \xrightarrow{d} \mathcal{N}(0, \sigma_m^2).$$

So now we can show that $\frac{1}{T}\sum E[z_t^2|\mathcal{F}_{\sqcup-\infty}] = \frac{1}{T}E[\varepsilon_t^2]\sum x_{t-1}^2$ using the LLN for iid data we get $E[\varepsilon_t^2]\frac{1}{T}\sum x_{t-1}^2 \xrightarrow{p} \frac{\sigma^4}{1-\rho^2}$. Using the CLT for MDS we can show the distribution of the numerator as written above.

Since our statistic is given by:

$$LR = [\sqrt{T}(\hat{\rho} - \rho_0)] * \frac{\frac{1}{\sqrt{T}}\sum x_t^2}{\sigma^2}$$

Using what we have seen so far, we get $LR \xrightarrow{d} \chi_1^2$.

1.5.2 Recursive Bootstrap AR(1)

When moving to Bootstrap AR(1), we would like to generate a Bootstrap scheme that fully mimics the main characteristics of the underlying classical AR(1). Therefore, we start replicating the model in a bootstrap world and decide which schemes to follow for replicating the AR(1):

$$\begin{cases} x_t^* = \underset{(a)}{\rho^*} x_{t-1}^* + \underset{(b)}{\varepsilon_t^*} \\ \underset{(c)}{x_0^*} \end{cases}$$

As we can observe from the model above one should carefully choose three elements before proceeding:

- **(a)** ρ^* is unknown so we can either choose for the *Unrestricted version* $\hat{\rho}$ or for the restricted version ρ_0 under the null hypothesis.
- **(b)** ε^* can also be restricted from *unrestricted* $\hat{\varepsilon}_t = x_t - \hat{\rho} x_{t-1}$ or *restricted* $\tilde{\varepsilon}_t = x_t - \rho_0 x_{t-1}$
- **(c)** x_0^* is generally chosen to be equal to the sample initial value

In our bootstrap world, we can therefore identify two layers of probability: the first one comes from the sample data we have, and the second comes from the resampling scheme we use in our bootstrap $\{\tilde{\varepsilon}\}_{t=1}^T \to \varepsilon_t^*$. Our bootstrap scheme would imply we resample with replacement our bootstrap residuals with replacement from the recentered ones in fact:

$$\varepsilon^* \sim_{iid} \{(\varepsilon_t - \bar{\varepsilon})\}_{t=1}^T$$

and given the resampling scheme we have that $P(\varepsilon^* = \tilde{\varepsilon}_1) = \frac{1}{T}$, therefore $E^*[\varepsilon^*] = \frac{1}{T}\sum(\varepsilon_t - \bar{\varepsilon}) = \bar{\varepsilon} - \bar{\varepsilon} = 0$. However we can also show that:

$$E^*[\varepsilon_t^{*2}] = \frac{1}{T}\sum(\varepsilon_t - \bar{\varepsilon})^2 \neq \sigma^2 \xrightarrow{p} \sigma^2$$

so it converges in probability hence is not equal to σ^2.

1.5.3 Law of Large Numbers for Recursive AR(1) Bootstrap

As in classical inference theory, we also seek asymptotic results in bootstrap theory in order to establish the validity of a bootstrap scheme. We recall the standard notion of *bootstrap convergence in probability*:

We write $X_T^* \xrightarrow{p^*}_p c$ if, for every $\delta > 0$,

$$P^*(|X_T^* - c| > \delta) \xrightarrow{p} 0,$$

where $P^*(\cdot)$ denotes probability *conditional on the original sample* (the bootstrap probability measure).

A convenient sufficient condition is obtained from Chebyshev's inequality in the bootstrap world:

$$P^*(|X_T^* - c| > \delta) \;\leq\; \frac{\mathbb{E}^*\left[(X_T^* - c)^2\right]}{\delta^2}, \qquad \forall \delta > 0. \tag{1.1}$$

Hence, to show $X_T^* \xrightarrow{p^*}_p c$, it is enough to prove $\mathbb{E}^*[(X_T^* - c)^2] \xrightarrow{p} 0$.

Let

$$X_T^* \;=\; \frac{1}{T}\sum_{t=1}^{T}\varepsilon_t^*,$$

where, conditionally on the data, $\{\varepsilon_t^*\}_{t=1}^T$ are i.i.d. draws from the *centered* empirical distribution of residuals $\{\varepsilon_t - \bar{\varepsilon}\}_{t=1}^T$. Then $\mathbb{E}^*(\varepsilon_t^*) = 0$, so the natural centering constant is $c = 0$. Using (1.1),

$$P^*(|X_T^*| > \delta) \;\le\; \frac{\mathbb{E}^*\big[(X_T^*)^2\big]}{\delta^2}.$$

Now compute the bootstrap second moment:

$$\begin{aligned}
\mathbb{E}^*\big[(X_T^*)^2\big] &= \mathbb{E}^*\left[\left(\frac{1}{T}\sum_{t=1}^{T}\varepsilon_t^*\right)^2\right] = \frac{1}{T^2}\,\mathbb{E}^*\left[\left(\sum_{t=1}^{T}\varepsilon_t^*\right)^2\right] \\
&= \frac{1}{T^2}\,\mathbb{E}^*\left[\sum_{t=1}^{T}(\varepsilon_t^*)^2 + 2\sum_{1\le s<t\le T}\varepsilon_t^*\varepsilon_s^*\right] \\
&= \frac{1}{T^2}\left(\sum_{t=1}^{T}\mathbb{E}^*\big[(\varepsilon_t^*)^2\big] + 2\sum_{1\le s<t\le T}\mathbb{E}^*[\varepsilon_t^*\varepsilon_s^*]\right).
\end{aligned}$$

Since the bootstrap draws are i.i.d. and $\mathbb{E}^*(\varepsilon_t^*) = 0$, for $s \neq t$ we have

$$\mathbb{E}^*[\varepsilon_t^*\varepsilon_s^*] = \mathbb{E}^*(\varepsilon_t^*)\,\mathbb{E}^*(\varepsilon_s^*) = 0,$$

so all cross-terms vanish. Moreover, by construction (sampling with equal probability),

$$\mathbb{E}^*\big[(\varepsilon_t^*)^2\big] = \frac{1}{T}\sum_{i=1}^{T}(\varepsilon_i - \bar{\varepsilon})^2, \qquad \forall t.$$

Therefore,

$$\begin{aligned}
\mathbb{E}^*\big[(X_T^*)^2\big] &= \frac{1}{T^2}\sum_{t=1}^{T}\frac{1}{T}\sum_{i=1}^{T}(\varepsilon_i - \bar{\varepsilon})^2 \\
&= \frac{1}{T}\left(\frac{1}{T}\sum_{i=1}^{T}(\varepsilon_i - \bar{\varepsilon})^2\right).
\end{aligned}$$

If $\frac{1}{T}\sum_{i=1}^{T}(\varepsilon_i - \bar{\varepsilon})^2 \xrightarrow{p} \sigma^2 < \infty$ (e.g. by a LLN under suitable conditions), then

$$\mathbb{E}^*\big[(X_T^*)^2\big] = \frac{1}{T}\,O_p(1) \xrightarrow{p} 0.$$

Plugging into Chebyshev yields, for every $\delta > 0$,

$$P^*(|X_T^*| > \delta) \;\le\; \frac{\mathbb{E}^*[(X_T^*)^2]}{\delta^2} \xrightarrow{p} 0,$$

which proves $X_T^* \xrightarrow{p^*}_p 0$.

1.5.4 Central Limit Theorem in Recursive AR(1) Bootstrap

We now turn to the Central Limit Theorem, which is also essential for proving inference in bootstrap settings. Recall that $X_T \xrightarrow{d} X$ if:

$$P(X_T \leq x) \to P(X \leq x)$$

for all continuity points of $F_X(x) = P(X \leq x)$. Likewise, this statement also holds in the bootstrap version, therefore

$X_T^* \xrightarrow{d^*}_p X$ if:

$$P^*(X_T^* \leq x) \xrightarrow{p} P(X \leq x)$$

We should be careful when evaluating the bootstrap central limit theorem, since each bootstrap scheme imposes particular conditions.

Now the important question is **How do we show the CLT?**

- **Empirically** we can in principle simulate the data such that $X \sim N(0, \sigma^2)$, then draw n observations and compute $\{X_N = \frac{1}{n}\sum_i^n x_i\}_{m=1}^M$, m times and then plot it.
- **Mathematically**, we rely on the characteristic function (see B.1).

Now given $X_t \to X$ iff $\phi_T(s) \to \phi(s)$. Recall that $i \in \mathbb{C}$ with $i^2 = -1$ and that the Taylor expansion of the exponential around zero is

$$e^x = 1 + x + \frac{x^2}{2} + o(x^2).$$

Let

$$X_T^* = \frac{1}{\sqrt{T}} \sum_{t=1}^{T} \varepsilon_t^*,$$

where $\{\varepsilon_t^*\}$ are i.i.d. under the bootstrap probability P^*, with

$$\mathbb{E}^*(\varepsilon_t^*) = 0, \qquad \mathbb{E}^*\left[(\varepsilon_t^*)^2\right] = \hat{\sigma}_T^2.$$

We compute the bootstrap characteristic function:

$$\begin{aligned}\phi_T^*(s) &= \mathbb{E}^*[\exp(isX_T^*)] \\ &= \mathbb{E}^*\left[\exp\left(is\frac{1}{\sqrt{T}}\sum_{t=1}^{T}\varepsilon_t^*\right)\right].\end{aligned}$$

Using independence under P^*,

$$\begin{aligned}\phi_T^*(s) &= \prod_{t=1}^{T} \mathbb{E}^*\left[\exp\left(is\frac{\varepsilon_t^*}{\sqrt{T}}\right)\right] \\ &= \left(\mathbb{E}^*\left[\exp\left(is\frac{\varepsilon_1^*}{\sqrt{T}}\right)\right]\right)^T.\end{aligned}$$

Now apply the Taylor expansion to the inner exponential. For small $1/\sqrt{T}$,

$$\exp\left(is\frac{\varepsilon_1^*}{\sqrt{T}}\right) = 1 + is\frac{\varepsilon_1^*}{\sqrt{T}} + \frac{(is)^2}{2}\frac{(\varepsilon_1^*)^2}{T} + o\left(\frac{1}{T}\right).$$

Taking bootstrap expectation and using $\mathbb{E}^*(\varepsilon_1^*) = 0$,

$$\begin{aligned}\mathbb{E}^*\left[\exp\left(is\frac{\varepsilon_1^*}{\sqrt{T}}\right)\right] &= 1 + \frac{(is)^2}{2}\frac{\mathbb{E}^*\left[(\varepsilon_1^*)^2\right]}{T} + o\left(\frac{1}{T}\right) \\ &= 1 - \frac{s^2}{2}\frac{\hat{\sigma}_T^2}{T} + o\left(\frac{1}{T}\right),\end{aligned}$$

since $i^2 = -1$.

Therefore,

$$\phi_T^*(s) = \left(1 - \frac{s^2}{2}\frac{\hat{\sigma}_T^2}{T} + o\left(\frac{1}{T}\right)\right)^T.$$

Now use the standard limit

$$\left(1 + \frac{a_T}{T}\right)^T \to e^a \quad \text{if } a_T \to a.$$

If $\hat{\sigma}_T^2 \xrightarrow{p} \sigma^2$, then

$$\phi_T^*(s) \xrightarrow{p} \exp\left(-\frac{s^2\sigma^2}{2}\right).$$

But this is exactly the characteristic function of

$$\mathcal{N}(0, \sigma^2).$$

Hence, by Lévy's continuity theorem,

$$X_T^* \xrightarrow{d^*}_p \mathcal{N}(0, \sigma^2).$$

Chapter 2

Cointegration

2.1 An Introduction to Cointegration

We begin by recalling the notions of **stationarity** and **integration**, which provide the basic framework for understanding cointegration. Consider a general linear process:

$$Y_t = \sum_{i=0}^{\infty} C_i \, \varepsilon_{t-i} = C(L)\varepsilon_t.$$

Assuming that $\sum_{i=0}^{\infty} C_i^2 < \infty$, the process is *integrated of order zero*, denoted $I(0)$, and hence stationary. Within the class of $I(0)$ processes, the filter $C(L)$ must satisfy the additional condition that its long-run multiplier is non-zero:

$$C = C(1) = \sum_{i=0}^{\infty} C_i \neq 0.$$

This condition ensures that $C(L)$ does not contain the unit-root factor $(1-L)$, so that Y_t remains stationary. If instead $C(1) = 0$, the polynomial $C(L)$ factors through $(1 - L)$ and the process fails to be $I(0)$.

Defining $X_t = \sum_{i=1}^{t} Y_i$ as the partial sum of an $I(0)$ process Y_t yields an $I(1)$ process, because each innovation ε_i enters the cumulative sum with equal weight and never dies out. The relationship between integration orders under summation and differencing is:

$$I(0) \xrightarrow{\Sigma} I(1) \xrightarrow{\Sigma} I(2), \qquad I(0) \xleftarrow{\Delta} I(1) \xleftarrow{\Delta} I(2).$$

Examples.

Example 1. Let $Y_t = e_t$, so $C(1) = C_0 = 1 \neq 0$ and $Y_t \sim I(0)$. The partial sum $X_t = \sum_{i=1}^{t} e_i$ is a non-stationary random walk, hence $X_t \sim I(1)$.

Example 2. Let $Y_t = e_t - e_{t-1}$, so $C(1) = 1 - 1 = 0$ and $Y_t \not\sim I(0)$. The partial sum telescopes: $X_t = \sum_{i=1}^{t}(e_i - e_{i-1}) = e_t - e_0$, which is stationary. The unit-root factor $(1-L)$ in $C(L)$ cancels the summation and prevents X_t from being $I(1)$.

A stochastic process $X_t \in \mathbb{R}^p$ is integrated of order d, written $X_t \sim I(d)$, if $\Delta^d X_t \sim I(0)$. The definition extends naturally to vectors: $X_t = (X_{1t}, \ldots, X_{pt})' \in \mathbb{R}^p$ is $I(1)$ if at least one component is $I(1)$ and no component requires more than one difference to achieve stationarity.

Within this multivariate framework, the vector process X_t is said to be **cointegrated** if there exists a nonzero matrix $\beta \in \mathbb{R}^{p \times r}$ such that

$$\beta' X_t \sim I(0).$$

Although the individual components of X_t are non-stationary, there exist r linearly independent combinations of them that are stationary. Each column of β defines one such *cointegrating relation*, representing a long-run equilibrium to which the system tends to return.

The Beveridge–Nelson Decomposition

To study the stochastic structure of $I(1)$ processes analytically, it is useful to decompose the lag polynomial $C(L)$. Starting from the finite-order example:

$$\begin{aligned} Y_t &= C_0\varepsilon_t + C_1\varepsilon_{t-1} + C_2\varepsilon_{t-2} + C_3\varepsilon_{t-3} \\ &= (C_0 + C_1 + C_2 + C_3)\varepsilon_t - (C_1 + C_2 + C_3)\Delta\varepsilon_t - (C_2 + C_3)\Delta\varepsilon_{t-1} - C_3\Delta\varepsilon_{t-2} \\ &= C(1)\varepsilon_t + C_0^*\Delta\varepsilon_t + C_1^*\Delta\varepsilon_{t-1} + C_2^*\Delta\varepsilon_{t-2}, \end{aligned}$$

where $C_j^* = -(C_{j+1} + C_{j+2} + \cdots)$. In general, any stationary linear process admits the decomposition:

$$Y_t = C(L)\varepsilon_t = C(1)\varepsilon_t + C^*(L)\Delta\varepsilon_t.$$

Forming the partial sum $X_t = X_0 + \sum_{i=1}^{t} Y_i$ and substituting:

$$\begin{aligned} X_t &= X_0 + C(1)\sum_{i=1}^{t}\varepsilon_i + \sum_{i=1}^{t} C^*(L)\Delta\varepsilon_i \\ &= X_0 + C(1)\sum_{i=1}^{t}\varepsilon_i + S_t - S_0, \end{aligned}$$

where $S_t = C^*(L)\varepsilon_t$ is stationary. This gives the fundamental representation of an $I(1)$ process:

$$X_t = \underbrace{(X_0 - S_0)}_{\text{initial conditions}} + \underbrace{C\sum_{i=1}^{t}\varepsilon_i}_{\text{stochastic trend}} + \underbrace{S_t}_{\text{stationary}}, \tag{2.1}$$

where $C \equiv C(1)$. The process is thus decomposed into initial conditions, a random-walk stochastic trend driven by all past shocks with equal weight, and a mean-reverting stationary component.

For $\beta' X_t$ to be stationary, the linear combination must eliminate the stochastic trend in (2.1). This requires:

$$\beta' C = 0. \tag{2.2}$$

Hence cointegration implies that the long-run impact matrix C has *reduced rank*. Specifically, if $X_t \in \mathbb{R}^p$ and there are r cointegrating relations, then $\operatorname{rank}(C) = p - r$. Full-rank C corresponds to the absence of cointegration: all p stochastic trends are linearly independent and no stationary combination exists.

2.1.1 Unit Roots in VAR Models

Consider a VAR(3) model:

$$X_t = \Pi_1 X_{t-1} + \Pi_2 X_{t-2} + \Pi_3 X_{t-3} + \varepsilon_t.$$

The dynamic properties are encoded in the characteristic polynomial $A(z) = I_p - \Pi_1 z - \Pi_2 z^2 - \Pi_3 z^3$. Evaluating at $z = 1$:

$$|A(1)| = |I_p - \Pi_1 - \Pi_2 - \Pi_3| = 0$$

signals the presence of at least one unit root, making the matrix $\Pi \equiv I_p - \Pi_1 - \Pi_2 - \Pi_3$ singular with rank $r < p$.

Re-parametrising the VAR in **Vector Error Correction** (VECM) form:

$$\Delta X_t = \Pi X_{t-1} + \Gamma_1 \Delta X_{t-1} + \varepsilon_t,$$

where $\Pi = \Pi_1 + \Pi_2 - I_p$ and $\Gamma_1 = -\Pi_2$. A unit root implies $\Pi = -A(1)$ has reduced rank. When $\text{rank}(\Pi) = r < p$, we decompose:

$$\Pi = \alpha\beta', \quad \alpha,\, \beta \in \mathbb{R}^{p\times r},$$

where $\text{col}(\alpha)$ is the adjustment space and $\text{col}(\beta)$ is the cointegration space. The three cases are:

- $r = 0$: $\Pi = 0$ and the VAR is estimated in first differences; no linear combination of X_t is stationary.
- $0 < r < p$: reduced-rank $\Pi = \alpha\beta'$; exactly r stationary cointegrating relations $\beta' X_t \sim I(0)$ exist alongside $p - r$ common stochastic trends.
- $r = p$: Π is of full rank and all components of X_t are stationary; the standard VAR in levels applies.

2.2 The Granger Representation Theorem

Theorem (Granger Representation). *Given a VAR(k) model in VECM form*

$$\Delta X_t = \Pi X_{t-1} + \sum_{i=1}^{k-1} \Gamma_i \, \Delta X_{t-i} + \varepsilon_t, \quad \varepsilon_t \overset{\text{i.i.d.}}{\sim} (0, \Omega),$$

suppose $\text{rank}(\Pi) = r$ *and* $\Pi = \alpha\beta'$ *with* $\alpha, \beta \in \mathbb{R}^{p\times r}$. *If the characteristic polynomial* $A(z)$ *has exactly* $p - r$ *roots at* $z = 1$ *and all remaining roots outside the unit circle, then* X_t *is an* $I(1)$ *process admitting the representation*

$$X_t = C \sum_{i=1}^{t} \varepsilon_i + C^*(L)\varepsilon_t + A,$$

where

$$\Gamma = I_p - \Gamma_1 - \cdots - \Gamma_{k-1}, \qquad C = \beta_\perp (\alpha_\perp' \, \Gamma \, \beta_\perp)^{-1} \alpha_\perp',$$

and the coefficients $C_j^* \to 0$ *exponentially fast.*

2.2.1 Understanding the Granger Representation Decomposition

Before presenting the steps that lead to the Granger representation theorem, we introduce two algebraic tools that will prove essential throughout the derivation: the **Orthogonal Projection Identity** and the **Alternative Projection Identity**. The underlying idea is to decompose the space $\mathbb{R}^n$ into two complementary subspaces — one spanned by β, along which linear combinations of X_t are stationary, and one spanned by $\beta_\perp$, along which the stochastic trends propagate. Formally, any vector in $\mathbb{R}^n$ can be written as a combination of components in these two directions, since

$$\text{col}(\beta) \oplus \text{col}(\beta_\perp) = \mathbb{R}^n,$$

and analogously for α and $\alpha_\perp$. The two identities below make this decomposition explicit and operational, allowing us to isolate the r cointegrating directions from the $n - r$ directions along which the common stochastic trends drive the non-stationarity of the system.

Orthogonal Projection Identity

We can start choosing $\beta \in \mathbb{R}^{p\times r}$ and $\beta_\perp \in \mathbb{R}^{p\times(p-r)}$, therefore we coudl write:

$$\mathbf{I}_p = \beta(\beta'\beta)^{-1}\beta' + \beta_\perp(\beta'_\perp\beta_\perp)^{-1}\beta'_\perp$$

so if we post multiply by a X we get:

$$X = \underbrace{\beta(\beta'\beta)^{-1}\beta'}_{\text{proj. onto } \beta} X + \underbrace{\beta_\perp(\beta'_\perp\beta_\perp)^{-1}\beta'_\perp}_{\text{proj. onto } \beta'} X$$

The intuition is essentially correct, but we sharpen it slightly. The **Orthogonal Projection Identity** states that the identity matrix I_n can be decomposed as

$$I_n = \underbrace{\beta(\beta'\beta)^{-1}\beta'}_{\text{projection onto col}(\beta)} + \underbrace{\beta_\perp(\beta'_\perp\beta_\perp)^{-1}\beta'_\perp}_{\text{projection onto col}(\beta_\perp)}, \tag{2.3}$$

so that any vector $X \in \mathbb{R}^n$ decomposes as

$$X = \beta\underbrace{(\beta'\beta)^{-1}\beta'X}_{v_1} + \beta_\perp\underbrace{(\beta'_\perp\beta_\perp)^{-1}\beta'_\perp X}_{v_2}, \tag{2.4}$$

where the two components βv_1 and $\beta_\perp v_2$ are orthogonal by construction, since $\beta'\beta_\perp = 0$.

Regarding the role of $(\beta'\beta)^{-1}$: while it may be tempting to read it as a simple scaling factor, it is more precisely a *metric correction*. It accounts for the fact that the columns of β are not necessarily orthonormal. If they were — that is, if $\beta'\beta = I_r$ — then $v_1 = \beta'X$ directly and no correction would be needed. In general, $(\beta'\beta)^{-1}\beta'$ is the **left inverse** of β, ensuring that projecting and re-expanding recovers exactly the right component without distortion.

The correct interpretation is therefore the following: v_1 is the *coordinate* of X in the β direction, measuring how much of X lies in the stationary subspace col(β). The product βv_1 then reconstructs the actual *component* of X along col(β). Symmetrically, v_2 is the coordinate in the $\beta_\perp$ direction, and $\beta_\perp v_2$ recovers the trending component. The two together reconstruct X exactly — with no overlap and no residual.

Alternative Projection Matrix

Proceeding in the same spirit, we now choose matrices $\alpha, \beta \in \mathbb{R}^{p\times r}$ that are not mutually orthogonal but together span $\mathbb{R}^p$ when combined with their orthogonal complements $\alpha_\perp, \beta_\perp \in \mathbb{R}^{p\times(p-r)}$. The key advantage of the **Alternative Projection Identity** over the orthogonal one is that it incorporates both α and β simultaneously, thereby encoding not only the direction of the long-run equilibrium relationships but also the speed of adjustment toward them. Specifically, we have

$$\mathbf{I}_p = \underbrace{\alpha(\beta'\alpha)^{-1}\beta'}_{\text{error correction: speed of adjustment}} + \underbrace{\beta_\perp(\alpha'_\perp\beta_\perp)^{-1}\alpha'_\perp}_{\text{common trends: stochastic direction}}, \tag{2.5}$$

where the first term projects onto col(α), capturing the mean-reverting dynamics, and the second term projects onto col($\beta_\perp$), spanning the $(p-r)$ directions along which the common stochastic trends propagate.

2.2.2 Derivation of the Granger Decomposition

Starting from the VAR(1) VECM:

$$\Delta X_t = \alpha\beta' X_{t-1} + \varepsilon_t, \qquad H_r : \text{rank}(\Pi) \le r.$$

Non-stationary component. Pre-multiplying by $\alpha'_\perp$ eliminates the error-correction term since $\alpha'_\perp \alpha = 0$:

$$\alpha'_\perp \Delta X_t = \underbrace{\alpha'_\perp \alpha}_{=0} \beta' X_{t-1} + \alpha'_\perp \varepsilon_t = \alpha'_\perp \varepsilon_t.$$

Shocks therefore accumulate permanently along the directions orthogonal to the adjustment space $\mathrm{col}(\alpha)$, generating $p - r$ common stochastic trends.

Stationary component. Pre-multiplying by β' and using $X_t = X_{t-1} + \Delta X_t$ gives:

$$\beta' X_t = (I_r + \beta'\alpha)\, \beta' X_{t-1} + \beta' \varepsilon_t.$$

The process $\beta' X_t$ therefore follows a stable VAR(1) in $\mathbb{R}^r$, provided that all eigenvalues of $(I_r + \beta'\alpha)$ lie strictly inside the unit circle — equivalently, that all eigenvalues of $\beta'\alpha$ have negative real parts. Under this condition, iterating forward yields the stationary moving-average representation:

$$\beta' X_t = \sum_{i=0}^{t-1} (I_r + \beta'\alpha)^i\, \beta' \varepsilon_{t-i} + (I_r + \beta'\alpha)^t\, \beta' X_0,$$

where the initial-condition term decays to zero exponentially.

Full decomposition. Using the identity $I_p = \beta(\beta'\beta)^{-1}\beta' + \beta_\perp(\beta'_\perp\beta_\perp)^{-1}\beta'_\perp$ to project X_t onto the cointegration and orthogonal complement spaces simultaneously yields:

$$\begin{aligned} X_t = {} & \underbrace{\beta_\perp(\alpha'_\perp\beta_\perp)^{-1}\alpha'_\perp \sum_{i=1}^{t} \varepsilon_i}_{\text{common stochastic trends}} + \underbrace{\beta_\perp(\alpha'_\perp\beta_\perp)^{-1}\alpha'_\perp X_0}_{\text{initial condition}} \\ & + \underbrace{\sum_{i=0}^{t-1} \alpha(\beta'\alpha)^{-1}(I_r + \beta'\alpha)^i\, \beta'\varepsilon_{t-i} + \alpha(\beta'\alpha)^{-1}(I_r + \beta'\alpha)^t \beta' X_0}_{\text{stationary component}} \\ = {} & C \sum_{i=1}^{t} \varepsilon_i + C X_0 + \text{stationary component}, \end{aligned}$$

confirming the Granger representation with $C = \beta_\perp(\alpha'_\perp\beta_\perp)^{-1}\alpha'_\perp$.

2.3 Deterministic Components

We now extend the framework to account for deterministic terms. Unlike stationary VAR models, where the impulse response function decays as $\Psi^h \to 0$, in an $I(1)$ system both stochastic shocks and deterministic components can accumulate through the permanent component $C \sum \varepsilon_i$. This makes the specification of deterministic terms a substantive modelling choice, not merely a cosmetic one.

Impulse Response Function. Using the Granger representation, the effect of a shock $\partial\varepsilon_t$ on X_{t+h} is:

$$\frac{\partial X_{t+h}}{\partial \varepsilon_t} = C + C^*_h.$$

Since $C_h^* \to 0$ exponentially, the long-run multiplier is:

$$\lim_{h\to\infty} \frac{\partial X_{t+h}}{\partial \varepsilon_t} = C.$$

The permanent effect of any shock is therefore fully characterised by the matrix C. Notably, the long-run cointegrating relations are unaffected by any shock: $\beta' C = \beta'\beta_\perp(\cdots)\alpha'_\perp = 0$, since $\beta'\beta_\perp = 0$ by definition. A shock in the direction of the adjustment space ($\partial\varepsilon_t = \alpha c$) also leaves levels unchanged in the long run, since $C\alpha = \beta_\perp(\alpha'_\perp\Gamma\beta_\perp)^{-1}\underbrace{\alpha'_\perp\alpha}_{=0}\, c = 0$.

2.3.1 Constant Term

Consider the VECM with an unrestricted constant:

$$\Delta X_t = \alpha\beta' X_{t-1} + \Gamma_1 \Delta X_{t-1} + \mu + \varepsilon_t.$$

Substituting into the Granger representation, the deterministic contribution to the levels is:

$$C\sum_{i=1}^{t}\mu + (C_0^* + C_1^* + \cdots)\mu = C\mu\, t + C^*(1)\mu,$$

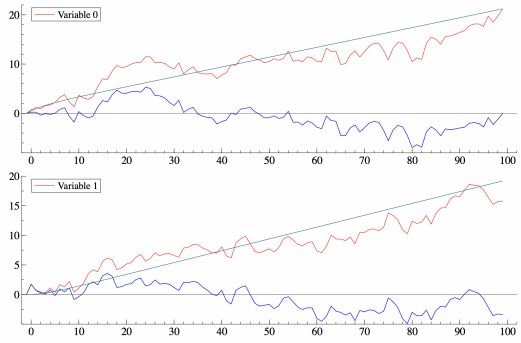
yielding a linear drift $C\mu\, t$ in the levels of X_t. Using the orthogonal projection decomposition:

$$\mu = \alpha\mu_c + \alpha_\perp\gamma_c,$$

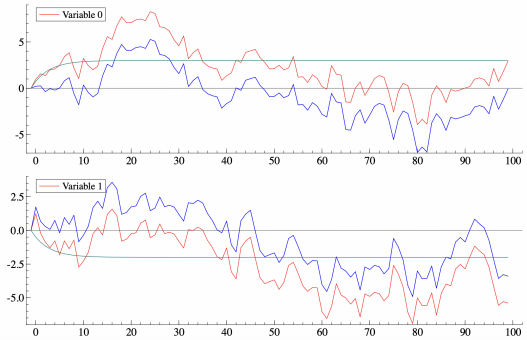
only the component $\alpha_\perp\gamma_c$ — the part of μ orthogonal to the adjustment space — contributes to the stochastic trend through $C\mu\, t$, since $C\alpha = 0$. The component $\alpha\mu_c$ affects only the stationary dynamics. To eliminate the linear drift, we impose the **restricted model**:

$$H_{r,c}:\ \Pi = \alpha\beta', \quad \mu = \alpha\mu_c,$$

so that $C\mu = C\alpha\mu_c = 0$ and no deterministic trend appears in the levels.



(a) Levels of the cointegrated system: a linear drift is present when μ is not proportional to α.



(b) Stationary cointegrating combination $\beta' X_t$, confirming mean reversion.

Figure 2.1: Simulated cointegrated VAR under an unrestricted constant. The drift in the levels disappears once the constant is restricted to lie in the adjustment space col(α).

2.3.2 Trend Term

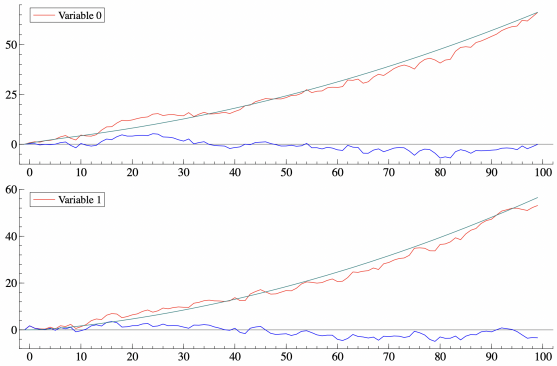
Adding an unrestricted linear trend to the VECM:

$$\Delta X_t = \alpha\beta' X_{t-1} + \Gamma_1 \Delta X_{t-1} + \mu + \tau\, t + \varepsilon_t,$$

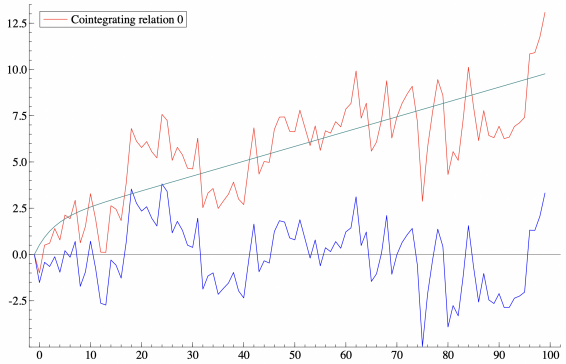
the deterministic contribution to the levels becomes:

$$C\sum_{i=1}^{t}(\mu + \tau\, i) = C\mu\, t + C\tau \cdot \frac{t(t+1)}{2} = \frac{1}{2}C\tau\, t^2 + \left(C\mu + \frac{1}{2}C\tau\right) t.$$

An unrestricted linear trend in the VECM therefore generates a *quadratic* deterministic component in the levels. To eliminate this, we require $C\tau = 0$, which holds when $\tau = \alpha\tau_l$ lies in the adjustment space.



(a) Levels X_{1t} and X_{2t}: both series exhibit a common stochastic trend together with a deterministic linear drift.



(b) Cointegrating relation $\beta' X_t$: the deterministic linear trend is visible when τ is unrestricted.

Figure 2.2: Simulated cointegrated VAR with unrestricted constant and trend. The left panel shows the level series; the right panel shows the cointegrating combination. The quadratic trend disappears in the levels when $\tau = \alpha\tau_l$.

2.3.3 Balanced Models and Deterministic Terms

A systematic treatment of deterministic components is provided by the *balanced model* framework. Write:

$$X_t = X_t^* + \phi D_t,$$

where X_t^* is the purely stochastic component satisfying the standard VECM and D_t collects deterministic terms (constants, trends, impulse and step dummies). Substituting into the VECM and rearranging yields:

$$\Delta X_t = \alpha \begin{pmatrix} \beta \\ \tau_D \end{pmatrix}' \begin{pmatrix} X_{t-1} \\ D_{t-1} \end{pmatrix} + \Gamma_1 \Delta X_{t-1} + \phi_0 \Delta D_t + \phi_1 \Delta D_{t-1} + \varepsilon_t.$$

This formulation is *balanced* in the sense that every deterministic component enters the model consistently across both the long-run and short-run parts, without artificially inducing stochastic trends. Large one-time shocks are captured by impulse dummies $d_t = \mathbf{1}(t = T_0)$; changes in equilibrium levels by step dummies $D_t = \mathbf{1}(t \geq T_0)$. After determining the cointegration rank r, restrictions such as $\tau_D = 0$ can be tested to assess whether equilibrium relations are stable. Finally, the asymptotic distribution of all rank tests depends on which deterministic terms are

included, making correct specification of deterministic components essential for valid inference on r.

2.4 Likelihood Ratio Test for the Cointegration Rank

Consider the VECM:

$$\Delta X_t = \Pi X_{t-1} + \sum_{i=1}^{k-1} \Gamma_i \, \Delta X_{t-i} + \varepsilon_t, \qquad H_r : \; \Pi = \alpha\beta', \quad \alpha, \beta \in \mathbb{R}^{p \times r}.$$

The parameter vector is $\theta = \{\alpha, \beta, \Gamma_1, \ldots, \Gamma_{k-1}\}$. Under Gaussianity the log-likelihood is:

$$\log L(\theta, \Omega) = -\frac{Tp}{2} \log(2\pi) - \frac{T}{2} \log |\Omega| - \frac{1}{2} \sum_{t=1}^{T} \varepsilon_t(\theta)' \Omega^{-1} \varepsilon_t(\theta).$$

Since the nuisance parameters $\{\Gamma_i\}$ enter linearly, we apply the **Frisch–Waugh–Lovell theorem** to partial them out. Define residual vectors:

$$W_{0t} = \Delta X_t, \quad W_{1t} = X_{t-1}, \quad W_{2t} = \Delta X_{t-1},$$

and project both W_{0t} and W_{1t} on W_{2t}:

$$W_{0t} = \hat{A}_0 W_{2t} + R_{0t}, \qquad W_{1t} = \hat{A}_1 W_{2t} + R_{1t}.$$

The concentrated regression is $R_{0t} = \alpha\beta' R_{1t} + \varepsilon_t$, with moment matrices $S_{ij} = T^{-1} \sum_t R_{it} R_{jt}'$. Given β, the OLS estimator of α is:

$$\hat{\alpha}(\beta) = S_{01}\beta(\beta' S_{11} \beta)^{-1},$$

and the concentrated residual covariance is:

$$\hat{\Omega}(\beta) = S_{00} - S_{01}\beta(\beta' S_{11}\beta)^{-1}\beta' S_{10}.$$

Maximising the likelihood is equivalent to minimising $|\hat{\Omega}(\beta)|$. Applying the matrix determinant lemma:

$$|\hat{\Omega}(\beta)| = |S_{00}| \cdot \frac{|\beta'(S_{11} - S_{10}S_{00}^{-1}S_{01})\beta|}{|\beta' S_{11} \beta|}.$$

Since $|S_{00}|$ is constant in β, the optimisation reduces to finding the directions β that minimise the ratio of residual to total variation in R_{1t}. This is a **generalised eigenvalue problem**:

$$|\lambda \, S_{11} - S_{10} S_{00}^{-1} S_{01}| = 0,$$

whose solutions $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_p \geq 0$ are the squared **canonical correlations** between R_{0t} and R_{1t}. Each $\hat{\lambda}_j$ measures how predictable the j-th linear combination of X_{t-1} is from the short-run dynamics; as $\hat{\lambda}_j \to 0$ the corresponding direction contributes nothing to prediction and is not a cointegrating relation. The optimal β collects the eigenvectors corresponding to the r largest eigenvalues:

$$\hat{\beta} = (\hat{V}_1, \ldots, \hat{V}_r), \qquad \hat{\alpha} = S_{01}\hat{\beta}.$$

2.5 Likelihood Ratio Testing for the Rank

Under H_r, models form a nested sequence:

$$H_0 \subset H_1 \subset \cdots \subset H_r \subset \cdots \subset H_{p-1} \subset H_p.$$

$$\underbrace{\text{VAR in } \Delta X}_{H_0} \qquad \underbrace{\text{Cointegrated VAR}}_{H_1 \subset \cdots \subset H_{p-1}} \qquad \underbrace{\text{Unrestricted VAR in } X}_{H_p}$$

Any pair $H_r \subset H_p$ can be tested using the **trace statistic**:

$$\text{LR}(H_r \mid H_p) = -T \sum_{i=r+1}^{p} \log(1 - \hat{\lambda}_i),$$

which aggregates the evidence that eigenvalues $\hat{\lambda}_{r+1}, \ldots, \hat{\lambda}_p$ are zero — equivalently, that there are no more than r cointegrating relations. Inference is conducted sequentially from H_0 upward, stopping at the first rank that is not rejected.

In practice, determining the cointegration rank requires triangulating several sources of evidence.

- Apply the trace test sequentially from H_0 to H_p, keeping in mind that the test may suffer from size distortions and low power in small samples.
- Verify that the selected model implies $p - r$ unit roots in the companion form; empirical unit roots in excess of $p - r$ suggest over-rejection of rank.
- Inspect the t-values of $\hat{\alpha}$ to identify which variables contribute actively to error correction. This is informal but informative, noting that standard errors rely on non-standard asymptotics.
- Examine the time series of $\hat{\beta}' X_t$ to assess visually whether the estimated cointegrating relations are indeed mean-reverting.

2.6 Asymptotics for Cointegration Analysis

We start now considering a univariate random walk such that $x_t \in \mathbb{R}^1$ such that:

$$x_t = x_{t-1} + \varepsilon_t$$

we assume for simplicity that $x_0 = 0$ and fixed. Now we could rewrite the entire process recursively and obtain a MA(∞) representation, therefore:

$$x_t = \sum_{i=1}^{t} \varepsilon_i$$

here we consider: *(i)* $x_0 = 0$, *(ii)* $E[x_t] = E[\sum_{i=1}^{t} \varepsilon_i] = 0$a and $V[x_t] = t\sigma_\varepsilon^2$, *(iii)* $\Delta x_t \perp \Delta x_{t-k}$ for any $k \neq 0$. We observe that so far the model is in discrete time, we could in principle define the continuous-time equivalent of a random walk. Therefore we introduce the Brownian motion:

The **Brownian motion** $\mathcal{B}_u$ with variance σ^2 defined on $u \in [0,1]$ is a stochastic process with the properties:

1. **Initial:** $\mathcal{B}_0 = 0$.
2. **Distribution:** $\mathcal{B}_u$ is $\mathrm{N}(0, u\sigma^2)$ distributed for all $u \in [0,1]$.
3. **Increments** $(\mathcal{B}_{u_2} - \mathcal{B}_{u_1}), \ldots, (\mathcal{B}_{u_k} - \mathcal{B}_{u_{k-1}})$ are independent.
4. $\mathcal{B}_u$ is continuous as a function of u $\big(\in C(0,1)\big)$.

The key idea for moving from the discrete time to the continuous time is given by change in scale, therefore:

$$X_u^T := \frac{1}{\sqrt{T}} \sum_{i=1}^{\lfloor Tu \rfloor} \varepsilon_i, \quad u \in [0,1]$$

SEE Theorem 3.2/3.3

now as $T \to \infty$ we can apply the **Donsker's Theorem** and obtain:

Donsker's Theorem

$$X_u^T := \frac{1}{\sqrt{T}} \sum_{i=1}^{\lfloor Tu \rfloor} \varepsilon_i \to \mathcal{B}_u$$

This hlds for i.i.d ε_t with $E|\varepsilon_t|^2 < \infty$ and ε_t as MDS. *Proof*: We can prove this by ensuring that CLT holds at any point in time and ensuring tightness.

- **Finite Dimensional Convergence**: for any $0 \le u_1 < \ldots < u_k \le 1$ then $(X_{i_1}^T, \ldots, X_{i_k}^T) \xrightarrow{d} (\mathcal{B}_{u_1}, \ldots, \mathcal{B}_{u_k})$. We show this by rewriting:

$$\begin{aligned} X_u^T &:= \frac{1}{\sqrt{T}} \sum_{i=1}^{\lfloor Tu \rfloor} \varepsilon_i \\ &= \underbrace{\frac{\sqrt{Tu}}{\sqrt{T}}}_{\xrightarrow{t\to\infty} \sqrt{u}} \underbrace{\left(\frac{1}{\sqrt{Tu}} \sum_{i=1}^{\lfloor Tu \rfloor} \varepsilon_i \right)}_{\xrightarrow{d} N(0,\sigma^2)} \xrightarrow{d} N(0, u\sigma^2) \end{aligned}$$

- **Tightness**: if for $0 \le r \le s \le u \le 1$,

$$\boxed{E\left[\left(X_s^T - X_r^T\right)^2 \left(X_u^T - X_s^T\right)^2 \right] \le c\,(u-r)^2.}$$

  It follows that e.g. $c = 4\sigma^2$ here, using that ϵ_i are independent.

2.6.1 Asymptotics of AR(1)

We now proceed with the asymptotics of an AR(1) therefore we consider:

$$X_t = \rho X_{t-1} + \varepsilon_t \qquad \Delta X_t = \pi X_{t-1} + \varepsilon_t$$

Now the maximum likelihood estimator gives us:

$$\hat{\rho} - \rho_0 = \left(\sum_{t=1}^{T} \varepsilon_t X_{t-1}\right)\left(\sum_{t=1}^{T} X_{t-1}^2\right)^{-1} = S \times \mathcal{I}^{-1} \tag{2.6}$$

We can therefore decompose both and obtain what follows:

- **Information** $\mathcal{I} = T^{-2}\sum_{t-1}^{T} X_{t-1}^2$ so we get:

$$\begin{aligned} T^{-2}\sum_{t-1}^{T} X_{t-1}^2 &= \frac{1}{T}\sum_{u=1/T}^{1}\left(\frac{1}{\sqrt{T}}\sum_{i=1}^{[Tu]}\varepsilon_i\right)^2 \\ &= \frac{1}{T}\sum_{u=1/T}^{[Tu]} f\left(\frac{t}{T}\right), \text{ we recognize Riemann sum} \\ &= \int_0^1 \left(\frac{1}{\sqrt{T}}\sum_{i=1}^{[Tu]}\varepsilon_i\right)^2 du \xrightarrow{d} \int_0^1 \mathcal{B}^2(u)du \end{aligned}$$

- **Score** $\mathcal{S} = T^{-1}\sum_{t-1}^{T} X_{t-1}\varepsilon_t$ now we get:

rewrite in terms of X_t assuming rho =1 so x_t = X_t-1 +espsilon = Delta X_t = epsilon_t

$$\begin{aligned} T^{-1}\mathcal{S} &= T^{-1}\sum_{t=1}^{T}\left(\sum_{i=1}^{t-1}\epsilon_i\right)\epsilon_t \\ &= \sum_{u=1/T}^{1}\left(\frac{1}{\sqrt{T}}\sum_{i=1}^{[Tu]-1}\epsilon_i\right)\Delta\left(\frac{1}{\sqrt{T}}\sum_{i=1}^{[Tu]}\epsilon_i\right) \\ &\simeq \sum_{u=1/T}^{1}\mathcal{B}_u\,\Delta\mathcal{B}_u \xrightarrow{\text{“}D\text{”}} \int_0^1 \mathcal{B}\,d\mathcal{B}. \end{aligned}$$

Finally we obtain that $T(\hat{\rho}-1) = T\hat{\pi} = T(\mathcal{S}\times\mathcal{I}^{-1}) \to_d \frac{\int_0^1 \mathcal{B}\,d\mathcal{B}}{\int_0^1 \mathcal{B}^2(u)du}$ and we notice that the estimator is also super consistent.

The LR test for the hypothesis that $\rho_0 = 1$ is given by,

$$LR(\rho_0 = 1) = T\log(1+W_T), \qquad W_T = T(\hat{\rho}-1)^2\frac{\sum_{t=1}^{T} X_{t-1}^2}{\hat{\sigma}^2}.$$

and this is asymptotically Dickey-Fuller (DF) type distributed,

$$LR(\rho_0 = 1) \xrightarrow{D} \frac{\left(\int_0^1 \mathcal{W}_u\,d\mathcal{W}\right)^2}{\int_0^1 \mathcal{W}_u^2\,du} \qquad \text{as} \quad T\to\infty.$$

Multivariate Donsker and the Rank Test

Multivariate Donsker's Theorem

In terms of cointegration we test $H_0 : \Pi = 0$, where the model is

$$\boxed{\Delta X_t = \Pi X_{t-1} + \varepsilon_t,} \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} N_p(0, \Omega).$$

Here $\Pi = A - I_p$, and $H_0 : \operatorname{rank}(\Pi) = 0 \Leftrightarrow \Pi = 0$, so under the DGP we have

$$\Delta X_t = \varepsilon_t \qquad \Longrightarrow \qquad X_t = X_0 + \sum_{i=1}^{t} \varepsilon_i,$$

a **multivariate random walk**. From the univariate result we know that

$$\frac{1}{\sqrt{T}} \sum_{i=1}^{[Tu]} e_i \xrightarrow{d} \mathcal{B}(u), \qquad e_i \in \mathbb{R}.$$

We want to extend this to the **bivariate case**, i.e. show that

$$\frac{1}{\sqrt{T}} \sum_{i=1}^{[Tu]} \begin{pmatrix} \varepsilon_{1i} \\ \varepsilon_{2i} \end{pmatrix} \xrightarrow{d} \underbrace{\begin{pmatrix} \mathcal{B}_1(u) \\ \mathcal{B}_2(u) \end{pmatrix}}_{=: \mathcal{B}(u)},$$

where $\mathcal{B}(u)$ is a bivariate Brownian motion with $\mathcal{B}(u) \sim N_2(0, u\Omega)$. The key idea is to **reduce the multivariate problem to a univariate one** via an arbitrary linear combination. Fix $\lambda = (\lambda_1, \lambda_2)' \in \mathbb{R}^2$ and define the scalar process

$$e_t := \lambda' \varepsilon_t = \underbrace{\lambda_1 \varepsilon_{1t} + \lambda_2 \varepsilon_{2t}}_{\text{univariate scalar}}.$$

Since $\varepsilon_t \overset{\text{i.i.d.}}{\sim} N_2(0, \Omega)$, we have $e_t \overset{\text{i.i.d.}}{\sim} N(0, \lambda' \Omega \lambda)$, and the **univariate Donsker theorem** applies:

$$\frac{1}{\sqrt{T}} \sum_{i=1}^{[Tu]} \underbrace{(\lambda_1 \varepsilon_{1i} + \lambda_2 \varepsilon_{2i})}_{= \lambda' \varepsilon_i} \xrightarrow{d} \mathrm{BM}\big(\lambda_1^2 \sigma_{11} + \lambda_2^2 \sigma_{22} + 2\lambda_1 \lambda_2 \sigma_{12}\big) = \mathrm{BM}(\lambda' \Omega \lambda),$$

where $\Omega = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}$. Compactly:

$$\boxed{\frac{1}{\sqrt{T}} \sum_{i=1}^{[Tu]} \lambda' \varepsilon_i \xrightarrow{d} \mathrm{BM}(\lambda' \Omega \lambda) \quad \forall\, \lambda \in \mathbb{R}^2.}$$

The **Cramér–Wold theorem** states that a sequence of random vectors $X_n \in \mathbb{R}^p$ satisfies $X_n \xrightarrow{d} N_p(0, \Sigma)$ if and only if $\lambda' X_n \xrightarrow{d} N(0, \lambda' \Sigma \lambda)$ for every fixed $\lambda \in \mathbb{R}^p$.

Since we have shown above that $\lambda' \Big(\frac{1}{\sqrt{T}} \sum_{i=1}^{[Tu]} \varepsilon_i \Big) \xrightarrow{d} N(0, u\lambda' \Omega \lambda)$ for *all* λ, the Cramér–Wold theorem gives:

$$\frac{1}{\sqrt{T}} \sum_{i=1}^{[Tu]} \varepsilon_i \xrightarrow{d} \mathcal{B}(u) \sim N_2(0,\, u\Omega). \qquad \blacksquare$$

We now consider the eigenvalue problem given by:

$$\left|\lambda S_{11} - S_{10}S_{00}^{-1}S_{01}\right| = 0 \quad \text{or} \quad \left|\lambda I_{p-r} - S_{11}^{-1/2}S_{10}S_{00}^{-1}S_{01}S_{11}^{-1/2}\right| = 0.$$

We recall that the statistic is given by:

$$LR(H_0 \mid H_p) = -T\sum_{i=1}^{p}\log(1-\hat{\lambda}_i) \approx T\sum_{i=1}^{p}\hat{\lambda}_i,$$

where the approximation $-\log(1-x) \approx x$ holds because $\hat{\lambda}_i \to 0$ under H_0. Using the trace identity from before:

$$T\sum_{i=1}^{p}\hat{\lambda}_i = T\,\mathrm{tr}\left(S_{10}S_{00}^{-1}S_{01}S_{11}^{-1}\right) = T\,\mathrm{tr}\left(S_{00}^{-1}S_{01}S_{11}^{-1}S_{10}\right).$$

Asymptotic Limits of the Moment Matrices

Under $H_0 : \Pi = 0$ the DGP is the multivariate random walk $X_t = X_0 + \sum_{i=1}^{t}\varepsilon_i$, and by the multivariate Donsker theorem and the LLN:

$$\boxed{\begin{aligned} S_{00} &= T^{-1}\sum_{t=1}^{T}\varepsilon_t\varepsilon_t' \xrightarrow{p} \Omega, \\ S_{10} &= T^{-1}\sum_{t=1}^{T}x_{t-1}\varepsilon_t' \xrightarrow{D} \int_0^1 \mathcal{B}\,d\mathcal{B}', \\ T^{-1}S_{11} &= T^{-2}\sum_{t=1}^{T}x_{t-1}x_{t-1}' \xrightarrow{D} \int_0^1 \mathcal{B}_u\mathcal{B}_u'\,du. \end{aligned}} \tag{2.7}$$

Note the different normalisation: S_{10} requires only T^{-1} (cross-product of an $I(1)$ and an $I(0)$ object), whereas S_{11} requires T^{-2} (product of two $I(1)$ objects).

Asymptotic Distribution under H_0 (Full Rank Case)

Substituting (1) into the trace expression and applying the Continuous Mapping Theorem:

$$LR(H_0 \mid H_p) \xrightarrow{D} \mathrm{tr}\left(\Omega^{-1}\int_0^1 d\mathcal{B}\mathcal{B}'\left(\int_0^1 \mathcal{B}_u\mathcal{B}_u'\,du\right)^{-1}\int_0^1 \mathcal{B}_u\,d\mathcal{B}'\right).$$

Setting $\mathcal{W} = \Omega^{-1/2}\mathcal{B}$, which is a standard p-dimensional Brownian motion, this is the **$\mathbf{DF^2(p)}$ distribution**.

General Case: Testing H_r against H_p

When testing $H_r : \mathrm{rank}(\Pi) \le r$ against $H_p : \mathrm{rank}(\Pi) = p$, only the $p-r$ smallest eigenvalues $\hat{\lambda}_{r+1}, \ldots, \hat{\lambda}_p$ are relevant. The statistic becomes:

$$LR(H_r \mid H_p) = -T\sum_{i=r+1}^{p}\log(1-\hat{\lambda}_i) \simeq T\sum_{i=r+1}^{p}\hat{\lambda}_i.$$

Under H_r, the r cointegrating directions are absorbed into the stationary part of the model, and only the $p-r$ non-stationary directions drive the asymptotics. Projecting onto the orthogonal complement $\alpha_\perp$ of the loading matrix, the relevant moment matrices become:

$$\boxed{\begin{aligned} \alpha_\perp' S_{10} &= T^{-1}\sum_{t=1}^{T} \alpha_\perp' x_{t-1}\varepsilon_t' \xrightarrow{D} \int_0^1 \mathcal{B}_{p-r}\, d\mathcal{B}_{p-r}', \\ T^{-1}\alpha_\perp' S_{11}\alpha_\perp &= T^{-2}\sum_{t=1}^{T} \alpha_\perp' x_{t-1}x_{t-1}'\alpha_\perp \xrightarrow{D} \int_0^1 \mathcal{B}_{p-r}\mathcal{B}_{p-r}'\, du, \end{aligned}} \tag{2.8}$$

where $\mathcal{B}_{p-r}$ is a $(p-r)$-dimensional Brownian motion. The asymptotic distribution of the test statistic is therefore:

$$LR(H_r \mid H_p) \xrightarrow{D} \operatorname{tr}\left(\int_0^1 d\mathcal{W}_{p-r}\mathcal{W}_{p-r}' \left(\int_0^1 \mathcal{W}_{p-r}\mathcal{W}_{p-r}'\, du\right)^{-1} \int_0^1 \mathcal{W}_{p-r}\, d\mathcal{W}_{p-r}'\right),$$

which is the **$DF^2(p-r)$ distribution**, where $\mathcal{W}_{p-r} = (\alpha_\perp'\Omega\alpha_\perp)^{-1/2}\mathcal{B}_{p-r}$ is a standard $(p-r)$-dimensional Brownian motion.

Testing Sequence

In practice the rank r is unknown and must be determined sequentially. For the VAR model $\Delta X_t = \Pi X_{t-1} + \varepsilon_t$ with $r = \operatorname{rank}(\Pi)$, the procedure is:

1. Test $H_0 : \operatorname{rank}(\Pi) = 0$ vs. H_p: use $DF^2(p)$ 95% quantile. If rejected, proceed.
2. Test $H_1 : \operatorname{rank}(\Pi) \le 1$ vs. H_p: use $DF^2(p-1)$ 95% quantile. If rejected, proceed.

$\vdots$

r. Test H_r : $\operatorname{rank}(\Pi) \le r$ vs. H_p: use $DF^2(p-r)$ 95% quantile. **Accept** r at first non-rejection.

The key insight is that each step uses a *different* critical value from the $DF^2(p-r)$ distribution, since under H_r there are only $p-r$ non-stationary directions remaining, and the dimension of the limiting Brownian motion shrinks accordingly.

2.7 Distribution of Cointegration Vectors

We consider the following model:

$$\Delta X_t = \alpha\beta' X_{t-1} + \varepsilon_t$$

where we define $\Pi = \alpha\beta'$ with $\alpha = (a_1,\, a_2)'$ and $\beta = (1,\, b)'$. The matrix Π is identified from the data via the regression of ΔX_t on X_{t-1}, but the decomposition into α and β separately is *not unique*: for any scalar $c \in \mathbb{R} \setminus \{0\}$, setting $\alpha^* = \alpha c$ and $\beta^* = c^{-1}\beta$ yields

$$\alpha^*\beta^{*\prime} = (\alpha c)(c^{-1}\beta)' = \alpha\beta' = \Pi,$$

so that the pair (α^*, β^*) is observationally equivalent to (α, β). This one-dimensional indeterminacy arises because the data can only identify the *cointegrating space* $\mathrm{sp}(\beta)$, not a specific vector within it. To resolve this, we impose the normalisation $\beta = (1,\, b)'$, which pins the first component to unity and uniquely selects a representative from the cointegrating space, rendering β — and consequently $\alpha = \Pi(\beta'\beta)^{-1}\beta$ — identifiable from the data. The goal is therefore to characterise the distribution of the estimator of this normalised β.

What is the distribution of b? As we will show, we have that the distribution of the b is a *Mixed Gaussian* meaning that it follows $N(0, \tilde{\sigma}^2)$ where the variance is a random variable.

We no proceed by specifying the true cointegration vector such that $\beta = (1,\, b)'$ and we would like to know the distribution of $\hat{\beta} - \beta_0$ such hence we get:

$$\hat{\beta} - \beta_0 = \begin{pmatrix} 1 \\ \hat{b} \end{pmatrix} - \begin{pmatrix} 1 \\ b_0 \end{pmatrix} = \begin{pmatrix} 0 \\ \hat{b} - b_0 \end{pmatrix}$$

Before proceeding, we highlight two key properties of the system that will drive the asymptotic analysis:

- The error-correction term $\beta' X_t$ is stationary by definition of cointegration, so a Law of Large Numbers applies and $T^{-1}\sum_{t=1}^{T} \beta' X_{t-1} X_{t-1}' \beta \xrightarrow{p} \Sigma_{\beta\beta} < \infty$.

- The second component $X_{2,t} = (0 \;\; 1) X_t$ is $I(1)$, representing the common stochastic trend of the system:

  $$X_{2,t} = (\alpha_\perp'\beta_\perp)^{-1}\alpha_\perp' \sum_{i=1}^{t} \varepsilon_i,$$

  driven by the cumulated shocks in the direction orthogonal to the adjustment mechanism α.

So basically here we are saying the stationary part and non-stationary part converge at two different speeds, meaning that the non-stationary is supercosistent converging at a rate of T (at a cost of Brownian motion distribution) while the stationary part $\sqrt{T}$, therefore the β requires a mixed regime of convergence.

In our setting, we assume the $E[\varepsilon_t \varepsilon_t'] = \Omega$ to be known as well as the α vector. We can now starting writing down the likelihood function as follows:

$$L(\beta) = L(b) = -\frac{1}{2}\sum_{t=1}^{T} \left[\log\det(\Omega) + (\Delta X_t - \alpha\beta' X_{t-1})'\,\Omega^{-1}\,(\Delta X_t - \alpha\beta' X_{t-1})\right]$$

so we can derive the score and information, respectively $\mathcal{S}$ and $\mathcal{I}$ as

$$S(b) = \frac{\partial L(b)}{\partial b} = -\frac{1}{2}\sum_{t=1}^{T} \frac{\partial}{\partial b}\left(\Delta X_t - \alpha\beta' X_{t-1}\right)' \Omega^{-1}\left(\Delta X_t - \alpha\beta' X_{t-1}\right)$$

$$= -\frac{1}{2}\sum_{t=1}^{T} 2\underbrace{\frac{\partial}{\partial b}\left(\Delta X_t - \alpha\beta' X_{t-1}\right)'}_{=-\alpha' X_{2t-1}} \Omega^{-1}\left(\Delta X_t - \alpha\beta' X_{t-1}\right)$$

$$\boxed{= \sum_{t=1}^{T}\left(\Delta X_t - \alpha\beta' X_{t-1}\right)' \Omega^{-1}\alpha\, X_{2t-1}}$$

and

$$\mathcal{I}(b) = -\frac{\partial^2 L(b)}{\partial b^2} = -\frac{\partial}{\partial b}\sum_{t=1}^{T}\left(\Delta X_t - \alpha\beta' X_{t-1}\right)' \Omega^{-1}\alpha\, X_{2t-1}$$

$$= -\sum_{t=1}^{T}\underbrace{\frac{\partial}{\partial b}\left(\Delta X_t - \alpha\beta' X_{t-1}\right)'}_{=-X_{2t-1}\,\alpha'} \Omega^{-1}\alpha\, X_{2t-1}$$

$$\boxed{= \sum_{t=1}^{T} X_{2t-1}^2\, \alpha'\Omega^{-1}\alpha}$$

So now we can move and analyse the $X_{2,t}$ as composed by a random walk and a stationary component hence:

$$X_{2t} = (0,1)X_t = (0,1)\,\beta_{0\perp}\left(\alpha'\beta_{0\perp}\right)^{-1}\alpha' \sum_{i=1}^{t}\epsilon_i + \text{stationary} \sim I(1)$$

So we have that:

$$\boxed{\frac{X_{2,[Tu]}}{\sqrt{T}} = (0,1)X_t = (0,1)\,\beta_{0\perp}\left(\alpha'\beta_{0\perp}\right)^{-1}\alpha'\mathcal{B}(u)}$$

so we show that $X_{2,t}$ converges to a Brownian motion, therefore we can exploit that for inserting this information in the score and information:

$$\boxed{\mathcal{I}/T^2 = \left[T^{-2}\sum_{t=1}^{T} X_{2t-1}^2\right]\alpha'\Omega^{-1}\alpha \xrightarrow{D} \left[\int \mathcal{B}_2^2(u)\,du\right]\alpha'\Omega^{-1}\alpha}$$

$$\boxed{S/T = \left[T^{-1}\sum_{t=1}^{T} X_{2t-1}\varepsilon_t'\right]\Omega^{-1}\alpha \xrightarrow{D} \left[\int \mathcal{B}_2\, d\mathcal{B}'\right]\Omega^{-1}\alpha}$$

We finally obtain:

$$T(\hat{b} - b_0) \xrightarrow{d} \left[\int \mathcal{B}_2^2(u)\,du\right]^{-1}\left[\int \mathcal{B}_2\, d\mathcal{B}'\right]\Omega^{-1}\alpha\,(\alpha'\Omega^{-1}\alpha)^{-1}$$

where $\mathcal{B}_2 = (\alpha_\perp'\beta_{0\perp})^{-1}\alpha_\perp'\mathcal{B}$ is the Brownian motion associated with the common stochastic

trend, and is independent of $\alpha'\Omega^{-1}\mathcal{B}$ since

$$\mathbb{E}\big(\alpha'_\perp \mathcal{B}\mathcal{B}'\Omega^{-1}\alpha\big) = \alpha'_\perp\, \Omega\Omega^{-1}\alpha = \alpha'_\perp\alpha = 0,$$

where the last equality follows from the definition of $\alpha_\perp$. This orthogonality, combined with joint Gaussianity of $\mathcal{B}$, implies full independence between $\mathcal{B}_2$ and $\alpha'\Omega^{-1}\mathcal{B}$.

The limiting distribution is therefore *Mixed Gaussian* with *random information*: conditioning on $\mathcal{B}_2$ — equivalently on the realized path of the stochastic trend — the distribution of $T(\hat{b} - b_0)$ is Gaussian with a random variance $\big[\int \mathcal{B}_2^2(u)\,du\big]^{-1}$. Unconditionally, the distribution is non-Gaussian, as the information matrix $\mathcal{I}/T^2 \to \big[\int \mathcal{B}_2^2\,du\big]\,\alpha'\Omega^{-1}\alpha$ is itself a random functional of Brownian motion rather than a fixed constant. This is the fundamental departure from classical OLS asymptotics: the randomness of the information is not noise that vanishes asymptotically, but a permanent feature of the limit experiment driven by the $I(1)$ regressor X_{2t-1}.

Chapter 3

Mathematics of the Portfolio Frontier

This chapter develops the mathematical foundations of mean-variance portfolio theory. Starting from a constrained optimisation problem over N risky assets, we derive the full portfolio frontier, characterise its hyperbolic geometry in mean–standard deviation space, and extend the framework to include a risk-free asset. The central result — that the entire frontier can be spanned by any two distinct frontier portfolios — is the celebrated two-fund separation theorem*, which underlies virtually every classical asset-pricing model.*

3.1 Setting and Assumptions

We work in a static, single-period economy. All returns are net (gross minus one). The following conditions are maintained throughout.

Maintained Assumptions.

[(A1)]

1. There are $N \geq 2$ risky assets traded in a frictionless economy (no taxes, no transaction costs, and shares are perfectly divisible).
2. Short selling is unrestricted; portfolio weights $x_i \in \mathbb{R}$ for all i.
3. Asset expected returns are finite and *not all equal*: $E \equiv \mathbb{E}[\mathbf{r}] \in \mathbb{R}^N$ with $E \neq c\,\mathbf{1}$ for any scalar c.
4. The N asset return vectors are linearly independent.
5. The variance-covariance matrix $V \equiv \mathrm{Var}(\mathbf{r})$ is symmetric positive definite, hence invertible.

Assumption (A5) is the critical structural condition: it rules out exact linear combinations among returns, ensuring that no asset is redundant and that every portfolio has strictly positive variance. Positive definiteness also guarantees a unique global solution to the frontier problem. Assumption (A3) ensures that $D = BC - A^2 > 0$ by the Cauchy-Schwarz inequality, a condition that the entire solution rests on.

Notation.

- $\mathbf{r} \in \mathbb{R}^N$: vector of asset net returns.

- $x \in \mathbb{R}^N$: portfolio weight vector, satisfying $x'\mathbf{1} = 1$ (i.e. $\sum_{i=1}^{N} x_i = 1$, the budget constraint).
- $R_p \equiv x'\mathbf{r}$: portfolio net return.
- $E \equiv \mathbb{E}[\mathbf{r}]$: N-vector of expected returns.
- V: $N \times N$ variance-covariance matrix of returns.
- $\mathbf{1}$: N-vector of ones.

The central problem is the following.

Frontier Problem. *For any target expected return $R \in \mathbb{R}$, find the portfolio weight vector x that minimises portfolio variance $\mathrm{Var}(R_p) = x'Vx$, subject to (i) achieving the target return and (ii) the budget constraint.*

Solving this problem for every admissible R traces out the *portfolio frontier* — the complete set of mean-variance combinations available to investors.

3.2 Derivation of the Portfolio Frontier

3.2.1 The Constrained Optimisation Problem

The frontier problem is formalised as the quadratic programme:

$$\min_{x} \ \frac{1}{2}\, x'Vx \qquad \text{subject to} \qquad x'E = R, \quad x'\mathbf{1} = 1. \tag{3.1}$$

The factor $\frac{1}{2}$ is a normalisation that simplifies the first-order conditions without affecting the solution. Two Lagrange multipliers λ and γ enforce the two linear constraints, yielding the Lagrangian:

$$\mathcal{L}(x, \lambda, \gamma) = \frac{1}{2}\, x'Vx + \lambda\big(R - x'E\big) + \gamma\big(1 - x'\mathbf{1}\big).$$

3.2.2 First-Order Conditions

Differentiating with respect to each argument and setting to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = Vx - \lambda E - \gamma \mathbf{1} = \mathbf{0}, \tag{3.2}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = R - x'E = 0, \tag{3.3}$$

$$\frac{\partial \mathcal{L}}{\partial \gamma} = 1 - x'\mathbf{1} = 0. \tag{3.4}$$

Since V is positive definite it is invertible, so (3.2) gives immediately:

$$x^* = V^{-1}(\lambda E + \gamma \mathbf{1}). \tag{3.5}$$

3.2.3 The Efficient Frontier Scalars

Substituting (3.5) into the two binding constraints and solving the resulting 2×2 linear system (see Appendix) yields the Lagrange multipliers in terms of the four scalar quantities:

$$A = \mathbf{1}'V^{-1}E, \qquad B = E'V^{-1}E, \qquad C = \mathbf{1}'V^{-1}\mathbf{1}, \qquad D = BC - A^2 > 0. \tag{3.6}$$

The multipliers are then:

$$\lambda = \frac{CR - A}{D}, \qquad \gamma = \frac{B - AR}{D}. \tag{3.7}$$

Economic interpretation. C is a precision-weighted measure of the "size" of the asset universe in the direction of the unit vector; B is the precision-weighted sum of squared expected returns; A is a cross-term capturing the inner product between expected returns and the covariance-implied all-ones direction. The positivity of $D = BC - A^2$ follows from the Cauchy-Schwarz inequality together with assumption (A3).

Note that λ is *increasing in* R: a higher target return demands a higher shadow price on the return constraint, pushing the portfolio toward higher-yielding but riskier assets. At the global minimum-variance portfolio, λ attains the value $C \cdot (A/C) - A)/D = 0$, meaning the return constraint is non-binding there.

3.2.4 The Frontier Portfolio Weights

Substituting (3.7) back into (3.5):

$$x^*(R) = \mathbf{g} + \mathbf{h}\, R, \tag{3.8}$$

where

$$\mathbf{g} = \frac{1}{D}\Big[B\, V^{-1}\mathbf{1} - A\, V^{-1}E\Big], \tag{3.9}$$

$$\mathbf{h} = \frac{1}{D}\Big[C\, V^{-1}E - A\, V^{-1}\mathbf{1}\Big]. \tag{3.10}$$

Equation (3.8) is the cornerstone of the entire theory: the weight vector of *every* frontier portfolio is an affine function of the target return R. The vector $\mathbf{g}$ gives the weights of the frontier portfolio with zero expected return, while $\mathbf{h}$ is the direction in weight space along which we move as R increases.

Budget feasibility. We verify that $\mathbf{g}$ is itself a legitimate portfolio:

$$\mathbf{1}'\mathbf{g} = \frac{1}{D}\Big[B(\mathbf{1}'V^{-1}\mathbf{1}) - A(\mathbf{1}'V^{-1}E)\Big] = \frac{BC - A^2}{D} = \frac{D}{D} = 1.$$

Since $\mathbf{1}'x^*(R) = 1$ for all R, differentiating with respect to R gives $\mathbf{1}'\mathbf{h} = 0$: the vector $\mathbf{h}$ is a *zero-net-investment* (self-financing) tilt, not a portfolio in its own right.

3.3 Two-Fund Separation

The affine structure (3.8) implies a dramatic dimensionality reduction.

[Two-Fund Separation] The entire portfolio frontier can be generated by forming affine combinations of *any* two distinct frontier portfolios.

Let p_1 and p_2 be two distinct frontier portfolios with returns $R_1 \neq R_2$ and weights $x^*(R_k) = \mathbf{g} + \mathbf{h}R_k$ for $k = 1, 2$. For any $\alpha \in \mathbb{R}$:

$$\begin{aligned} \alpha\, x^*(R_1) + (1-\alpha)\, x^*(R_2) &= \alpha(\mathbf{g} + \mathbf{h}R_1) + (1-\alpha)(\mathbf{g} + \mathbf{h}R_2) \\ &= \mathbf{g} + \mathbf{h}\big[\alpha R_1 + (1-\alpha)R_2\big] \\ &= x^*\big(\alpha R_1 + (1-\alpha)R_2\big). \end{aligned}$$

As α ranges over $\mathbb{R}$, the target return $\alpha R_1 + (1-\alpha)R_2$ takes every value in $\mathbb{R}$, so every frontier portfolio is reachable.

Economic significance. Two-fund separation means that an investor facing the full N-asset universe can confine attention to two frontier mutual funds, regardless of her risk preferences. She need only decide how to allocate between the two funds — a reduction in dimensionality from N to 1. This result is the theoretical pillar of the CAPM: in equilibrium, if all investors hold frontier portfolios, the market portfolio must itself lie on the frontier.

3.4 Geometry of the Portfolio Frontier

3.4.1 Frontier Variance

For the frontier portfolio $x^*(R) = \mathbf{g} + \mathbf{h}R$, a direct computation gives:

$$\sigma^2(R) \;\equiv\; x^*(R)'\, V\, x^*(R) \;=\; \frac{1}{C} + \frac{C}{D}\left(R - \frac{A}{C}\right)^2. \tag{3.11}$$

This is a *parabola* in (σ^2, R) space. In (σ, R) space it becomes a *hyperbola*.

3.4.2 Covariance Between Two Frontier Portfolios

More generally, for any two frontier portfolios with expected returns R_p and R_q:

$$\mathrm{Cov}(R_p, R_q) = \frac{1}{C} + \frac{C}{D}\left(R_p - \frac{A}{C}\right)\left(R_q - \frac{A}{C}\right). \tag{3.12}$$

Setting $R_p = R_q$ recovers (3.11). This formula will be essential later when deriving the zero-beta CAPM: every frontier portfolio has a unique companion frontier portfolio with which it has zero covariance.

3.4.3 The Hyperbolic Shape

Rearranging (3.11):

$$\frac{\sigma^2(R)}{1/C} \;-\; \frac{(R - A/C)^2}{D/C^2} \;=\; 1, \tag{3.13}$$

which is the standard equation of a hyperbola centred at $\big(\sigma = 0,\; R = A/C\big)$. In (σ, R) space the asymptotes satisfy:

$$R = \frac{A}{C} \;\pm\; \sigma\sqrt{\frac{D}{C}}, \tag{3.14}$$

meaning the frontier becomes increasingly linear as $\sigma \to \infty$.

Key points on the frontier.

- *Global minimum variance portfolio (MVP).* Setting $\mathrm{d}\sigma^2/\mathrm{d}R = 0$ in (3.11) gives $R_{\mathrm{MVP}} = A/C$ and $\sigma^2_{\mathrm{MVP}} = 1/C$. The MVP is the leftmost point of the hyperbola; no portfolio can have lower variance.

- *Efficient frontier.* The upper branch $R \geq A/C$ is the mean-variance efficient frontier. For any given σ, these portfolios deliver strictly higher expected return than their lower-branch counterparts.

- *Inefficient frontier.* The lower branch $R < A/C$ is dominated and would never be chosen by a non-satiated investor.

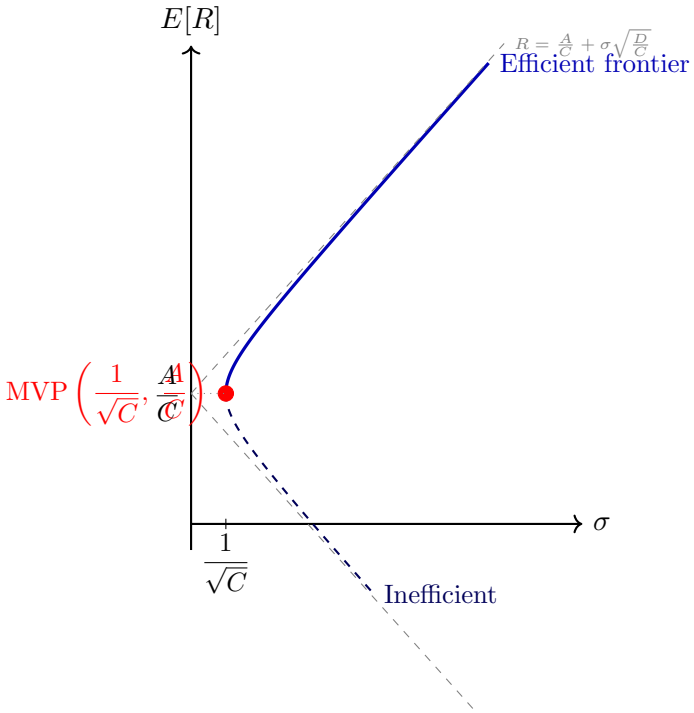


Figure 3.1: Portfolio frontier in mean–standard deviation space. The frontier is a hyperbola whose leftmost point is the global minimum-variance portfolio (MVP) at $\sigma_{\min} = 1/\sqrt{C}$ and $E[R] = A/C$. The efficient upper branch (solid) dominates the inefficient lower branch (dashed) at every level of risk. The dashed lines are the asymptotes along which the frontier becomes linear as $\sigma \to \infty$.

3.5 Including a Risk-Free Asset

We now augment the risky-asset universe with a riskless asset yielding gross return R_f with certainty. This fundamentally changes the frontier geometry.

3.5.1 The Modified Problem

Let $x \in \mathbb{R}^N$ denote weights on risky assets only. The fraction $(1 - x'\mathbf{1})$ is invested in the risk-free asset, so the portfolio return is:

$$R_p = x'\mathbf{r} + (1 - x'\mathbf{1})R_f = R_f + x'(\mathbf{r} - R_f\mathbf{1}).$$

The budget constraint is automatically satisfied. The frontier problem reduces to:

$$\min_x \frac{1}{2} x'Vx \qquad \text{subject to} \qquad x'(E - R_f\mathbf{1}) = R - R_f. \tag{3.15}$$

Only one multiplier is needed. Solving (see Appendix):

$$x^*(R) = V^{-1}(E - R_f\mathbf{1})\,\frac{R - R_f}{H}, \tag{3.16}$$

where

$$H \;\equiv\; (E - R_f\mathbf{1})'V^{-1}(E - R_f\mathbf{1}) \;>\; 0. \tag{3.17}$$

The tangency portfolio. All frontier portfolios with a risk-free asset are proportional to $V^{-1}(E - R_f\mathbf{1})$. Normalising to unit budget weight gives the *tangency portfolio* T, the unique

risky portfolio at which the Capital Market Line is tangent to the risky-only hyperbola.

Interpretation of H. H is the squared Sharpe ratio of the tangency portfolio. It measures the maximum excess return per unit of risk achievable from the risky-asset universe alone, and determines the slope of the entire efficient frontier.

3.5.2 Frontier Variance and the Capital Market Line

Substituting (3.16) into $x'Vx$:

$$\sigma^2(R_p) \;=\; \frac{(R_p - R_f)^2}{H}, \tag{3.18}$$

so that in terms of standard deviation:

$$E[R_p] \;=\; R_f \;+\; \sqrt{H}\,\sigma(R_p), \qquad R_p \geq R_f. \tag{3.19}$$

This is the *Capital Market Line* (CML). The frontier is now a pair of half-lines in $(\sigma, E[R])$ space, both emanating from $(0, R_f)$: the upper half-line (slope $+\sqrt{H}$, efficient) corresponds to going long T, and the lower (slope $-\sqrt{H}$, inefficient) to shorting T.

3.5.3 Two Parametric Cases

The geometry depends critically on whether R_f lies below or above A/C, the expected return of the global minimum-variance portfolio.

Case 1: $R_f < A/C$

This is the empirically relevant regime. The CML from $(0, R_f)$ is tangent to the *upper* (efficient) branch of the risky frontier, at a tangency portfolio T with $E[R_T] > A/C$. All efficient portfolios are convex combinations of T and the risk-free asset for $\alpha \in [0, 1]$, or levered positions in T (borrowing at R_f) for $\alpha > 1$. This is exactly the one-fund result embedded in the CAPM.

Case 2: $R_f > A/C$

When the risk-free rate exceeds the MVP return, the CML is tangent to the *lower* (inefficient) branch of the risky frontier. Efficient allocations now involve *short-selling* the tangency portfolio and holding the proceeds in the risk-free asset. No investor with standard preferences would voluntarily hold risky assets in this regime, making it empirically unusual but theoretically important for understanding the complete structure of the frontier.

3.6 Summary of Key Results

The following table collects the central formulae derived in this chapter.

Object	Formula	Key property
Frontier weights	$x^*(R) = \mathbf{g} + \mathbf{h}R$	Affine in R
Frontier variance	$\sigma^2(R) = \dfrac{1}{C} + \dfrac{C}{D}\left(R - \dfrac{A}{C}\right)^2$	Parabola in (σ^2, R)
MVP return / variance	$R_{\text{MVP}} = A/C, \quad \sigma^2_{\text{MVP}} = 1/C$	Lower bound on variance
CML (with R_f)	$E[R_p] = R_f + \sqrt{H}\,\sigma_p$	Linear; slope = max Sharpe ratio

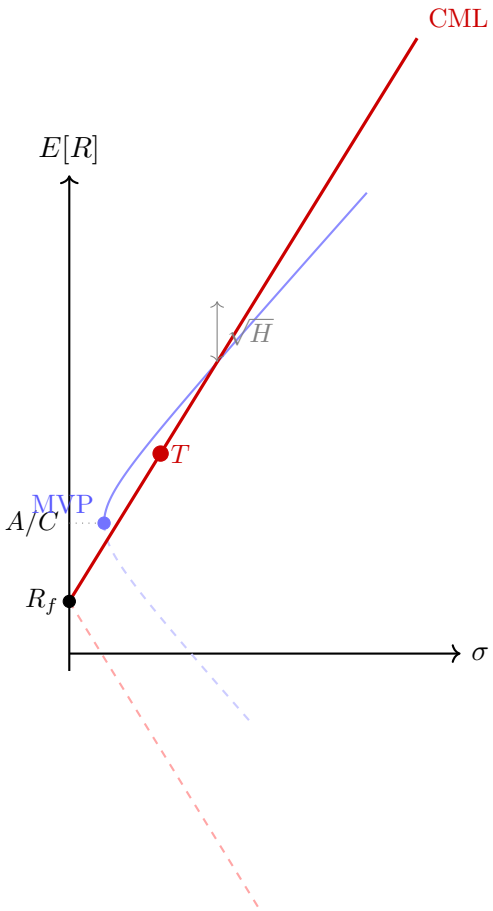


Figure 3.2: Capital Market Line when $R_f < A/C$. The CML (red solid) is tangent to the risky frontier (blue) at the tangency portfolio T. Efficient investors hold T combined with the risk-free asset: long T and lending for moderate risk, levered T (borrowing) for higher risk. The dashed red line below R_f is inefficient.

Looking ahead. The mathematics of this chapter is entirely *positive*: it characterises what is achievable, without imposing preferences. The CAPM adds an *equilibrium* argument: if all investors are mean-variance optimisers and markets clear, the market portfolio must coincide with the tangency portfolio T. This transforms the engineering result (3.18) into the pricing equation

$$\mathbb{E}[R_i] - R_f \;=\; \beta_i\big(\mathbb{E}[R_M] - R_f\big), \qquad \beta_i = \frac{\mathrm{Cov}(R_i, R_M)}{\sigma^2(R_M)},$$

where R_M is the market return — connecting portfolio mathematics directly to the cross-section of expected asset returns.

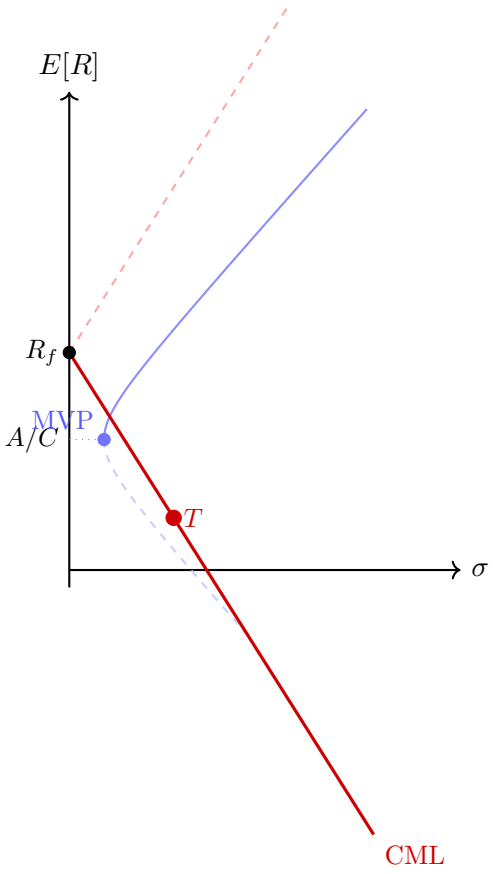


Figure 3.3: Capital Market Line when $R_f > A/C$. The tangency portfolio T lies on the inefficient lower branch. The CML now slopes *downward*: efficient investors short-sell risky assets and hold large positions in the risk-free asset. This case is rarely observed empirically.

Chapter 4

Covariance Estimation and Multivariate GARCH

4.1 Multivariate GARCH models

We now move from a constant covariance matrix Ω to the modelling of a *time-varying* variance-covariance matrix $\Omega_t(\theta)$. This extension is essential for dynamic portfolio optimisation: as new information arrives at each date t, the optimal weights should be updated to reflect the current conditional distribution of returns. The family of Multivariate GARCH models achieves this by specifying a recursion for $\Omega_t(\theta)$ as a function of past data and past conditional covariances, each parametrisation representing a different trade-off between generality and parsimony.

The common starting point for all M-GARCH models is the factor representation:

$$X_t = \Omega_t^{1/2}(\theta)\, z_t, \qquad t = 1, \dots, T, \tag{4.1}$$

where $z_t \overset{\text{i.i.d.}}{\sim} N(0, I_p)$ and $\Omega_t(\theta) \in \mathcal{F}_{t-1}$, meaning that the conditional covariance matrix is measurable with respect to the information set available at time $t-1$. The different M-GARCH models differ in how they parametrise the evolution of $\Omega_t(\theta)$ over time.

4.2 Why We Parametrize at All

When it comes to multivariate ARCH and GARCH models, after specifying the model we move to estimation, relying on the classical tools of maximum or quasi-maximum likelihood (MLE/QMLE). Denoting the parameter vector of interest as $\boldsymbol{\alpha} \in \boldsymbol{\Theta}$, where $\boldsymbol{\Theta}$ is the constrained parameter space, estimation amounts to finding the peak of the log-likelihood surface $\ell(\boldsymbol{\alpha})$ over $\boldsymbol{\Theta}$:

$$\hat{\boldsymbol{\alpha}} =_{\boldsymbol{\alpha} \in \boldsymbol{\Theta}} \ \ell(\boldsymbol{\alpha}).$$

However, certain regions of $\boldsymbol{\Theta}$ may be difficult to explore numerically, since the model imposes constraints on the parameters — for instance, positive definiteness of the conditional covariance matrix $\Sigma_t \succ 0$. A natural remedy is *reparametrization*: we introduce a new parameter vector $\boldsymbol{\theta} \in \mathbb{R}^k$, related to $\boldsymbol{\alpha}$ through a transformation $g : \mathbb{R}^k \to \boldsymbol{\Theta}$, and optimize the reparametrized log-likelihood

$$\hat{\boldsymbol{\theta}} =_{\boldsymbol{\theta} \in \mathbb{R}^k} \ \ell^*(\boldsymbol{\theta}), \qquad \ell^*(\boldsymbol{\theta}) \equiv \ell(g(\boldsymbol{\theta}))\,.$$

This changes the coordinate system underneath the likelihood surface — stretching, compressing, or flipping it — without altering the surface itself. Provided g is a *bijection*, every point $\boldsymbol{\theta} \in \mathbb{R}^k$

This is very important

corresponds to exactly one $\boldsymbol{\alpha} \in \boldsymbol{\Theta}$, and vice versa, so that

$$\sup_{\boldsymbol{\alpha} \in \boldsymbol{\Theta}} \ell(\boldsymbol{\alpha}) = \sup_{\boldsymbol{\theta} \in \mathbb{R}^k} \ell^*(\boldsymbol{\theta}),$$

and the two optimizers are related by $\hat{\boldsymbol{\alpha}} = g(\hat{\boldsymbol{\theta}})$. The optimization is thus carried out freely over $\mathbb{R}^k$, with no constraints, and the estimate of interest is recovered via the inverse map g^{-1}.

4.2.1 Core Discussion

The fundamental challenge in multivariate volatility modeling is deceptively simple to state but technically demanding to solve: a conditional covariance matrix Ω_t must be **symmetric positive definite at every point in time**, for every realization of the data. In the univariate case, ensuring $\sigma_t^2 > 0$ is straightforward — constraining a scalar to stay positive requires only $\omega > 0$ and $\alpha, \beta \geq 0$ in a GARCH(1,1). In the multivariate case, positive definiteness of a $p \times p$ matrix is a *global, nonlinear* constraint on all $p(p+1)/2$ unique parameters simultaneously, and it cannot be handled by simple sign restrictions on individual entries.

More precisely, $\Omega > 0$ (positive definite) means:

$$v'\Omega v > 0 \quad \text{for all } v \in \mathbb{R}^p,\ v \neq 0, \tag{4.2}$$

which equivalently says that every portfolio $v'X_t$ has strictly positive variance. This is an economically meaningful restriction: no linear combination of assets can be riskless unless a perfect hedge exists by construction.

The log-likelihood for an MGARCH model with Gaussian innovations takes the form

$$\mathcal{L}(\theta) = -\frac{T}{2} \log |\Omega_t(\theta)| - \frac{1}{2} \sum_{t=1}^{T} X_t' \Omega_t(\theta)^{-1} X_t, \tag{4.3}$$

and numerical maximization requires that $\Omega_t(\theta) > 0$ throughout the entire optimization path, not just at the solution. A parametrization $\Omega = \Omega(\theta)$ with $\theta \in \mathbb{R}^n$ freely varying — imposing positive definiteness *by construction* — therefore serves three purposes simultaneously:

1. It eliminates the need for constrained optimization, replacing a geometrically complicated feasible set with all of $\mathbb{R}^n$.

2. It ensures that the recursive dynamics of Ω_t remain valid covariance matrices at every time step, not only in expectation.

3. It provides a structured language for proposing dynamic MGARCH models, because each reparametrization of the static Ω naturally suggests a dynamic extension in which one or more of its components are made time-varying.

The remainder of this section reviews the four main parametrizations in detail, discussing for each: its construction, whether positive definiteness is guaranteed, the number of free parameters, and the dynamic MGARCH model it gives rise to. Throughout, Ω is a $p \times p$ symmetric positive definite covariance matrix with $n = p(p+1)/2$ unique entries, and $X_t \mid \mathcal{F}_{t-1} \sim N_p(0, \Omega_t)$, where $\mathcal{F}_{t-1} = (X_{t-1}, X_{t-2}, \ldots)$.

4.3 The Direct vech Parametrization

4.3.1 Construction

The most natural way to parametrize Ω is to collect its unique entries into a vector using the *vech* operator, which stacks the lower triangular part of a matrix column by column:

$$\text{vech}(\Omega) = (\sigma_{11}, \sigma_{21}, \ldots, \sigma_{p1}, \sigma_{22}, \ldots, \sigma_{pp})' \in \mathbb{R}^n, \quad n = \frac{p(p+1)}{2}. \tag{4.4}$$

The parameters $\theta = (\theta_1, \ldots, \theta_n)'$ are set equal to the elements of $\text{vech}(\Omega)$, and $\Omega(\theta)$ is recovered by the inverse operation $\text{unvech}(\theta)$, which fills in both triangles to restore symmetry.

4.3.2 Positive Definiteness

Symmetry is guaranteed by construction, but positive definiteness is *not*. For the bivariate case ($p = 2$), the constraints reduce to

$$\theta_1 > 0, \quad \theta_3 > 0, \quad \theta_1\theta_3 > \theta_2^2, \tag{4.5}$$

which are easy to state but hard to enforce during unconstrained numerical optimization. For $p > 2$ the analogous constraints involve all leading principal minors, becoming rapidly intractable. Individual positivity constraints can be handled by reparametrizing $\theta_i = \exp(\tilde{\theta}_i)$, making $\theta_i > 0$ automatic for all $\tilde{\theta}_i \in \mathbb{R}$, but the cross-constraints have no analogous scalar trick.

4.3.3 Parameter Count and Dynamic Extension

The vech parametrization has $n = p(p+1)/2$ parameters for the static covariance. When extended to the dynamic **vech-GARCH(1,1)** model,

$$\text{vech}(\Omega_t) = C + A\ \text{vech}(X_{t-1}X_{t-1}') + B\ \text{vech}(\Omega_{t-1}), \tag{4.6}$$

the matrices A and B are each $(n \times n)$-dimensional, giving a total of $\dim(\theta) = n(2n+1)$ free parameters — already 78 for $p = 3$ and 210 for $p = 4$. The vech-GARCH is therefore rarely estimated in its unrestricted form. It plays an important theoretical role, however, because any MGARCH model can be rewritten as a restricted vech-GARCH, enabling unified probabilistic analysis via stochastic recurrence equation (SRE) theory.

[Stationarity condition] For the vech-GARCH(1,1) with $C_{\text{vech}} > 0$, stationarity and geometric ergodicity with finite second moments hold if and only if

$$\varrho(A_{\text{vech}} + B_{\text{vech}}) < 1, \tag{4.7}$$

where $\varrho(\cdot)$ denotes the spectral radius. This is the direct multivariate analogue of the univariate condition $\alpha + \beta < 1$.

4.4 The Cholesky Parametrization

4.4.1 Construction

Every symmetric positive definite matrix Ω admits a unique decomposition

$$\Omega = LL', \tag{4.8}$$

where L is a lower triangular matrix with strictly positive diagonal entries $\ell_{ii} > 0$. The parameters are $\theta = \text{vech}(L) \in \mathbb{R}^n$, with $n = p(p+1)/2$. The diagonal positivity constraint is

enforced by reparametrizing $\ell_{ii} = \exp(\tilde{\ell}_{ii})$, $\tilde{\ell}_{ii} \in \mathbb{R}$, while all off-diagonal entries ℓ_{ij}, $i > j$, remain unconstrained. For $p = 2$ the factorization is explicit:

$$\Omega = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix} = \begin{pmatrix} \ell_{11} & 0 \\ \ell_{21} & \ell_{22} \end{pmatrix} \begin{pmatrix} \ell_{11} & \ell_{21} \\ 0 & \ell_{22} \end{pmatrix}, \tag{4.9}$$

giving $\ell_{11} = \sigma_{11}^{1/2}$, $\ell_{21} = \sigma_{12}/\sigma_{11}^{1/2}$, and $\ell_{22} = (\sigma_{22} - \sigma_{12}^2/\sigma_{11})^{1/2}$.

4.4.2 Positive Definiteness

Positive definiteness is **guaranteed by construction**: for any $\theta \in \mathbb{R}^n$ with $\ell_{ii} > 0$, the matrix $\Omega(\theta) = L(\theta)L(\theta)'$ is positive definite. This makes the Cholesky factorization one of the most numerically stable parametrizations available.

4.4.3 Interpretation and Dynamic Extension

The entries of L do not carry direct interpretations as variances or covariances, with the single exception that $\hat{\ell}_{11} = \hat{\sigma}_{11}^{1/2}$ is the estimated standard deviation of X_{1t}. Despite this lack of interpretability, the Cholesky parametrization is the standard device for specifying the constant term in the **BEKK-GARCH(1,1)**:

$$\Omega_t(\theta) = LL' + AX_{t-1}X_{t-1}'A' + B\Omega_{t-1}(\theta)B', \tag{4.10}$$

where positive definiteness of Ω_t follows because $LL' > 0$ and $AX_{t-1}X_{t-1}'A' \geq 0$ for any matrix A.

[Non-uniqueness] The Cholesky factor is unique only when diagonal entries are required to be strictly positive. Without this restriction, multiplying any row of L by -1 yields another valid factorization, creating 2^p observationally equivalent representations. The convention $\ell_{ii} > 0$ resolves this identification problem.

4.5 The Correlation Parametrization

4.5.1 Construction

The correlation parametrization factorizes Ω into marginal standard deviations and pairwise correlations:

$$\Omega = D\,\Gamma\,D, \tag{4.11}$$

where $D = \mathrm{diag}(\sigma_1, \ldots, \sigma_p)$ with $\sigma_i > 0$, and $\Gamma = (\rho_{ij})_{i,j=1,\ldots,p}$ is a correlation matrix, i.e. a symmetric positive definite matrix with unit diagonal. The parameter vector contains p variances and $p(p-1)/2$ correlations, for a total of $n = p(p+1)/2$ parameters.

4.5.2 Positive Definiteness

Positive definiteness of Ω reduces to positive definiteness of Γ. For $p = 2$, the single condition $|\rho_{12}| < 1$ is both necessary and sufficient, enforced by

$$\rho_{12} = \frac{\exp(\tilde{\rho}) - 1}{\exp(\tilde{\rho}) + 1}, \quad \tilde{\rho} \in \mathbb{R}. \tag{4.12}$$

For $p > 2$, the conditions $|\rho_{ij}| < 1$ are necessary but *not sufficient*. A general solution is the hyper-spherical parametrization $\Gamma(\theta) = G(\theta)G(\theta)'$, where $G(\theta)$ is built from trigonometric rotation matrices involving $\cos(\theta_{ij})$ and $\sin(\theta_{ij})$, guaranteeing $\Gamma > 0$ for any $\theta \in \mathbb{R}^{p(p-1)/2}$.

4.5.3 Interpretation and Dynamic Extension

This parametrization is economically transparent: D captures asset-level volatility and Γ captures co-movement. The natural dynamic extension is the **CCC-MGARCH** model, where each σ_{it}^2 follows a univariate GARCH specification while Γ remains constant:

$$\Omega_t(\theta) = D_t(\theta)\,\Gamma(\theta)\,D_t(\theta), \quad \sigma_{it}^2 = \omega_i + \alpha_i X_{it-1}^2 + \beta_i \sigma_{it-1}^2. \tag{4.13}$$

Making Γ time-varying via a BEKK-type recursion on a proxy matrix Q_t yields the **DCC-MGARCH** of Engle (2002):

$$Q_t(\theta) = (1-\alpha-\beta)\bar{Q} + \alpha \tilde{X}_{t-1}\tilde{X}_{t-1}' + \beta Q_{t-1}(\theta), \tag{4.14}$$

$$\Gamma_t(\theta) = \operatorname{diag}(Q_t)^{-1/2}\, Q_t\, \operatorname{diag}(Q_t)^{-1/2}, \tag{4.15}$$

where $\tilde{X}_t = D_t^{-1} X_t$ are the standardized returns and $\bar{Q}$ is the unconditional covariance of $\tilde{X}_t$.

[Two-step estimation] A computationally attractive feature of the correlation parametrization is that it admits a two-step estimator: first estimate each $\hat{\sigma}_{it}^2$ from univariate GARCH models, then estimate Γ from the standardized residuals. Under standard regularity conditions this two-step estimator is equivalent to the full MLE.

4.6 The Eigenvalue–Eigenvector (Spectral) Parametrization

4.6.1 Construction

The spectral decomposition of any symmetric positive definite Ω is

$$\Omega = V \Lambda V', \quad V'V = VV' = I_p, \quad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_p), \quad \lambda_1 \geq \cdots \geq \lambda_p > 0. \tag{4.16}$$

The parametrization separates Ω into:

- **Eigenvalues** $(\lambda_i)_{i=1}^p$: freely parametrized as $\lambda_i = \exp(\tilde{\lambda}_i)$, $\tilde{\lambda}_i \in \mathbb{R}$.
- **Eigenvectors** $V(\theta)$: the orthonormal rotation matrix, parametrized via $p(p-1)/2$ rotation angles $(\theta_{ij})_{i<j}$ using products of Givens rotation matrices $V = \prod_{i=1}^{p-1} \prod_{j=i+1}^{p} R(i,j;\,\theta_{ij})$.

The total parameter count is $n = p + p(p-1)/2 = p(p+1)/2$.

4.6.2 Positive Definiteness

Positive definiteness is **guaranteed by construction** for any $(\tilde{\lambda}_i, \theta_{ij}) \in \mathbb{R}^n$, since $\exp(\tilde{\lambda}_i) > 0$ and $VV' = I_p$ by construction.

4.6.3 Interpretation and Dynamic Extension

The key analytical advantage is that the log-likelihood *decouples* under this parametrization. Defining the rotated series $\tilde{X}_t = V(\theta)' X_t$, one obtains

$$\mathcal{L}(\theta) = -\frac{T}{2} \sum_{i=1}^{p} \Big(\log \lambda_i + S_{\tilde{x}_i \tilde{x}_i} / \lambda_i \Big), \quad S_{\tilde{x}_i \tilde{x}_i} = T^{-1} \sum_{t=1}^{T} \tilde{X}_{it}^2, \tag{4.17}$$

so that Ω is estimated as p independent univariate variance problems on the rotated series, connected only through $V(\theta)$. This structure motivates the **λ-MGARCH** model of Hetland, Pedersen, and Rahbek (2021), in which the eigenvectors V are held constant while the eigenvalues

λ_{it} follow univariate GARCH-type dynamics:

$$\Omega_t(\theta) = V(\theta)\,\Lambda_t(\theta)\,V(\theta)', \quad \Lambda_t = \mathrm{diag}(\lambda_{1t}, \ldots, \lambda_{pt}), \tag{4.18}$$

$$\lambda_t(\theta) = C + A\big(V(\theta)'X_{t-1}\big)^2 + B\,\lambda_{t-1}(\theta), \tag{4.19}$$

where $(\cdot)^2$ denotes element-wise squaring. The model may thus be viewed as p univariate GARCH models for the principal components $\tilde{X}_{it} = v_i'X_t$, a structure closely related to the GO-GARCH and Rotated-GARCH models.

[Stationarity condition] For the λ-MGARCH, the process X_t is stationary with finite second moments if and only if

$$\varrho(A + B) < 1, \tag{4.20}$$

a clean scalar condition directly analogous to the univariate GARCH requirement $\alpha + \beta < 1$, and considerably simpler than the corresponding BEKK condition $\varrho\big((A \otimes A) + (B \otimes B)\big) < 1$.

4.7 The Matrix Logarithm Parametrization

4.7.1 Construction

Using the spectral decomposition $\Omega = V\Lambda V'$, one defines the matrix logarithm as

$$\log(\Omega) = V\ \mathrm{diag}\big(\log(\lambda_i)\big)_{i=1}^{p}\,V', \tag{4.21}$$

which is a well-defined symmetric matrix for any $\Omega > 0$, since $\log(\lambda_i) \in \mathbb{R}$ whenever $\lambda_i > 0$. The reparametrization sets

$$\log(\Omega(\theta)) = V(\theta)\ \mathrm{diag}(\delta_i)_{i=1}^{p}\,V(\theta)', \quad \delta_i \in \mathbb{R}, \tag{4.22}$$

so that $\Omega(\theta) = \exp(\log\Omega(\theta)) = V(\theta)\ \mathrm{diag}(e^{\delta_i})\,V(\theta)'$, and all $n = p(p+1)/2$ parameters vary freely in $\mathbb{R}^n$.

4.7.2 Positive Definiteness

Positive definiteness is **guaranteed by construction**, because $\exp(\delta_i) > 0$ for all $\delta_i \in \mathbb{R}$, and the matrix exponential preserves positive definiteness:

$$\Omega(\theta) = \exp\big(\log\Omega(\theta)\big) > 0 \quad \text{for all } \theta \in \mathbb{R}^n. \tag{4.23}$$

4.7.3 Interpretation and Dynamic Extension

Similar to the spectral parametrization, the log-parametrization decouples the likelihood into p univariate problems on the rotated series $\tilde{X}_t = V(\theta)'X_t$:

$$\mathcal{L}(\theta) = -\frac{T}{2}\sum_{i=1}^{p}\Big(\delta_i + S_{\tilde{x}_i\tilde{x}_i}\,e^{-\delta_i}\Big), \tag{4.24}$$

where each $\delta_i = \log\lambda_i$ plays the role of $\log\sigma_i^2$ in a univariate log-variance model. The natural dynamic extension replaces the constant δ_i with a time-varying δ_{it} following, for example, a GARCH-type recursion on $\log\lambda_{it}$, yielding a log-linear formulation of the λ-MGARCH that is numerically well-behaved and straightforward to estimate.

4.8 Comparative Summary

Table 4.1 collects the key properties of the five parametrizations reviewed above.

Table 4.1: Comparison of covariance parametrizations and their dynamic MGARCH extensions.

Parametrization	**PD guaranteed**	**# params**	**Dynamic model**	**Key feature**
vech	No	$p(p+1)/2$	vech-GARCH	Theoretical tool; curse of dim.
Cholesky	Yes	$p(p+1)/2$	BEKK-GARCH	Stable; entries not interpretable
Correlation	Yes*	$p(p+1)/2$	CCC / DCC	Transparent; two-step feasible
Eigenvalue	Yes	$p(p+1)/2$	λ-MGARCH	Decoupled likelihood; PCA link
Matrix log	Yes	$p(p+1)/2$	Log-GARCH	Log-linear; unconstrained

* Requires hyper-spherical parametrization of Γ for $p > 2$.

A unifying theme is that all five parametrizations contain exactly $n = p(p+1)/2$ free parameters — equal to the number of unique entries in Ω — and that each leads to the same MLE $\hat{\Omega} = S_{xx}$ in the static i.i.d. Gaussian setting. Their differences lie entirely in how they handle the positive definiteness constraint during optimization, and in how naturally they suggest dynamic extensions of the covariance structure.

4.9 Multivariate ARCH: The BEKK Model

The **BEKK** model — named after Baba, Engle, Kraft, and Kroner (1990, 1995) — is the multivariate analogue of the univariate ARCH model. In its simplest form it specifies:

$$\Omega_t(\theta) = \underbrace{\Omega}_{\substack{\text{baseline}\\ \text{covariance}}} + \underbrace{A\, X_{t-1} X_{t-1}'\, A'}_{\substack{\text{shock spillover}\\ \text{effects}}}, \tag{4.25}$$

where Ω is a $(p \times p)$ positive definite matrix and $A \in \mathbb{R}^{p \times p}$ is an unrestricted parameter matrix. The parameter vector is $\theta = (\Omega, A)$.

Why is $\Omega_t > 0$ guaranteed? The positive definiteness of Ω_t follows directly from the algebraic structure of (4.25), without imposing any additional constraints on A. To see this, note that the spillover term can be written as the outer product of the vector AX_{t-1} with itself:

$$A\, X_{t-1} X_{t-1}'\, A' = (AX_{t-1})(AX_{t-1})'.$$

For any nonzero vector $v \in \mathbb{R}^p$:

$$v'\big(AX_{t-1}X_{t-1}'A'\big)v = \|A'v \cdot X_{t-1}\|^2 \;\geq\; 0,$$

so the spillover term is always positive semidefinite. Since $\Omega > 0$ by assumption, the sum $\Omega_t \geq \Omega > 0$ is guaranteed positive definite at every t, regardless of the value taken by X_{t-1}. This is the central advantage of the BEKK parametrisation over, for example, the VEC model, which requires additional restrictions to prevent Ω_t from becoming non-positive definite. For $p = 2$, the matrix $A = (a_{ij})_{i,j=1,2}$ and the product AX_{t-1} is:

$$AX_{t-1} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} X_{1,t-1} \\ X_{2,t-1} \end{pmatrix} = \begin{pmatrix} a_{11}X_{1,t-1} + a_{12}X_{2,t-1} \\ a_{21}X_{1,t-1} + a_{22}X_{2,t-1} \end{pmatrix}.$$

Forming the outer product gives the spillover matrix:

$$AX_{t-1}X_{t-1}'A' = \begin{pmatrix} \big(a_{11}X_{1,t-1} + a_{12}X_{2,t-1}\big)^2 & * \\ * & \big(a_{21}X_{1,t-1} + a_{22}X_{2,t-1}\big)^2 \end{pmatrix}, \tag{4.26}$$

where the off-diagonal entry is:

$$* \;=\; \big(a_{11}X_{1,t-1} + a_{12}X_{2,t-1}\big)\big(a_{21}X_{1,t-1} + a_{22}X_{2,t-1}\big).$$

Expanding the diagonal elements reveals the key feature of the BEKK model. The conditional variance of asset 1, the $(1,1)$ entry of Ω_t, is:

$$\sigma_{11,t} = \omega_{11} + a_{11}^2\, X_{1,t-1}^2 + 2\, a_{11}a_{12}\, X_{1,t-1}X_{2,t-1} + a_{12}^2\, X_{2,t-1}^2.$$

Three effects are present: the own squared shock $X_{1,t-1}^2$, the foreign squared shock $X_{2,t-1}^2$, and — crucially — the *cross-product* $X_{1,t-1}X_{2,t-1}$. This cross-product term means that when the two assets move in opposite directions, their contributions partially cancel in the variance equation. The full BEKK with a free matrix A therefore does *not* reduce to a collection of univariate ARCH(1) models for the individual assets; the marginal dynamics of each asset are richer than standard ARCH(1). If the empirical goal is to preserve a clean univariate ARCH(1) interpretation for each individual asset while still allowing for cross-asset dynamics in the conditional covariance, the matrix A must be restricted. The natural choice is to use two

diagonal factor matrices:

$$\Omega_t(\theta) = \Omega + A_1\, X_{t-1}X_{t-1}'\, A_1' + A_2\, X_{t-1}X_{t-1}'\, A_2', \qquad A_1 = \begin{pmatrix} \sqrt{\alpha_1} & 0 \\ 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\alpha_2} \end{pmatrix}. \tag{4.27}$$

Direct computation yields:

$$A_1\, X_{t-1}X_{t-1}'\, A_1' + A_2\, X_{t-1}X_{t-1}'\, A_2' = \begin{pmatrix} \alpha_1\, X_{1,t-1}^2 & 0 \\ 0 & \alpha_2\, X_{2,t-1}^2 \end{pmatrix}.$$

Now the diagonal entries of Ω_t are exactly $\omega_{ii} + \alpha_i X_{i,t-1}^2$, which are proper univariate ARCH(1) equations. The off-diagonal elements, however, are inherited from Ω alone, meaning that conditional correlations remain constant in this restricted specification.

4.9.1 The Full BEKK(p,q) and Scalar Restriction

Adding GARCH-type persistence and allowing for multiple lags and K component matrices gives the general **BEKK(p, q)** model:

$$\Omega_t(\theta) = \Omega + \sum_{j=1}^{p}\sum_{k=1}^{K_p} A_k\, X_{t-j}X_{t-j}'\, A_k' + \sum_{l=1}^{q}\sum_{k=1}^{K_q} B_k\, \Omega_{t-l}\, B_k'. \tag{4.28}$$

The matrices B_k play the same role as the GARCH coefficient in the univariate model: they allow the conditional covariance to depend on its own past values, generating the persistence in volatility that is characteristic of financial time series. Positive definiteness is again guaranteed by construction, since each term is of the quadratic form $M(\cdot)M'$.

The dimensionality problem. The practical difficulty with the general BEKK is the number of free parameters. Even the simplest BEKK(1,1) with a single component matrix ($K_p = K_q = 1$) requires:

- $\frac{p(p+1)}{2}$ parameters in the symmetric matrix Ω,
- p^2 parameters in A,
- p^2 parameters in B,

for a total of $2p^2 + \frac{p(p+1)}{2}$ parameters. For $p = 5$ assets this is already 65 parameters; for $p = 10$ it is 255. The curse of dimensionality manifests in three related ways. First, *overfitting*: with many parameters relative to the time-series length T, the likelihood surface becomes nearly flat in many directions. Second, *identification failure*: different pairs (A, B) can generate the same Ω_t trajectory, since only the products AA' and BB' appear in (4.28), leaving the sign and rotation of A and B unidentified. Third, *numerical instability*: gradient-based optimisation in high-dimensional parameter spaces is sensitive to starting values and scaling.

The scalar BEKK. The dominant solution in empirical work is the **scalar BEKK**, which replaces the full matrices A and B with scalar multiples of the identity:

$$\Omega_t(\theta) = \Omega + \alpha\, X_{t-1}X_{t-1}' + \beta\, \Omega_{t-1}, \qquad \alpha,\, \beta \geq 0. \tag{4.29}$$

This reduces the effective dimension of the ARCH and GARCH components to just two scalars, α and β, regardless of p. The structural interpretation is clean: α controls the uniform sensitivity of all conditional variances and covariances to past shocks, while β governs the common speed of mean-reversion. The cost is the strong homogeneity assumption that all elements of Ω_t respond

identically to shocks and decay at the same rate — an assumption that is often rejected in data on heterogeneous asset classes.

Positive definiteness of (4.29) is preserved provided $\Omega > 0$ and $\alpha + \beta < 1$, the latter being the stationarity condition ensuring that the unconditional covariance matrix $\bar{\Omega} = \Omega/(1-\alpha-\beta)$ is finite.

Estimation. The BEKK-MGARCH can be estimated through the maximization of the likelihood function given $L_{BEKK}(\theta)$ where we identify:

$$L_{\text{BEKK}}(\theta) = \frac{1}{T}\sum_{t=1}^{T}\left(\log|\Omega_t(\theta)| + X_t'\Omega_t^{-1}(\theta)X_t\right)$$

Choosing between specifications. In practice, the choice between the full BEKK, diagonal BEKK, and scalar BEKK involves a standard bias-variance trade-off. The full BEKK allows the most flexible cross-asset dynamics but at the risk of poor finite-sample estimation. The scalar BEKK is the most parsimonious and numerically stable but imposes strong homogeneity restrictions. Likelihood ratio tests or information criteria (AIC, BIC) can guide model selection, though one must be cautious about the boundary behaviour of test statistics when parameters such as α are constrained to be non-negative.

4.10 CCC-GARCH

The CCC-GARCH also defiend as constant conditional correlation GARCH has been developed by Bollerslev and relies on the decomposition $\Omega = D\,\Gamma\,D$, in particular we have:

$$\Omega_t(\theta) = D_t(\theta)\,\Gamma(\theta)\,D_t(\theta)$$

where $D_t(\theta) = diag((\sigma_{i,t})_{i=1}^{p})$ where the $\sigma_{it}^2(\theta)$ can be specified by any ARCH-GARCH model. Now considering $\Gamma(\theta) = (\rho_{i,j})_{i,j=1,\dots,p}$ with $\rho_{ii} = 1$. So basically the structure of $\Omega(\theta)_t$ is given as:

$$\Omega_t(\theta) = \begin{pmatrix} \sigma_{1t}^2(\theta) & \sigma_{1t}(\theta)\,\sigma_{2t}(\theta)\,\rho \\ \sigma_{1t}(\theta)\,\sigma_{2t}(\theta)\,\rho & \sigma_{2t}^2(\theta) \end{pmatrix}.$$

In this setting the main goal is to ensure $\Gamma(\theta) > 0$ therefore, we rely on the parametrisation given by:

$$\Gamma(\theta) = G(\theta)G(\theta) = \begin{pmatrix} 1 & 0 \\ \cos(\theta) & \sin(\theta) \end{pmatrix}\begin{pmatrix} 1 & \cos(\theta) \\ 0 & \sin(\theta) \end{pmatrix} = \begin{pmatrix} 1 & \cos(\theta) \\ \cos(\theta) & 1 \end{pmatrix}$$

Here, the estimation procedure is given by the so-called **Two-Step Estimation** which is an alternative to plain maximum likelihood estimation.

Two-Step Estimation (CCC-GARCH). Step 1. Estimate the p univariate GARCH/ARCH models and obtain conditional variances $\hat{\sigma}_{it}^2$, forming

$$\hat{D}_t = \text{diag}(\hat{\sigma}_{1t}, \dots, \hat{\sigma}_{pt}), \quad \hat{X}_t = \hat{D}_t^{-1}X_t.$$

Step 2. Given $\hat{D}_t$, estimate the correlation matrix by maximizing the concentrated likelihood

$$L_{TS}(\theta) = \log|\Gamma(\theta)| + \text{tr}\big(\Gamma(\theta)^{-1}S_{\hat{x}\hat{x}}\big),$$

where

$$S_{\hat{x}\hat{x}} = \frac{1}{T}\sum_{t=1}^{T} \hat{D}_t^{-1} X_t X_t' \hat{D}_t^{-1}.$$

The estimator of the correlation matrix is

$$\hat{\Gamma}_{TS} = S_{\hat{x}\hat{x}}.$$

Final estimator:

$$\hat{\theta}_{TS,CCC} = (\hat{\theta}_{\text{GARCH}}, \hat{\Gamma}_{TS}),$$

where $\hat{\theta}_{\text{GARCH}}$ are the parameters from the univariate GARCH estimations.

4.11 DCC-MGARCH

The CCC model is extended by Engle (2002) to allow for *time-varying* conditional correlations. The simplest version parametrises the correlation matrix as $\Gamma_t(\theta) = G_t(\theta)G_t(\theta)'$, where:

$$G_t(\theta) = \begin{pmatrix} 1 & 0 \\ \cos(\gamma_t) & \sin(\gamma_t) \end{pmatrix}, \qquad \gamma_t(\theta) \in (0,\, \pi/2),$$

so that $\Gamma_t(\theta) > 0$ is guaranteed by construction. While elegant, this parametrisation is difficult to interpret directly since the dynamics enter through angles γ_t rather than correlations themselves. Moreover, it does not generalise readily beyond the bivariate case.

The DCC reparametrisation. A more tractable and general approach is to work directly with a time-varying positive definite matrix $Q_t(\theta)$, and recover the correlation matrix by standardising its diagonal:

$$\Gamma_t(\theta) = \operatorname{diag}\big(Q_t(\theta)\big)^{-1/2} Q_t(\theta) \operatorname{diag}\big(Q_t(\theta)\big)^{-1/2}. \tag{4.30}$$

Since $Q_t(\theta) > 0$ by construction (shown below), the diagonal entries are strictly positive and (4.30) delivers a valid correlation matrix: $\Gamma_t(\theta) > 0$ with unit diagonal entries. The interpretation is immediate — $(\Gamma_t)_{ij}$ is simply the conditional correlation between assets i and j at time t.

Dynamics of $Q_t(\theta)$. Define the *standardised residuals* $X_{D,t} = D_t(\theta)^{-1} X_t$, where $D_t(\theta) = \operatorname{diag}(\Omega_t(\theta))^{1/2}$ strips out the individual conditional standard deviations. By construction, each component of $X_{D,t}$ has unit conditional variance, so $\mathbb{E}[X_{D,t}X_{D,t}' \mid \mathcal{F}_{t-1}] = \Gamma_t(\theta)$.

The matrix $Q_t(\theta)$ follows a **BEKK-ARCH**(1) process on $X_{D,t}$:

$$Q_t(\theta) = \underbrace{\bar{Q} - A\bar{Q}A'}_{\text{intercept (variance targeting)}} + \underbrace{A\, X_{D,t-1}X_{D,t-1}'\, A'}_{\text{shock spillovers}}, \tag{4.31}$$

where $A \in \mathbb{R}^{p\times p}$ is a parameter matrix and $\bar{Q}$ is the unconditional covariance matrix of the standardised residuals, estimated as $\bar{Q} = T^{-1}\sum_{t=1}^{T} X_{D,t}X_{D,t}'$. The intercept $\bar{Q} - A\bar{Q}A'$ is chosen by *variance targeting*: it ensures that the unconditional mean of Q_t equals $\bar{Q}$, since $\mathbb{E}[Q_t] = (\bar{Q} - A\bar{Q}A') + A\bar{Q}A' = \bar{Q}$. Positive definiteness of Q_t is inherited from the BEKK structure: the shock term $AX_{D,t-1}X_{D,t-1}'A'$ is always positive semidefinite, and the intercept $\bar{Q} - A\bar{Q}A' > 0$ provided the spectral radius of A is small enough.

Remark on identification. The matrix $\bar{Q}$ satisfies the relation $\bar{Q} = \operatorname{diag}(\bar{Q})^{1/2}\,\bar{\Gamma}\,\operatorname{diag}(\bar{Q})^{1/2}$, where $\bar{\Gamma} = \operatorname{diag}(\bar{Q})^{-1/2}\bar{Q}\operatorname{diag}(\bar{Q})^{-1/2}$ is the unconditional correlation matrix. The normalisation

(4.30) therefore maps the dynamics of Q_t into the unit-diagonal correlation space at every t, without imposing constant correlations.

4.12 λ-MGARCH

Another multivariate model for the conditional variance-covariance matrix is the **λ-MGARCH**, also known as *Eigenvalue GARCH*. Its starting point is the spectral decomposition of $\Omega_t(\theta)$:

$$\Omega_t(\theta) = V(\theta)\,\Lambda_t(\theta)\,V(\theta)', \tag{4.32}$$

where $V(\theta) = (V_1, \ldots, V_p) \in \mathbb{R}^{p\times p}$ is a matrix of eigenvectors satisfying the orthogonality constraint

$$V(\theta)V(\theta)' = V(\theta)'V(\theta) = I_p,$$

and $\Lambda_t(\theta) = \operatorname{diag}\big((\lambda_{i,t}(\theta))_{i=1}^p\big)$ is a diagonal matrix of time-varying eigenvalues. Positive definiteness of $\Omega_t(\theta)$ reduces to the condition $\lambda_{i,t}(\theta) > 0$ for all i and t.

Parametrising $V(\theta)$: the rotation matrix. The orthogonality constraint $V'V = I_p$ must be maintained for every value of the parameter vector θ. For $p = 2$, any orthogonal matrix with $\det(V) = 1$ is a rotation and can be written as:

$$V(\theta) = \begin{pmatrix} \cos(\gamma) & -\sin(\gamma) \\ \sin(\gamma) & \cos(\gamma) \end{pmatrix}, \qquad \gamma = \gamma(\theta) \in (0,\, \pi/2). \tag{4.33}$$

One can verify directly that $V'V = I_2$ for any γ, so the constraint is satisfied automatically for every value of γ — no additional restrictions on the entries of V are needed. The single scalar γ parametrises the *angle of rotation* that aligns the original coordinate system with the eigenvector basis of Ω_t. The columns $V_1 = (\cos\gamma, \sin\gamma)'$ and $V_2 = (-\sin\gamma, \cos\gamma)'$ are the two eigenvectors, orthogonal by construction. For $p > 2$, orthogonal matrices are parametrised analogously using a product of Givens (planar) rotations, one angle per pair of coordinates, for a total of $p(p-1)/2$ angles.

Eigenvalue dynamics. The model places all time variation in the eigenvalue vector $\lambda_t(\theta) = (\lambda_{1,t}(\theta), \ldots, \lambda_{p,t}(\theta))'$, which follows the recursion:

$$\lambda_t(\theta) = C + A\left(V(\theta)'X_{t-1}\right)^{\odot 2} + B\,\lambda_{t-1}(\theta), \tag{4.34}$$

where $C = (c_1, \ldots, c_p)'$ with $c_i > 0$, and the matrices A and B have non-negative entries. The operator $(\,\cdot\,)^{\odot 2}$ denotes elementwise squaring: the i-th element of $(V'X_{t-1})^{\odot 2}$ is $(V_i'X_{t-1})^2$, the squared projection of the return vector X_{t-1} onto the i-th eigenvector.

This has a natural interpretation. The transformation $V'X_{t-1}$ rotates the raw return vector into the eigenvector coordinate system. Squaring each component gives the *contribution of each principal direction* to the previous period's shock. The eigenvalue $\lambda_{i,t}$ therefore responds to shocks in its own principal direction, governed by the i-th row of A, and to past eigenvalues through B in a GARCH-type feedback.

Positive definiteness of Ω_t is guaranteed by construction: since $C > 0$ componentwise and $A, B \geq 0$ elementwise, the recursion (4.34) keeps all $\lambda_{i,t}$ strictly positive for every t, given $\lambda_0 > 0$.

Simplified log-likelihood. The Gaussian log-likelihood involves two terms: $\log|\Omega_t|$ and $\operatorname{tr}(\Omega_t^{-1}X_tX_t')$. The spectral structure reduces both considerably. Since V is orthogonal, $|V| =$

± 1, and:

$$\log|\Omega_t| = \log|V\Lambda_t V'| = \log|\Lambda_t| = \sum_{i=1}^{p} \log \lambda_{i,t}.$$

Similarly, using $\Omega_t^{-1} = V\Lambda_t^{-1}V'$ and the cyclic property of the trace:

$$\mathrm{tr}\big(\Omega_t^{-1}X_tX_t'\big) = \mathrm{tr}\big(V\Lambda_t^{-1}V'X_tX_t'\big) = \mathrm{tr}\big(\Lambda_t^{-1}V'X_tX_t'V\big) = \sum_{i=1}^{p} \frac{(V_i'X_t)^2}{\lambda_{i,t}}.$$

The concentrated log-likelihood thus becomes:

$$-\frac{2}{T}L_\lambda(\theta) = T^{-1}\sum_{t=1}^{T}\sum_{i=1}^{p}\left(\log\lambda_{i,t}(\theta) + \frac{(V_i(\theta)'X_t)^2}{\lambda_{i,t}(\theta)}\right). \tag{4.35}$$

Each summand is precisely the log-likelihood of a *univariate* GARCH model for the projected series $V_i'X_t$ with conditional variance $\lambda_{i,t}$. The λ-MGARCH therefore decomposes the multivariate problem into p scalar problems along the eigenvector directions, which is the main computational appeal of the model. The coupling across i arises solely through the shared eigenvector matrix $V(\theta)$, which must be estimated jointly.

4.13 Properties and Asymptotics of BEKK

When introducing the BEKK, we need to take into account that we are evaluating a Multivariate model, which complicates the identification of the regularity conditions we usually need for asymptotics. Once those results are obtained, it is possible to show that we get very close to the results already obtained with the ARCH model.

We write here the model for the scalar-BEKK as follows:

$$\begin{cases} X_t = \Omega(\alpha)^{1/2}z_t, \quad z_t \sim_{i.i.d} N(0, I_p) \\ \Omega_t(\alpha) = \Omega + \alpha X_{t-1}X_{t-1}' \end{cases}$$

Here we see that the usual condition regarding stationarity is equal to the one we have in the univariate ARCH model, since the parameter we consider is given by $\theta = (\alpha, \Omega)$. However, for our exercise, we consider Ω_0 as known. Therefore, we can proceed in the usual way and write the log-likelihood:

$$L_{\mathrm{BEKK}}(\alpha) = -\frac{1}{2}\sum_{t=1}^{T}\left(\log|\Omega_t(\alpha)| + X_t'\Omega_t^{-1}(\alpha)X_t\right)$$

Here we want want to use, under regularity consitions, the classical properties of MGARCH to have:

$$CLT : s_T(\alpha) \xrightarrow{d} N(0,\eta), \quad s_T(\alpha) = \frac{1}{\sqrt{T}}\frac{\partial L_T(\alpha)}{\partial\theta}$$

and

$$LLN : h_t(\alpha) \xrightarrow{d} \eta > 0, \quad h_T(\alpha) = \frac{1}{T}\frac{\partial L_T^2(\alpha)}{\partial\theta\partial\theta'}$$

which, in our case, refers to a convergence in distribution of a scalar. Now using the proof reported in the Appendix, we show how to derive the different components for the score and for

the Information. So we are able to report:

$$\frac{\partial \Omega(\alpha)^{-1}}{\partial \alpha} = \Omega(\alpha)^{-1} \frac{\partial \Omega_t(\alpha)}{\partial \alpha} \Omega(\alpha)^{-1} = \Omega(\alpha)^{-1} X_{t-1} X_{t-1}' \Omega(\alpha)^{-1}$$
$$\frac{\partial \log |\Omega_t(\alpha)|}{\partial \alpha} = tr(\Omega(\alpha)^{-1} \partial \Omega(\alpha)/\partial \alpha) = X_{t-1}' \Omega_t(\alpha)^{-1} X_{t-1}$$

Now we can finally report the score as:

$$S_T(\alpha) = -\frac{1}{2} \sum_{t=1}^{T} m_t(\alpha)$$

where $m_t(\alpha)$ is given by the combination of the two results reached above, namely:

$$X_{t-1}' \Omega_t(\alpha)^{-1} X_{t-1} - X_t' \Omega(\alpha)^{-1} X_{t-1} X_{t-1}' \Omega(\alpha)^{-1} X_t$$

and obtain:

$$X_{t-1}' \Omega_t(\alpha)^{-1/2} (I - \underbrace{\Omega(\alpha)^{-1/2} X_t X_t' \Omega(\alpha)^{-1/2}}_{z_t' z_t}) \Omega_t^{-1/2} X_{t-1}$$

Score as a Martingale Difference Sequence

Evaluating at the true parameter α_0 and using $X_t = \Omega_t^{1/2} z_t$, we obtain:

$$m_t(\alpha_0) = X_{t-1}' \Omega_t^{-1/2} \left(I_p - z_t z_t' \right) \Omega_t^{-1/2} X_{t-1}$$

To verify the MDS property, we take the conditional expectation with respect to $\mathcal{F}_{t-1} = \sigma(z_{t-1}, z_{t-2}, \ldots)$:

$$\mathbb{E}\left[m_t(\alpha_0) \,|\, \mathcal{F}_{t-1} \right] = X_{t-1}' \Omega_t^{-1/2} \left(I_p - \mathbb{E}\left[z_t z_t' \right] \right) \Omega_t^{-1/2} X_{t-1} = 0,$$

since $\mathbb{E}[z_t z_t'] = I_p$ by assumption. Hence $\{m_t(\alpha_0)\}$ is a martingale difference sequence (MDS) with respect to $\mathcal{F}_{t-1}$.

CLT for the Score

By a standard CLT for MDS, defining

$$v_t = \mathbb{E}\left[m_t^2 \,\middle|\, \mathcal{F}_{t-1} \right],$$

the LLN gives $T^{-1} \sum_{t=1}^{T} v_t \xrightarrow{p} \sigma_s^2 = \mathbb{E}[m_t^2] > 0$, provided $\mathbb{E}\|X_t\|^2 < \infty$, i.e. $0 \leq \alpha_0 < 1$. It follows that:

$$-2T^{-1/2} s_T(\alpha_0) = T^{-1/2} \sum_{t=1}^{T} m_t(\alpha_0) \xrightarrow{d} N\left(0, \sigma_s^2\right).$$

LLN for the Information

The observed information is $h_T(\alpha) = -T^{-1} \partial^2 L_T(\alpha)/\partial \alpha^2$. Analogously to the univariate case, applying the LLN to the stationary sequence $\eta_t = \partial^2 \ell_t(\alpha_0)/\partial \alpha^2$ yields:

$$h_T(\alpha_0) \xrightarrow{p} \eta = \mathbb{E}[\eta_t] > 0,$$

provided $\mathbb{E}\|X_t\|^4 < \infty$, which requires $0 \leq \alpha_0 < 1/\sqrt{3} \simeq 0.5$.

Lemma 1 in Jensen and Rahbek (2004a), provided (i)-(iii)
(i)-(iii) hold if Xt is
geometrically ergodic and stationary, with E kXtk6 < 1

Asymptotic Normality of the MLE

Combining the CLT for the score and the LLN for the information via the standard delta-method argument, we obtain:

$$\sqrt{T}\,(\hat{\alpha}_T - \alpha_0) \xrightarrow{d} N\big(0,\, \sigma_\alpha^2\big)\,, \quad \sigma_\alpha^2 = \eta^{-1}\sigma_s^2\,\eta^{-1},$$

under the regularity conditions summarised below.

Summary: Scalar-BEKK

Property	Condition
X_t stationary and ergodic	$\alpha < \exp\big(-\mathbb{E}\log z_{it}^2\big) \simeq 3.56$
$\mathbb{E}\|X_t\|^2 < \infty$ (CLT for score)	$0 \le \alpha_0 < 1$
$\mathbb{E}\|X_t\|^4 < \infty$ (LLN for information)	$0 \le \alpha_0 < 1/\sqrt{3} \simeq 0.5$
$\sqrt{T}(\hat{\alpha}_T - \alpha_0) \xrightarrow{d} N(0, \sigma_\alpha^2)$	$\mathbb{E}\|X_t\|^4 < \infty \;\; (\ldots \mathbb{E}\|X_t\|^6 < \infty)$

These conditions are the multivariate analogue of those derived for the univariate ARCH(1) model, with the scalar α replacing the spectral radius $\varrho(A \otimes A)$ of the full BEKK.

4.14 Portfolio Selection Based on Estimated Conditional Mean and Variance from Multivariate GARCH models

The portfolio selection is based on ex-ante assumptions related to the institutional setting; in particular, the investor's strategy may differ depending on whether short-selling is permitted and whether a risk-free asset is available.

We proceed with Markowitz mean-variance analysis, optimising a portfolio over a single period based on μ and Ω, where these are estimated from historical data under a given statistical model. Specifically, we employ the sample estimators:

$$\hat{\mu} = \frac{1}{T}\sum_{t=1}^{T} R_t,$$

$$\hat{\Omega} = \frac{1}{T}\sum_{t=1}^{T}(R_t - \hat{\mu})(R_t - \hat{\mu})',$$

where $R_t \in \mathbb{R}^p$ denotes the vector of asset returns at time t.

Since $\Omega \in \mathbb{R}^{p\times p}$ is symmetric, estimation requires $p(p+1)/2$ free parameters, so the dimensionality of the problem grows at rate $O(p^2)$. As p increases relative to T, the sample covariance matrix becomes increasingly noisy and may even be singular. To address this, *shrinkage estimators* have been proposed; a prominent example is **Ledoit and Wolf (2003, 2004)**, who suggest:

$$\hat{\Omega}_{\text{shrink}} = (1-\delta)\,\hat{\Omega} + \delta\, S,$$

where $S \in \mathbb{R}^{p\times p}$ is a structured (low-parameter) target matrix — commonly the scaled identity $\hat{\sigma}^2 I_p$ — and $\delta \in [0,1]$ is the shrinkage intensity, which can be chosen optimally by minimising the expected quadratic loss under an asymptotic framework.

Turning to the statistical model, we may assume a **Multivariate GARCH** (MGARCH) specification, which jointly models the conditional mean and conditional covariance matrix. The

model takes the form:

$$R_t = \mu_t + \varepsilon_t,$$
$$\varepsilon_t = \Omega_t^{1/2} z_t,$$

where $z_t \overset{\text{i.i.d.}}{\sim} (0, I_p)$ is a standardised innovation vector, $\Omega_t^{1/2}$ is the conditional covariance matrix Ω_t, and both $\mu_t = \mathbb{E}[R_t \mid \mathcal{F}_{t-1}]$ and $\Omega_t = \mathrm{Var}(R_t \mid \mathcal{F}_{t-1})$ are measurable with respect to the information set $\mathcal{F}_{t-1}$. Since prices and their co-movements evolve dynamically, investors may wish to adjust portfolio weights over time. We therefore consider *dynamic* portfolio weights $v_t = v_t(\mathcal{F}_{t-1}) \in \mathbb{R}^p$, which are adapted to the investor's information set at time $t-1$. This requires a model for the conditional covariance matrix Ω_t; a natural candidate is the **BEKK model** Engle and Kroner, 1995:

$$\begin{aligned} R_t &= \mu + \varepsilon_t, \\ \varepsilon_t &= \Omega_t^{1/2} z_t, \quad z_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_p), \\ \Omega_t &= LL' + A\,\varepsilon_{t-1}\varepsilon_{t-1}'A' + B\,\Omega_{t-1}B', \end{aligned} \tag{4.36}$$

where $\mu \in \mathbb{R}^p$, $L \in \mathbb{R}^{p\times p}$ is lower triangular (so that LL' is symmetric positive semi-definite by construction), and $A, B \in \mathbb{R}^{p\times p}$. The parameter vector is $\theta = \big(\mu,\, \mathrm{vech}(L),\, A,\, B\big) \in \Theta \subseteq \mathbb{R}^k$, with

$$k = \underbrace{p}_{\mu} + \underbrace{\frac{p(p+1)}{2}}_{L} + \underbrace{2p^2}_{A,\,B},$$

which grows at rate $O(p^2)$. While the BEKK parameterisation guarantees positive definiteness of Ω_t at every t, it already imposes structure. A fully unrestricted alternative is the **VECH-GARCH** model, defined by applying the half-vectorisation operator $\mathrm{vech}(\cdot)$ to both sides of the covariance recursion:

$$\underbrace{\mathrm{vech}(\Omega_t)}_{\frac{p(p+1)}{2}\times 1} = \underbrace{c}_{\frac{p(p+1)}{2}\times 1} + \underbrace{\mathcal{A}}_{\frac{p(p+1)}{2}\times\frac{p(p+1)}{2}} \underbrace{\mathrm{vech}\big(\varepsilon_{t-1}\varepsilon_{t-1}'\big)}_{\frac{p(p+1)}{2}\times 1} + \underbrace{\mathcal{B}}_{\frac{p(p+1)}{2}\times\frac{p(p+1)}{2}} \underbrace{\mathrm{vech}(\Omega_{t-1})}_{\frac{p(p+1)}{2}\times 1}, \tag{4.37}$$

where $c \in \mathbb{R}^{p(p+1)/2}$ and $\mathcal{A}, \mathcal{B} \in \mathbb{R}^{p(p+1)/2\times p(p+1)/2}$. The total parameter count now grows at rate $O(p^4)$, rendering the VECH model computationally intractable even for moderate p, which motivates the use of more parsimonious specifications such as BEKK or factor-based GARCH models in empirical work.

4.14.1 Timing: Filtering vs Forecasting

While dynamic rebalancing allows the portfolio to adapt to evolving market conditions, a critical practical question concerns the *timing* of trades. A subtle but important identification issue arises in estimation: although the theoretical model is specified at time $t-1$, standard implementations often incorporate the time-t observation in estimation, which amounts to *filtering* rather than *forecasting*. To ensure that portfolio weights are genuinely out-of-sample, one should instead pursue a pseudo-real-time estimation of $\hat{\Omega}_{t|t-1}$, exploiting only information available at time $t-1$ to construct the conditional covariance forecast. Formally, the optimal weight vector is then

$$v_{t|t-1} \;=\; \frac{\hat{\Omega}_{t|t-1}^{-1}\,\iota}{\iota'\,\hat{\Omega}_{t|t-1}^{-1}\,\iota}, \tag{4.38}$$

so that portfolio construction relies exclusively on *forecasts*, not filters, preserving the out-of-sample validity of the exercise.

The investment horizon introduces a further layer of complexity. If the investor rebalances daily, it is natural to estimate the model on daily data and forecast $\Omega_{t+1|t}$ one step ahead. A weekly investor, by contrast, might instead work directly with weekly returns and produce weekly covariance forecasts, thereby avoiding the need to aggregate daily dynamics. The reason is that multi-period variance forecasting is both analytically demanding and computationally costly. Specifically, the H-period portfolio variance,

$$\mathrm{Var}\left(\sum_{h=1}^{H} v' R_{t+h} \;\middle|\; \mathcal{F}_t\right), \tag{4.39}$$

requires propagating the conditional covariance matrix through the model's recursive structure for H steps, accumulating cross-period covariance terms that vanish only under very restrictive assumptions. When returns exhibit GARCH-type dynamics, this multi-period propagation has no closed form beyond a few special cases (e.g. the integrated GARCH or the scalar BEKK), making direct temporal aggregation the more tractable strategy in practice.

Additional Open Issues

Whenever we need to deal with constructing portfolio weights we need to consider different elements:

- **Holding period**
- **Timing**
- **Turnover**

Especially turnover is important since we might be tempted to follow the weights forecasted by our model, and therefore we might end up changing asset weights in our portfolio continuously, generating high turnover:

$$\text{Turnover}_t = \sum_{i=1}^{p} |v_{i,t} - v_{i,t-1}|$$

This becomes an issue as we introduced transaction costs, which would lead to a potential loss which should be accounted for in our model. To reduce this excessive turnover once could choose:

1. **Rebalance less frequently**
2. **Use partial adjustment**: $v_t^{\text{trade}} = \lambda v_t^* + (1-\lambda) v_{t-1}$
3. **Impose no trade regions**
4. **Add penalty turnover**

4.14.2 DCC

We now introduce an example of a model that reports also time-varying correlation parameters therefore, we can state the model as follows:

$$\begin{aligned} R_t &= \mu + \varepsilon_t, \\ \varepsilon_t &= \Omega_t^{1/2} z_t, \\ \Omega_t &= D_t \, \Gamma_t \, D_t. \end{aligned}$$

here we can think of:

$$D_t = \begin{pmatrix} \sigma_{1t} & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \sigma_{pt} \end{pmatrix} \tag{4.40}$$

where the $\sigma_{i,t}$ are estimated using univariate garch models. Now we consider $\eta_t = D_t^{-1}\epsilon_t$ which is standardize but correlated since:

$$\mathbb{E}[D_t^{-1}\epsilon_t\epsilon_t' D_t^{-1}|\mathcal{F}_{t-1}] = D_t^{-1}(D_t\Gamma_t D_t)D_t^{-1} = \Gamma_t$$

which presents correlations. Because we need to consider dynamic varying correlations we need to provide a model for the dynamics of the correlations:

$$Q_t = Q(1-a-b) + a\eta_{t-1}\eta_{t-1}' + bQ_{t-1}$$

This is not a correlation matrix because we consider dynamics coming from the standardised but correlated returns and assume some recursive dynamics which revert to the mean. The diagonal entries are not ones since $Q_t^* = diag(\sqrt{q_{11,t}}, ..., \sqrt{q_{pp,t}})$ therefore we use:

$$\Gamma_t = Q_t^{*-1}Q_tQ^{*-1}$$

here we observe that a and b are scalars and all correlations move with the same speed/sensitivity.

Chapter 5

Asset Pricing

5.1 Capital Asset Pricing Model an Intorduction

Financial markets offer a wide array of assets, each characterized by different prices and expected returns. This naturally raises a fundamental and non-trivial question:

"Why do some assets yield higher expected returns than others?"

Answering this question is far from trivial. A deeper understanding of the determinants of expected returns allows us to uncover the underlying mechanisms of asset pricing and to identify the nature of the compensation that investors receive for holding different assets.

Starting from **Portfolio Optimisation**, we saw that a single agent can select a portfolio on the efficient frontier that maximises their utility, which is increasing in expected returns and decreasing in variance (or volatility). Portfolio optimisation is the first step toward understanding how assets are priced; by introducing additional assumptions, we can develop a model for asset pricing. Before doing so, it is important to consider two key elements: *risk* and *diversification*. Each asset carries its own risk, driven by an *idiosyncratic* component and a *systematic* component. For a portfolio of p assets with equal weights $\frac{1}{p}$, the portfolio variance decomposes as:

$$\bar{\sigma}_p^2 = \underbrace{\frac{1}{p}\left(\frac{1}{p}\sum_{i=1}^{p}\sigma_{ii}^2\right)}_{\text{average idiosyncratic variance}} + \underbrace{\frac{p(p-1)}{p}\left(\frac{1}{p(p-1)}\sum_{i=1}^{p}\sum_{\substack{j=1\\ j\neq i}}^{p}\sigma_{ij}\right)}_{\text{average covariance}}$$

As the number of assets p increases, the average idiosyncratic risk term is of order $O(1/p)$; therefore, as $p \to \infty$ it vanishes, whereas the average covariance term persists.

5.2 CAPM Model and Assumptions

We now start to introduce multiple investors and assume the following:

1. All investors maximise a mean-variance expected utility function.
2. Holding period equals the time horizon of the investment.
3. Same beliefs regarding relevant means and variances.
4. Heterogeneous agents in terms of risk aversion.
5. agents are infinitesimally small and cannot influence prices.

6. **Capital Market Equilibrium**: Risk-free asset with net zero supply
7. Perfect Competition and No Market Frictions
8. Short-selling is allowed

Now, with these assumptions, it is possible to conclude that aggregating all investors, we have portfolio holdings corresponding to the tangent portfolio, the one that allows the highest Sharpe ratio given the efficient frontier.

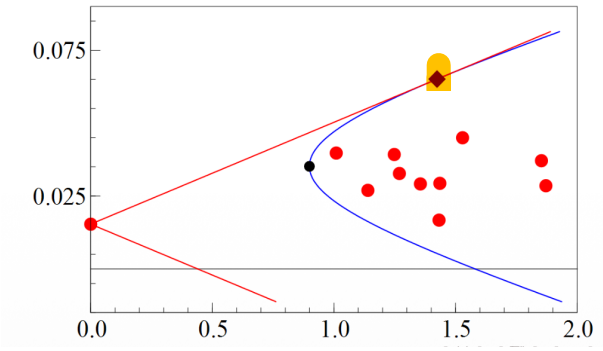


Figure 5.1

Any efficient portfolio will lie therefore on the Capital allocation line (red one), which is given by:

$$\bar{\mu}_b = R_f + \underbrace{\frac{\bar{\mu}_m - R_f}{\bar{\sigma}_m}}_{\text{Sharpe Ratio}} \bar{\sigma}_b$$

By the **Two-Fund Separation Theorem**, all investors hold a combination of the risk-free asset and the tangent portfolio, placing each investor at some point along the Capital Market Line depending on their risk aversion. Now the real question is **How much does the asset move with the market**, this is basically summarised by the measure of covariance, hence by β, which we will show later.

We now state the equations related to the mean and variance of a portfolio b given weights for the risky assets and the risk-free asset $w = (v', v_f')'$:

$$\hat{\mu}_b(v) = R_f + v'(\mu - R_f)$$
$$\hat{\sigma}_b^2(v) = v'\Omega v = \sum_{i=1}^{N}\sum_{j=1}^{N} v_i v_j \sigma_{ij}$$

The portfolio theory and CAPM deal with the trade-off between returns and risk, or mean-variance so we look at marginal return to risk which we summarize as:

$$RRR(i,b) = \frac{\partial \bar{\mu}_b}{\partial \bar{\sigma}_b} = \frac{\partial \bar{\mu}_b / \partial v_i \ (1)}{\partial \bar{\sigma}_b / \partial v_i \ (2)}$$

we proceed with (1) as follows:

$$\frac{\partial}{\partial v_i}\bar{\mu}_b(v) = \frac{\partial}{\partial v_i}\left(R_f + v'(\mu - R_f\mathbf{1})\right) = \mu_i - R_f$$

Taking the derivative of the portfolio variance with respect to weight v_i:

$$\frac{\partial}{\partial v_i}\bar{\sigma}_b^2(v) = \frac{\partial}{\partial v_i}\sum_{i=1}^{N}\sum_{j=1}^{N} v_i v_j \sigma_{ij} = 2\sum_{j=1}^{N} v_j \sigma_{ij} = 2e_i'\Omega v = 2\,\mathrm{cov}(R_i, R_b) = 2\sigma_{ib},$$

where the factor of 2 arises from the product rule, Ω is the covariance matrix, e_i the i-th unit vector, and $\sigma_{ib} = \mathrm{cov}(R_i, R_b)$ is the covariance between asset i and the portfolio $R_b = \sum_j v_j R_j$. Now we need the marginal contribution of the standard deviation and not the variance hence:

Taking the derivative of the portfolio variance with respect to weight v_i:

$$\begin{aligned}
\frac{\partial}{\partial v_i}\bar{\sigma}_b^2(v) &= \frac{\partial}{\partial v_i}\sum_{i=1}^{N}\sum_{j=1}^{N} v_i v_j \sigma_{ij} \\
&= 2\sum_{j=1}^{N} v_j \sigma_{ij} \\
&= 2\, e_i'\Omega v \\
&= 2\,\mathrm{cov}(R_i, R_b) \\
&= 2\,\sigma_{ib},
\end{aligned}$$

where the factor of 2 arises from the product rule, Ω is the covariance matrix, e_i denotes the i-th unit vector, and $\sigma_{ib} = \mathrm{cov}(R_i, R_b)$ is the covariance between asset i and the portfolio $R_b = \sum_j v_j R_j$.

We can combine (1) and (2) together and obtain:

$$RRR(i,b) = \frac{\mu_i - R_f}{\sigma_{ib}/\bar{\sigma}_b}$$

Now, for the market portfolio, it holds all assets that must have the same return-to-risk; otherwise, we could improve the total return without increasing the risk, which implies an arbitrage opportunity and contradicts the fact that m is a portfolio on the frontier. Therefore it holds that:

$$\frac{\mu_i - R_f}{\sigma_{im}/\bar{\sigma}_m} = \frac{\bar{\mu}_m - R_f}{\bar{\sigma}_m^2/\bar{\sigma}_m}.$$

Rewriting this equation, we find that

$$\begin{aligned}
\mu_i - R_f &= \frac{\sigma_{im}}{\bar{\sigma}_m^2}\,(\bar{\mu}_m - R_f) \\
&= \beta_i\,(\bar{\mu}_m - R_f),
\end{aligned}$$

So we can state the CAPM as follows:

$$\mu_i - R_f = \beta_i(\bar{\mu}_m - R_f)$$

where β_i measures exposure to market risk with no reference to idiosyncratic risk since $\beta_i = Cov(R_i, R_m)/Var(R_m)$ so no reference to the variance of asset i.

Now this can be seen and directly addressed as an OLS estimator of the time series regression:

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}$$

but here we need to be careful since the CAPM is a statement regarding the DGP so it helps us to understand how excess return of asset i is evaluated while the OLS regression is a

This is the Duhem-Quine problem (those are phylosopher not citeee!!!) applied to asset pricing — if you reject, you can never be sure whether CAPM is wrong or your statistical model is misspecified. It's one reason why empirical tests of CAPM remain deeply controversial even decades later.

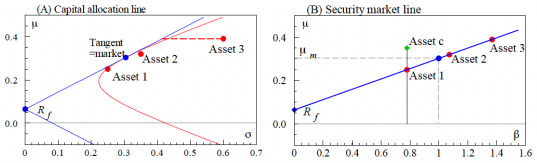


Figure 1: *Capital allocation line and security market line for Example 1.*

Figure 5.2

convenient way to estimate it.

The CAPM pricing condition admits two useful interpretations.

Interpretation I: Marginal risk pricing. Rewriting $\beta_i = \sigma_{im}/\bar{\sigma}_m^2$ as $\sigma_{im}/\bar{\sigma}_m$, the excess return decomposes as:

$$\mu_i - R_f = \frac{\sigma_{im}}{\bar{\sigma}_m} \cdot \frac{\bar{\mu}_m - R_f}{\bar{\sigma}_m} = \underbrace{\frac{\partial \bar{\sigma}_m}{\partial v_i}}_{\text{marginal risk}} \cdot \underbrace{S(m)}_{\text{price of risk}},$$

where $\partial\bar{\sigma}_m/\partial v_i = \sigma_{im}/\bar{\sigma}_m$ is the marginal contribution of asset i to portfolio volatility, and $S(m) = (\bar{\mu}_m - R_f)/\bar{\sigma}_m$ is the Sharpe ratio of the market portfolio.

Interpretation II: Covariance risk pricing. Alternatively, grouping σ_{im} separately yields:

$$\mu_i - R_f = \underbrace{\sigma_{im}}_{\text{covariance}} \cdot \underbrace{\lambda}_{\text{price of covariance risk}},$$

where $\lambda = (\bar{\mu}_m - R_f)/\bar{\sigma}_m^2$ is a single market-wide price of risk, common to all assets. Assets differ in their excess returns only through their covariance σ_{im} with the market.

5.3 Asset Pricing Formulation and derivation

The CAPM we have obtained is an equilibrium result whcih can hold at steady state, however financial markets evolve and change over time theredore also the measure of market risk given by the β might actually be considered time varyign and therefore we should add some time reference to it, which motivates the conditional CAPM approach.

The joint distribution of asset i and the market return is:

$$\begin{pmatrix} R_i \\ R_m \end{pmatrix} = \begin{pmatrix} \mu_i \\ \mu_m \end{pmatrix} + \begin{pmatrix} \varepsilon_i \\ \varepsilon_m \end{pmatrix}, \qquad \begin{pmatrix} \varepsilon_i \\ \varepsilon_m \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} \Omega_{ii} & \Omega_{im} \\ \Omega_{im} & \Omega_{mm} \end{pmatrix}\right).$$

The off-diagonal term $\Omega_{im} = \text{cov}(R_i, R_m) \neq 0$, reflecting that the two residuals are correlated. Applying the projection theorem for jointly normal variables, we condition on R_m and rewrite the system as:

$$\begin{pmatrix} R_i \\ R_m \end{pmatrix} = \begin{pmatrix} \tilde{\mu}_i \\ \mu_m \end{pmatrix} + \begin{pmatrix} \beta \\ 0 \end{pmatrix} R_{me} + \begin{pmatrix} \tilde{\varepsilon}_i \\ \varepsilon_m \end{pmatrix}, \qquad \begin{pmatrix} \tilde{\varepsilon}_i \\ \varepsilon_m \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} \tilde{\Omega}_{ii} & 0 \\ 0 & \Omega_{mm} \end{pmatrix}\right),$$

where $R_{me} = R_m - \mu_m$ is the demeaned market return and $\beta = \Omega_{im}\Omega_{mm}^{-1}$. The residual variance of asset i shrinks to:

$$\tilde{\Omega}_{ii} = \Omega_{ii} - \Omega_{im}\Omega_{mm}^{-1}\Omega_{im} = \Omega_{ii} - \beta^2\Omega_{mm},$$

which is the variance of R_i unexplained by the market. Crucially, the covariance matrix of the new residuals is *diagonal* — the off-diagonal term is exactly zero by construction of the projection. This means $\tilde{\varepsilon}_i$ is orthogonal to ε_m: all co-movement between R_i and R_m has been

absorbed into the βR_{me} term, leaving only idiosyncratic risk behind.

Now we could consider the explicit time reference and get:

$$\mu_{i,t+1|t} = E_t\left[\frac{p_{i,t+1}}{p_{i,t}} - 1\right]$$

now we can write the CAPM with the conditional expectations with respect to time and get:

$$E_t[R_{i,t+1}] - R_f = \frac{\text{cov}_t(R_{i,t+1}, R_{m,t+1})}{\text{Var}_t(R_{m,t+1})}\left(E_t[R_{m,t+1}] - R_f\right) = \beta_{i,t}\left(E_t[R_{m,t+1}] - R_f\right).$$

from which we obtain:

$$p_{i,t} = E_t\left[\frac{1}{1 + R_f + \beta_{i,t}\left(\mu_{m,t+1|t} - R_f\right)}\right] E_t[p_{i,t+1}].$$

which states that today's price is given by the conditional expectation of future price discounted $1 + R_f + \beta_{i,t}\left(\mu_{m,t+1|t} - R_f\right)$ which sums the rate of risk free asset plus the market related one. Alternatively we could write it as:

$$p_{i,t} = \frac{E_t[p_{i,t+1}] - \lambda_t \,\text{cov}_t(p_{i,t+1}, R_{m,t+1})}{1 + R_f}.$$

which can be understood as the present value of the asset's expected future payoff, adjusted for the relevant risks (i.e. discounted).

5.4 A General Approach to Asset pricing

The CAPM is one specific way to model asset prices, we can derive a more general and broader approach to asset pricing and derive how the CAPM arises as a general case.

In this model, we introduce Asset Pricing theory starting from consumption and investment in order to understand how they contribute to pricing assets. With Asset pricing we want to find the fair value of the stochastic stream of the cash flow.

An agent would like to maximize his lifetime utility given deciding how much to save, investimg into an asset:

$$\max_s [u(y_t - sp_t) + \mathbb{E}_t[\delta u(y_{t+1} + sX_{t+1})]$$

We know that X_{t+1} represents the payoff a specific asset pays at time $t+1$, and it is given by:

$$\underbrace{X_{t+1}}_{\text{payoff}} = \underbrace{p_{t+1}}_{\text{price asset}} + \underbrace{d_{t+1}}_{\text{dividend}}$$

Now performing the maximiziation of all this and assuming $c_t = y_t - sp_t$ we can get the **Basic Pricing Equation**:

$$u'(c_t)p_t = \mathbb{E}[\delta u'(c_{t+1})X_{t+1}]$$

Now what the equation is telling us is *The marginal loss of consuming less today and buying little more of an asset should equate the marginal gain of consuming a little more of an asset's payoff in the future.*

In case the first term is grater than the second it means that the loss of consuming less today is higher than the gains of consuming of little more of asste's payoff in the future therefore I increase my consumption today.

From the previous analysis we can retrive the basic pricing equation as follow:

$$p_t = E_t \left[\underbrace{\delta \frac{u'(c_{t+1})}{u'(c_t)}}_{\text{sdf}} X_{t+1} \right]$$

[1]

Now the interpretation we give to that is that we price our asset given the relation between the future Payoff of the asset X_{t+1}, which is stochastic, and the discounted ratio of the intertemporal marginal utility of consumption. We can see also interpret this formula following:

- **Portfolio choice**: given the price and the return we find the best or optimal behaviour of agent.
- **Asset Pricing**: we look at p_t, given X_{t+1}, m_{t+1}

Since we have that X_{t+1} and p_{t+1} moves together [2] we can normalize everything for the price and get:

$$1 = \mathbb{E}[m_{t+1} R_{t+1}]$$

Here R_{t+1} is the GROSS Return (ex. 1.05 instead of the excess return of 0.05).

Considering a situation withouth uncertainty then we can assess that $R_{t+1} = R_{t+2} = R_{t+3} = R^f$, tehrefore we have that:

$$R^f = \frac{1}{\mathbb{E}[m_{t+1}]}$$

In this setting, R^f is linked to both discount factor δ and risk aversion given by the utility function we are going to use.

Conversely, under uncertainty we get to the following so:

$$1 = \mathbb{E}[m_{t+1} R]$$

Subtracting the two we obtain:

$$\mathbb{E}[m(R - R^f)] = 0$$

Now we can consider $Cov(\phi, p) = E[\phi p] - E[\phi]E[p]$ therefore we get:

$$p_t = E_t[\phi_{t+1}]E_t[p_{t+1}] + cov(\phi_{t+1}, p_{t+1}) = \frac{E_t[p_{t+1}]}{1 + R_f} + cov(\phi_{t+1}, p_{t+1})$$

now we could derive the same implications for the CAPM and get:

$$E_t[1 + R_{i,t+1}] = R_f - \underbrace{\frac{cov(\phi_{t+1}, R_{i,t+1})}{V_t(\phi_{t+1})}}_{\beta^{\phi}_{i,t}} \underbrace{\frac{V_t(\phi_{t+1})}{E_t[\phi_{t+1}]}}_{\text{price of risk}}$$

[1] **SDF** stochastic discount factor

[2] If I have a stock of amazon than the payoff is high but also the price is high

5.5 From the General Pricing Formula to CAPM

We begin with the general asset pricing formula, which prices any asset by discounting its future payoff using the stochastic discount factor (SDF) ϕ_{t+1}:

$$p_t = E_t\big[\phi_{t+1} \cdot p_{t+1}\big].$$

CAPM restricts the SDF to be a *linear function of the market return*, reflecting the assumption that only market-wide risk drives the marginal rate of substitution:

$$\phi_{t+1} = \frac{1 - \lambda_t\big(R_{m,t+1} - E_t[R_{m,t+1}]\big)}{1 + R_f} = \frac{1 + \lambda_t E_t[R_{m,t+1}]}{1 + R_f} - \frac{\lambda_t}{1 + R_f} R_{m,t+1} = a_t + b_t R_{m,t+1},$$

where λ_t is the time-varying price of covariance risk, and a_t, b_t are defined by the last equality. Substituting into the general formula:

$$\begin{aligned} p_t &= E_t\big[\phi_{t+1} \cdot p_{t+1}\big] \\ &= E_t\left[\frac{1 - \lambda_t\big(R_{m,t+1} - E_t[R_{m,t+1}]\big)}{1 + R_f} \cdot p_{t+1}\right] \\ &= E_t\left[\frac{p_{t+1} - \lambda_t\big(R_{m,t+1} - E_t[R_{m,t+1}]\big)p_{t+1}}{1 + R_f}\right] \\ &= \frac{E_t[p_{t+1}] - \lambda_t \operatorname{cov}_t(p_{t+1},\, R_{m,t+1})}{1 + R_f}, \end{aligned}$$

where the last step uses the fact that $E_t\big[(R_{m,t+1} - E_t[R_{m,t+1}])p_{t+1}\big] = \operatorname{cov}_t(p_{t+1}, R_{m,t+1})$, since the deviation of $R_{m,t+1}$ from its conditional mean is orthogonal to any $\mathcal{F}_t$-measurable quantity. This is precisely the CAPM pricing formula: the current price equals the discounted expected future payoff, penalised by the covariance of the payoff with the market, scaled by λ_t.

The SDF $\phi_{t+1} = a_t + b_t R_{m,t+1}$ makes the CAPM restriction transparent: among all possible SDFs, CAPM selects the unique one that is *affine in the market return*. This embodies the core assumption that the only source of priced risk is co-movement with the market portfolio — idiosyncratic risk carries no risk premium, as it can be diversified away.

Finally, note that the price of covariance risk λ_t depends on conditional moments and is therefore time-varying in general, which connects this formulation directly to the conditional CAPM.

5.6 Asset Pricing Theory

We have seen how the CAPM arises as a nested special case of a more general asset pricing framework. Imposing assumptions on investor portfolio optimization — specifically mean-variance preferences and homogeneous expectations — yields the conditional CAPM:

$$\mathbb{E}_{t-1}[R_{i,t}] - R_f = \beta_{i,t}\big(\mathbb{E}_{t-1}[R_{m,t}] - R_f\big),$$

where the time-varying beta,

$$\beta_{i,t} = \frac{\operatorname{Cov}_{t-1}(R_{i,t},\, R_{m,t})}{\operatorname{Var}_{t-1}(R_{m,t})},$$

measures the sensitivity of asset i to systematic market risk. The equity risk premium $\mathbb{E}_{t-1}[R_{m,t}] - R_f$ compensates investors for bearing undiversifiable market-wide risk, while $\beta_{i,t}$ determines the share of that premium required by asset i. Crucially, the choice of market proxy directly affects the estimated betas and, consequently, the implied compensation for systematic risk — a source

of both empirical flexibility and well-known identification challenges.

It is natural to extend the model to accommodate multiple sources of systematic risk. We introduce a vector of factors $f_t \in \mathbb{R}^k$, where k denotes the number of market factors with $1 \leq k < N$. The model is then specified through two equations: a conditional equation for excess returns Y_t given the factors and the past filtration, and an equation describing the dynamics of the factors themselves. Formally, the **Factor Model** is:

$$\begin{cases} Y_t = \mu + Bf_t + \varepsilon_t, & \text{(conditional return equation)} \\ f_t = x_t - \mathbb{E}_{t-1}[x_t], & \text{(factor dynamics)} \end{cases} \tag{5.1}$$

where $\mu \in \mathbb{R}^N$ is the vector of unconditional means, $B \in \mathbb{R}^{N \times k}$ is the matrix of factor loadings (betas), and ε_t is an idiosyncratic error term orthogonal to f_t. The first equation decomposes excess returns into a systematic component Bf_t — representing compensation for exposure to each factor — and an asset-specific residual. The second equation defines each factor as the surprise component of a candidate observable x_t, i.e. the deviation from its conditional expectation, ensuring that f_t is a martingale difference sequence with $\mathbb{E}_{t-1}[f_t] = 0$.

5.6.1 The no arbitrage condition

We now derive the no-arbitrage restriction and its geometric meaning. No-arbitrage states that in an efficient market, there is no portfolio that generates a strictly positive expected return at zero cost and zero risk — any compensation must reflect exposure to systematic risk.

Consider an *arbitrage portfolio* $\omega \in \mathbb{R}^N$ satisfying two conditions:

- **Zero cost:** $\omega'\iota = \sum_{i=1}^N \omega_i = 0$, so no net capital is invested.
- **Zero systematic risk:** $\omega' B = 0$, meaning the portfolio has no exposure to any of the k factors.

When additionally the portfolio is well-diversified, the idiosyncratic component $\omega'\varepsilon_t \to 0$ as $N \to \infty$, so all risk vanishes. No-arbitrage then requires that such a portfolio also earns zero expected return:

$$\mathbb{E}_{t-1}(\omega' Y_t) = \omega'\mu = 0.$$

This gives us the key logical chain:

$$\omega' B = 0 \implies \omega'\mu = 0.$$

Geometric interpretation. Think of $B \in \mathbb{R}^{N \times k}$ as defining a k-dimensional subspace of asset space $\mathbb{R}^N$, the *factor space* $\mathcal{F} = \text{col}(B)$. The condition $\omega' B = 0$ means ω lies in $\mathcal{F}^\perp = \text{null}(B')$, the orthogonal complement — the subspace of portfolios with no factor exposure whatsoever. The implication above says that every such ω must also satisfy $\omega'\mu = 0$, i.e. μ is orthogonal to the entire null space of B'. By the Fundamental Theorem of Linear Algebra, this is only possible if:

$$\mu \in \text{col}(B),$$

meaning μ lies entirely within the factor space. If μ had any component *outside* $\text{col}(B)$ — in the idiosyncratic subspace — one could construct a zero-cost, zero-risk portfolio that earns a strictly positive return, a direct contradiction. Therefore, there must exist a vector $\lambda_k \in \mathbb{R}^k$ such that:

$$\boxed{\mu = B\lambda_k,}$$

where $\lambda_k = (\lambda_{k,1}, \ldots, \lambda_{k,k})'$ collects the *risk premia* associated with each factor. Each component $\lambda_{k,j}$ is the expected excess return per unit of exposure to factor j, and $B\lambda_k$ aggregates these

premia according to each asset's loading profile. This is the **APT restriction**: expected excess returns are completely spanned by the k risk factors, and any return not attributable to factor exposure would violate no-arbitrage. The CAPM is recovered as the special case $k = 1$, with $B = \beta$ and $\lambda_k = \mathbb{E}_{t-1}[R_{m,t}] - R_f$.

5.6.2 The simple IID Case

We now consider the unrealistic case where the excess returns and the factor are assumed to be iid, in particular, we assume $x_t \in \mathbb{R}^k$ and we specify a model for $(Y_t, x_t)'$ to be joint Gaussian. In particular we specify:

$$\begin{aligned} Y_t &= \mu + e_t \\ x_t &= \delta + f_t \end{aligned}$$

such that:

$$\begin{pmatrix} \varepsilon_t \\ f_t \end{pmatrix} \sim N\left(0, \begin{pmatrix} \Omega_e & \Omega_{e,f} \\ \cdot & \Omega_f \end{pmatrix}\right)$$

We may consider the conditional and the marginal models as: †

where $B = \Omega_e \Omega_f^{-1}$ hence obtaining e_t independent rom f_t after conditioning hence the parameter space is defined by:

$$\theta = \{\mu, B, \Omega_\varepsilon, \delta, \Omega_f\}$$

Since they do not share the same parameters, we can do MLE sequentially. Therefore, the MLE of the unrestricted model can be done either by simultaneous estimation or by considering the conditional model:

$$Y_t = \mu + B(x_t - \delta) + \epsilon_t$$

and the marginal model:

$$x_t = \delta + f_t$$

separately. The formal justification follows directly from the joint Gaussian structure. Suppose:

$$\begin{pmatrix} Y_t \\ x_t \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu \\ \delta \end{pmatrix}, \begin{pmatrix} \Omega_{YY} & \Omega_{Yx} \\ \Omega_{xY} & \Omega_x \end{pmatrix}\right).$$

By standard properties of the multivariate normal, the conditional distribution of Y_t given x_t is:

$$Y_t \mid x_t \sim \mathcal{N}\big(\mu + \Omega_{Yx}\Omega_x^{-1}(x_t - \delta),\ \Omega_{YY} - \Omega_{Yx}\Omega_x^{-1}\Omega_{xY}\big).$$

Defining $B = \Omega_{Yx}\Omega_x^{-1}$ and $\Omega_\varepsilon = \Omega_{YY} - \Omega_{Yx}\Omega_x^{-1}\Omega_{xY}$ (the Schur complement of Ω_x), we recover exactly:

$$Y_t = \mu + B(x_t - \delta) + \varepsilon_t, \qquad x_t = \delta + f_t,$$

with $\varepsilon_t \sim \mathcal{N}(0, \Omega_\varepsilon)$, $f_t \sim \mathcal{N}(0, \Omega_x)$, and $\varepsilon_t \perp f_t$ by construction. The joint likelihood then factorizes as:

$$L(\mu, B, \Omega_\varepsilon, \delta, \Omega_x) = L_{Y|x}(\mu, B, \Omega_\varepsilon) \cdot L_x(\delta, \Omega_x),$$

since $\{\mu, B, \Omega_\varepsilon\}$ enter only the conditional factor and $\{\delta, \Omega_x\}$ enter only the marginal factor. The two parameter sets are variation independent — they live in separate, unconstrained blocks of the likelihood — which justifies sequential MLE with no loss of efficiency.

Now we consider trying to add the no-arbitrage restriction, we would impose $\mu = B\lambda$ and obtain:

$$Y_t = B\lambda + BF_t + e_t$$
$$x_t = \delta + f_t$$

Here, we do not have variation independence anymore; therefore, we require a joint optimisation. We recall that the restriction we introduce is not linear, therefore we cannot run use OLS.

Once the joint model for $(Y_t', x_t')'$ is specified, inference on the no-arbitrage restriction $H_0 : \mu = B\lambda$ proceeds via classical likelihood-based tests. Under H_0 and standard regularity conditions,

$$LR \xrightarrow{d} \chi^2(N-k).$$

The pragmatic two-step approach. A fully efficient MLE requires joint optimisation of the restricted likelihood, since imposing $\mu = B\lambda$ breaks the variation independence between the conditional and marginal parameter blocks (see above). In practice, the literature often adopts a *pragmatic* two-step approach: in the first step, estimate the factor by demeaning the observed candidate factor,

$$\hat{f}_t \;=\; (R_{m,t} - R_{f,t}) \;-\; T^{-1}\sum_{i=1}^{T}(R_{m,t} - R_{f,t}),$$

treating $\hat{f}_t$ as observed. In the second step, run OLS of Y_t on $\hat{f}_t$ and a constant,

$$Y_t \;=\; \mu + \beta\,\hat{f}_t + \varepsilon_t.$$

This is a *plug-in* estimator: it replaces the unobserved $f_t = x_t - \delta$ with its sample analogue $\hat{f}_t = x_t - \bar{x}$. While computationally straightforward, this approach is **not** equivalent to the MLE. In the unrestricted case the two coincide by variation independence, but under the restriction $\mu = B\lambda$ the parameter δ (the mean of x_t) enters both the conditional and the marginal likelihood, so the plug-in step introduces a first-step estimation error that is not accounted for in the second step. As a consequence, LR-type statistics computed from two-step estimates are *pseudo*-LR statistics and their finite-sample distribution may deviate from the asymptotic χ^2 benchmark, particularly when N is large relative to T.

Empirical illustration. Consider $N = 10$ sector excess returns,

$$Y_i \;\in\; \{\text{Food, Beer, Smoke, Games, Books, Hshld, Clths, Hlth, Chems, Txtls}\},$$

on monthly data 1926–2009, with the market excess return $X_t = R_{m,t} - R_{f,t}$ as the sole risk factor ($k = 1$).

Parameter count: unrestricted model. The conditional model contains two free parameter vectors in the mean:

$$\theta^U \;=\; \{\underbrace{\mu_1,\ldots,\mu_N}_{N},\ \underbrace{\beta_1,\ldots,\beta_N}_{N},\ \Omega\},\qquad \text{so } 2N = 20 \text{ parameters in the conditional mean.}$$

Parameter count: restricted model. Imposing $\mu = \beta\lambda$ replaces the N free intercepts by a single scalar price of risk λ, while β remains unrestricted:

$$\theta^R \;=\; \{\underbrace{\beta_1,\ldots,\beta_N}_{N},\ \underbrace{\lambda}_{1},\ \Omega\},\qquad \text{so } N+1 = 11 \text{ parameters in the conditional mean.}$$

Degrees of freedom. The number of restrictions imposed equals

$$d_f \;=\; 2N - (N+1) \;=\; N - 1 \;=\; 9.$$

It is important to note that the count yields $N-1$ and *not* N, even though the restriction $\mu = \beta\lambda$ consists of N equations. The reason is that λ is itself an unknown scalar: the restriction does not pin the μ_i to known constants, but rather imposes the cross-sectional proportionality condition

$$\frac{\mu_1}{\beta_1} \;=\; \frac{\mu_2}{\beta_2} \;=\; \cdots \;=\; \frac{\mu_N}{\beta_N} \;=\; \lambda,$$

which constitutes $N-1$ independent equality constraints. Concretely, taking $\lambda = \mu_1/\beta_1$ from the first sector to identify the price of risk, the remaining $N-1=9$ intercepts are constrained by

$$\mu_i \;=\; \beta_i \,\frac{\mu_1}{\beta_1}, \qquad i = 2, \ldots, N,$$

leaving the $N+1 = 11$ free parameters $\{\mu_1, \beta_1, \beta_2, \ldots, \beta_N\}$ (plus Ω) to be estimated under H_0.

The general rule for k factors follows the same logic: the restriction $\mu = B\lambda$ pins the N-dimensional vector μ onto a k-dimensional subspace spanned by the columns of B, reducing the parameter count by $N-k$. Hence:

$$\boxed{d_f \;=\; N - k.}$$

For the present example, $N = 10$ and $k = 1$, giving $d_f = 9$ and therefore

$$LR \;\xrightarrow{d}\; \chi^2(9).$$

The estimated LR statistic is 14.8 with a p-value of 0.096, so the no-arbitrage restriction is borderline accepted at the 10% significance level.

5.6.3 ARCH Effects and Time-Varying Betas

A natural extension of the previous framework is to allow the innovations to display conditional heteroskedasticity. This matters because, as we show below, ARCH effects in the joint covariance structure directly induce *time variation in the conditional betas*, rendering the constant-beta assumption of the preceding sections inappropriate.

The joint model. Consider the joint system for excess returns $Y_t \in \mathbb{R}^N$ and the candidate factor $X_t \in \mathbb{R}^k$,

$$Y_t = \theta_y + \varepsilon_{y,t}, \tag{5.2}$$
$$X_t = \theta_x + \varepsilon_{x,t}, \tag{5.3}$$

where the innovation vector $\varepsilon_t = (\varepsilon'_{y,t}, \varepsilon'_{x,t})' \in \mathbb{R}^{N+k}$ follows a multivariate GARCH process,

$$\varepsilon_t \;=\; \Omega_t(\gamma)^{1/2}\,\eta_t, \qquad \eta_t \;\sim\; i.i.d.\; \mathcal{N}(0,\, I_{N+k}),$$

with $\Omega_t(\gamma)$ the conditional covariance matrix parametrised by $\gamma \in \mathbb{R}^G$ under some multivariate GARCH specification (e.g. DCC, CCC, or BEKK),

$$\Omega_t(\gamma) \;=\; \begin{pmatrix} \Omega_{yy,t}(\gamma) & \Omega_{yx,t}(\gamma) \\ \Omega_{xy,t}(\gamma) & \Omega_{xx,t}(\gamma) \end{pmatrix}.$$

The conditional model and time-varying beta. By standard properties of the multivariate Gaussian distribution, the conditional distribution of Y_t given X_t and $\mathcal{F}_{t-1}$ is:

$$Y_t \;=\; \theta_y - \beta_t(\gamma)\,\theta_x \;+\; \beta_t(\gamma)\,X_t \;+\; \varepsilon_t^*, \qquad \varepsilon_t^* \,|\, \mathcal{F}_{t-1} \;\sim\; \mathcal{N}(0,\; \Omega_{yy.x,t}(\gamma))\,,$$

where the *time-varying conditional beta* is

$$\beta_t(\gamma) \;=\; \Omega_{yx,t}(\gamma)\,\Omega_{xx,t}(\gamma)^{-1}, \tag{5.4}$$

and $\Omega_{yy.x,t}(\gamma) = \Omega_{yy,t}(\gamma) - \Omega_{yx,t}(\gamma)\,\Omega_{xx,t}(\gamma)^{-1}\Omega_{xy,t}(\gamma)$ is the Schur complement. Equation (5.4) makes clear that ARCH effects — which drive variation in $\Omega_{yx,t}$ and $\Omega_{xx,t}$ — are directly inherited by β_t, so that the conditional risk exposure of each asset to the factor fluctuates over time.

Unrestricted estimation. The unrestricted MLE is obtained by maximising the joint Gaussian log-likelihood,

$$L(\theta_y, \theta_x, \gamma) \;=\; -\frac{1}{2}\sum_{t=1}^{T}\Big[\log\det\big(\Omega_t(\gamma)\big) \;+\; (Z_t-\theta)'\,\Omega_t(\gamma)^{-1}\,(Z_t-\theta)\Big]\,, \tag{5.5}$$

where $Z_t = (Y_t', X_t')'$ and $\theta = (\theta_y', \theta_x')'$. Denoting the unrestricted MLE as $(\hat\theta_y, \hat\theta_x, \hat\gamma)$, the time-varying betas are recovered as

$$\hat\beta_t \;=\; \Omega_{yx,t}(\hat\gamma)\,\Omega_{xx,t}(\hat\gamma)^{-1}.$$

The no-arbitrage restriction and its fundamental difficulty. In the i.i.d. case, the no-arbitrage restriction $\theta_y = \beta\lambda$ was well-posed because β was a fixed matrix. Under ARCH, the conditional CAPM requires

$$E_{t-1}(Y_t) \;=\; \beta_t(\gamma)\,\lambda, \qquad \text{for all } t,$$

where λ is a constant price of risk. Since $E_{t-1}(Y_t) = \theta_y$ is a fixed vector, this imposes $\theta_y = \beta_t(\gamma)\lambda$ simultaneously for all t. This is *internally inconsistent*: the left-hand side is constant while the right-hand side varies over time unless β_t is itself constant — which would negate the time-varying specification entirely.

To make this precise, consider the attempted constrained model,

$$Y_t \;=\; \beta_t(\gamma)\big(X_t - \theta_x - \lambda\big) + \varepsilon_t^*.$$

The unrestricted and constrained models are **not nested in the standard sense**: in the unrestricted model β_t enters only the covariance structure, whereas in the constrained model it also enters the conditional mean. The resulting pseudo-LR statistic,

$$PLR \;=\; -2\Big[L\Big(\tilde\theta_x,\,\tilde\lambda,\,\tilde\gamma\Big) \;-\; L\Big(\hat\theta_y,\,\hat\theta_x,\,\hat\gamma\Big)\Big]\,,$$

is *not* a standard LR statistic: the two likelihoods being compared correspond to models with qualitatively different mean specifications, so the usual χ^2 asymptotic theory does not apply and the distribution of PLR under H_0 is unknown. As acknowledged in the literature, *how to formally test the no-arbitrage condition under time-varying betas remains an open problem.*

The BEW approach. Bollerslev, Engle and Wooldridge (1988) sidestep this difficulty by embedding the no-arbitrage condition directly into the model structure rather than testing it after estimation. Defining the market portfolio as $r_{m,t} = \omega_{t-1}'Y_t$ where $\omega_{t-1} \in \mathcal{F}_{t-1}$ are

predetermined portfolio weights, the GARCH-in-mean model is:

$$Y_t \;=\; \delta\,\Omega_t\,\omega_{t-1} \;+\; \varepsilon_t, \qquad \varepsilon_t \;=\; \Omega_t^{1/2} v_t, \quad v_t \sim i.i.d.\,\mathcal{N}(0, I),$$

where δ is a *scalar* aggregate relative risk aversion parameter. Since $\Omega_t\omega_{t-1} = \mathrm{Cov}_{t-1}(Y_t, r_{m,t})$, the CAPM relationship holds by construction:

$$E_{t-1}(Y_t) \;=\; \delta\,\Omega_t\,\omega_{t-1} \;=\; \beta_t\,E_{t-1}(r_{m,t}),$$

with $\beta_t = \Omega_t\omega_{t-1}/(\omega_{t-1}'\Omega_t\omega_{t-1})$. The no-arbitrage content is then encoded in the single scalar restriction that the same δ governs all N assets simultaneously, which is estimable by MLE without any separate testing step.

Chapter 6

Penalised QMLE - LASSO

Machine Learning is a field that is developing rapidly and introducing important features that enable the handling of high-dimensional settings. In fact, it is quite common for the number of regressors in our dataset to exceed the number of observations. We would like to take into account the variables that contain the most relevant information. We do face two main important questions then:

1. **What is Machine Learning**: Starting from a simple regression model where we consider $N > T$, so the number of variables exceeds the number of time observations:

   $$y_t = \alpha + \beta_1 x_{1,t} + \cdots + \beta_N x_{N,t} + \epsilon_t$$

   OLS is not going to work in this case. To see why, note that each observation t can be viewed as a row vector $(x_{1,t}, \ldots, x_{N,t}) \in \mathbb{R}^N$. These T row vectors span at most a T-dimensional subspace of the parameter space $\mathbb{R}^N$. Since $T < N$, this subspace is strictly smaller than $\mathbb{R}^N$, leaving a null space of dimension at least $N - T > 0$. Any vector δ in this null space satisfies $X\delta = 0$, so if $\hat{\beta}$ solves the normal equations, so does $\hat{\beta} + \delta$ for any such δ — producing identical fitted values. The data cannot identify these directions in parameter space, and OLS yields infinitely many solutions rather than a unique estimator. Equivalently, $X'X$ is an $N \times N$ matrix of rank at most $T < N$, hence singular and non-invertible. Now the main problem is to choose a number of parameters that would allow OLS to work by setting all the others to zero. In particular we need to identify:

   $$\begin{cases} \beta^Z \\ \beta^{NZ} \end{cases}$$

2. **Where I want to apply LASSO to?** Lasso can be applied in many different settings, in particular, in financial econometrics, we could apply Lasso to high-dimensional GARCH models where we have:

   $$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i x_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{j,t-j}^2$$

   As we can observe from the picture below, the GARCH$(1,1)$ imposes a specific, rigid geometric decay structure on the ACF of squared returns r_t^2, which gives high persistence but fails to match the true empirical ACF. We could in principle rely on ARCH(p) models to achieve a more flexible lag structure, however as p grows we face a high-dimensional estimation problem. This is where LASSO kicks in: by shrinking irrelevant lag coefficients exactly to zero, it selects the sparse set of lags that carry genuine information about future volatility, making estimation feasible even when p is large relative to T.

   When we consider

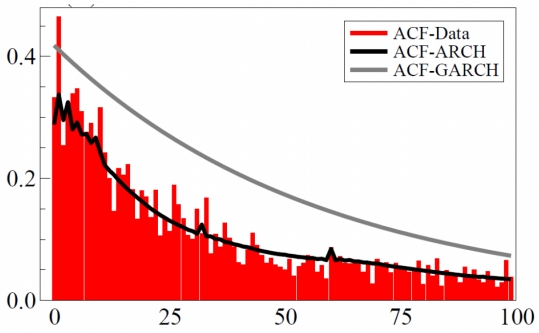


Figure 6.1

6.1 Intoduction

We start with a simple location model, where we specify $\beta \geq 0$ so we have:

$$x_t = \beta + \varepsilon_t$$

and $\varepsilon_t \sim_{i.i.d} N(0,1)$. We ca then compute the log-likelihood so we have:

$$L(\beta) = -n^{-1} \sum_{t=1}^{n} (X_t - \beta)^2$$

In this setting the $\hat{\beta}_{mle} = \max(0, \bar{x})$ where the estimator is given by $\bar{x} = n^{-1} \sum_{t=1}^{n} x_t$. We consider positive values of the estimate; otherwise, we set it to 0.

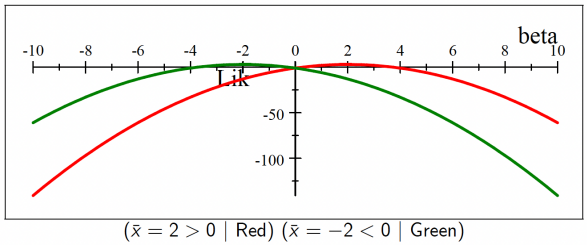


Figure 6.2

We now introduce LASSO penalisation. Since we are maximising the likelihood, subtracting a penalty term that grows with β tilts the objective downward as β increases, shifting the peak of the penalised likelihood toward zero. Coefficients with weak data support are pushed exactly to zero, while larger coefficients are shrunk by λ but remain positive. The penalised objective is:

$$Q(\beta) = L(\beta) - P(\beta), \qquad P(\beta) = n\lambda\beta,$$

so that maximising $Q(\beta)$ trades off data fit against the cost of having a large coefficient.

In the standard LASSO, the penalty is imposed on $|\beta|$, which introduces a kink at zero and hence non-differentiability. However, since we are operating in the constrained region $\beta \geq 0$, we have $|\beta| = \beta$ throughout the feasible domain, so the penalty reduces to a linear function of β. The non-differentiability is nonetheless still operative at the boundary: when the gradient of the likelihood at $\beta = 0$ is smaller in magnitude than λ, the optimum is pinned exactly at zero, producing sparsity. This gives the penalised QMLE:

$$\hat{\beta}_{\text{pQML}} = \max(0,\ \bar{x} - \lambda),$$

so that coefficients with insufficient signal are set exactly to zero, while those with strong signal survive but are shrunk toward zero by λ.

In particular, we obtain the following likelihood function:

$$Q(\beta) = -n^{-1} \sum_{t=1}^{n} (x_t - \beta)^2 - 2\lambda\beta$$

and here below we report a graphical representation of what LASSO does.

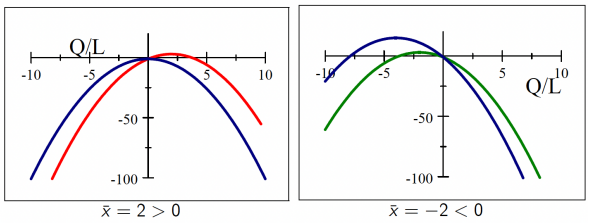


Figure 6.3

We can now consider another setting where we work with ARCH(p)-X(q) identified as:

$$x_t = \sigma_t(\theta) z_t$$
$$\sigma_t^2(\theta) = \gamma + \sum_{i=1}^{p} \beta_i x_{t-1-i} + \sum_{j=1}^{q} \beta_{p+i} v_{t-1-i}^2$$

here the dimension of the parameter is given by $dim(\beta) = p + q$ where $\beta \geq 0$ and $\gamma > 0$. Now two main questions arise:

1. How many coefficients are truly 0?

2. What are the properties of $\hat{\theta}_{pQML}$ where $\hat{\theta}_{pQML} = (\hat{\gamma}_{pQML}, \hat{\beta}_{pQML})$

6.2 Different Penalisations

Once again we define and formalise the penalised estimator as follows:

$$\hat{\theta}_{\text{pQML}} = \arg\max_{\theta} Q(\theta),$$

where $Q(\theta) = L(\theta) - P(\theta; \lambda)$ and the total penalty is separable across coefficients:

$$P(\beta; \lambda) = \sum_{j=1}^{\dim(\beta)} p(\beta_j; \lambda).$$

The separability is important: each coefficient is penalised independently, which makes the high-dimensional problem tractable.

Different choices of $p(\cdot\,; \lambda)$ lead to estimators with different properties. Under **LASSO** penalisation, $p(\beta_j; \lambda) = \lambda\beta_j$, the penalty grows linearly with β_j, so all coefficients are shrunk toward zero proportionally to their magnitude. Coefficients with insufficient data support are pushed exactly to zero, while larger coefficients survive but remain biased downward by λ. This persistent shrinkage of large coefficients is what prevents LASSO from achieving the oracle property.

An alternative is **hard thresholding**:

$$p(\beta_j;\lambda) = \lambda^2 - (\beta_j - \lambda)^2\,\mathbb{I}(\beta_j < \lambda).$$

Here the penalty is active and quadratic for $\beta_j < \lambda$, but becomes a constant λ^2 for $\beta_j \geq \lambda$, meaning penalisation stops entirely once a coefficient is large enough. The estimator therefore makes a binary decision: coefficients below the threshold are set exactly to zero, while those above it are retained without any shrinkage bias. This is precisely what allows hard thresholding to achieve the oracle property that LASSO cannot.

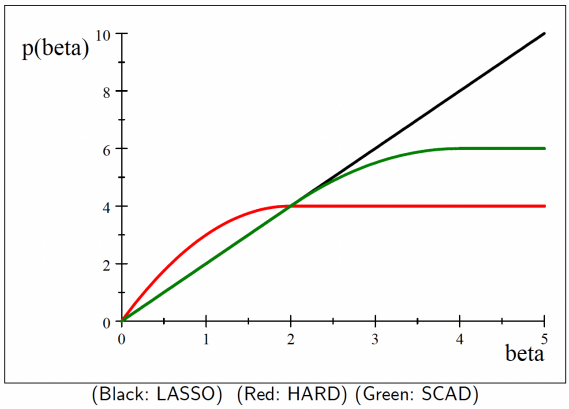


Figure 6.4

In both cases $\lambda > 0$ is a tuning parameter controlling the degree of penalisation. In practice its optimal value is selected by minimising an information criterion such as BIC over a grid of candidate values, or equivalently via cross-validation.

6.3 Theoretical Arguments

As usual, for all the derivations, we end up with two important objects that will ensure convergence in distribution of our estimator: the score S_n and the information matrix I_n. The argument proceeds in two steps. First, we perform a local Taylor expansion of $Q(\theta_0 + u/\sqrt{n})$ around the true parameter θ_0, which delivers the precise asymptotic behaviour of the penalised objective — this is where the score and information enter and where both the sparsity result and the oracle asymptotic normality are established. Second, a global argument shows that $Q(\theta_0) > Q(\theta_0 + u/\sqrt{n})$ for large $|u|$, ruling out the possibility that the estimator drifts far from θ_0 and thereby establishing consistency. This second step is cruder — it does not require precision about the shape of Q, only that the quadratic term in I_n eventually dominates and drives the objective to $-\infty$ away from θ_0. The local expansion does the real work; the global argument simply ensures the estimator never leaves the neighbourhood where that local analysis is valid.

- **First Expansion**: Through the first Taylor expansion, we need to impose regularity conditions on the score, information matrix, and third-order derivatives:

  $$S_n = \partial L_n(\theta_0)/\partial\theta \xrightarrow{D} N(0,\Sigma_S), \qquad I_n = -\partial^2 L_n(\theta_0)/\partial\theta^2 \xrightarrow{P} \Sigma_I,$$

  and

  $$\sup_{\theta\in\mathcal{N}(\theta_0)} \left|n^{-1}\partial^3 L_n(\theta)/\partial\theta^3\right| \leq c_n = O_P(1).$$

  These conditions ensure that the score admits a central limit theorem, the information matrix converges to a well-defined positive definite limit, and the remainder in the Tay-

lor expansion is asymptotically negligible. We further impose conditions on the tuning parameter λ as $n \to \infty$:

$$\lambda = \lambda_n \to 0, \qquad \sqrt{n}\lambda_n \to \infty.$$

The first condition ensures the penalty vanishes for the nonzero coefficients, recovering oracle asymptotic normality. The second ensures the penalty dominates the likelihood gradient at zero, producing exact sparsity.

- **Second Expansion**: The global consistency argument expands $L(\theta_0 + u/\sqrt{n}) - L(\theta_0)$ and shows that this quantity diverges to $-\infty$ as $|u| \to \infty$:

$$L\left(\theta_0 + \frac{u}{\sqrt{n}}\right) - L(\theta_0) = \frac{u}{\sqrt{n}} S(\theta_0) - \frac{u^2}{2n} I(\theta_0) + \dots$$

Completing the square in u:

$$= \left[(I(\theta_0)/n)^{-1} \left(S(\theta_0)/\sqrt{n}\right)\right]^2 \frac{I(\theta_0)/n}{2} - \left(u - (I(\theta_0)/n)^{-1} \left(S(\theta_0)/\sqrt{n}\right)\right)^2 \frac{I(\theta_0)/n}{2}$$

$$= O_p(1) - (u - O_p(1))^2 O_p(1) \;\to\; -\infty \quad \text{as } u \to \infty.$$

The first term is a constant in u depending only on the data. The second term is a negative quadratic in u that dominates for large $|u|$, ensuring the penalised objective at θ_0 strictly exceeds the objective at any point far from θ_0. The maximum $\hat{\theta}_n$ must therefore lie within a $\sqrt{n}$-neighbourhood of θ_0, establishing consistency.

In particular, we can establish consistency $\hat{\theta}_n \xrightarrow{p} \theta_0$ as $n \to \infty$ via the following argument. For C large, we can show that:

$$P\left\{\sup_{|u|=C} \left[Q(\theta_0) > Q\big(\theta_0 + |u|/\sqrt{n}\big)\right]\right\} = 1,$$

meaning that with probability tending to one, the penalised objective evaluated at the true parameter θ_0 strictly dominates the objective at every point on the sphere of radius $C/\sqrt{n}$ around θ_0. Intuitively, the second expansion shows that moving $u/\sqrt{n}$ away from θ_0 makes the objective worse, and the further you move the worse it gets — because the negative quadratic term in I_n eventually dominates for large $|u|$.

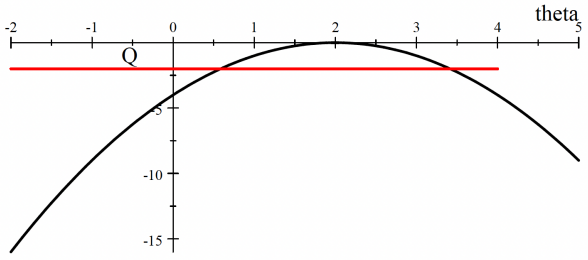


Figure 6.5

This implies that the maximum $\hat{\theta}_n$ must lie strictly inside this sphere, i.e.:

$$\hat{\theta}_n \in \Theta_n = \left\{\theta \,\middle|\, \theta = \theta_0 + |u|/\sqrt{n},\ |u|/\sqrt{n} \le 100\right\}.$$

Since $C/\sqrt{n} \to 0$ as $n \to \infty$, the estimator is confined to a shrinking neighbourhood of θ_0, which is precisely the statement of consistency. Crucially, this argument does not require knowledge

of the true parameter — it only uses the fact that the curvature of the likelihood, captured by I_n, is strictly positive and bounded away from zero.

Theorem
Under the stated assumptions, with $\hat{\beta}_{\text{pQML}} = (\hat{\beta}^N_{\text{pQML}}, \hat{\beta}^Z_{\text{pQML}})$, the following hold with probability tending to one:

- **Sparsity:**
$$\hat{\beta}^Z_{\text{pQML}} = \beta^Z_0 = \mathbf{0}_{\dim(\beta^Z)}$$
- **Oracle property:**
$$\sqrt{n}\left(\hat{\gamma}_{\text{pQML}} - \gamma_0,\, \hat{\beta}^N_{\text{pQML}} - \beta^N_0\right) \xrightarrow{D} N(0, \Omega), \qquad \Omega = \Sigma^{-1}_{I,N}\Sigma_{S,N}\Sigma^{-1}_{I,N}$$

A further regularity condition is required to avoid asymptotic bias:

$$\partial p(\beta^N_0, \lambda_n)/\partial\beta, \quad \partial^2 p(\beta^N_0, \lambda_n)/\partial\beta^2 = 0 \quad \text{for some } n > n^*.$$

This condition holds for HARD thresholding and SCAD, but **not** for LASSO. For LASSO, avoiding asymptotic bias would require $\sqrt{n}\lambda_n \to 0$, which directly contradicts the sparsity condition $\sqrt{n}\lambda_n \to \infty$.

6.4 Comments

It is important to highlight two key aspects of the LASSO penalisation in the asymptotic regime.

First, as $n \to \infty$, the condition $\lambda_n \to 0$ ensures that the penalty becomes negligible relative to the likelihood, which progressively dominates. For the truly zero parameters $\beta^Z_0 = 0$, the condition $\sqrt{n}\lambda_n \to \infty$ guarantees that the penalty still dominates the score at zero, since the score at zero is of order $1/\sqrt{n}$ while $\lambda_n \gg 1/\sqrt{n}$ by assumption. Therefore, asymptotically, the truly zero parameters are pushed exactly to zero with probability tending to one — the sparsity result holds.

Second, while LASSO correctly identifies the zero parameters, it comes at a cost for the nonzero parameters $\beta^N_0 > 0$. Since the LASSO penalty $p(\beta_j; \lambda) = \lambda\beta_j$ grows linearly for all β_j, it never stops penalising large coefficients. This introduces an asymptotic bias of order λ_n in the nonzero estimates, preventing LASSO from achieving the oracle property. Specifically, the condition required to eliminate asymptotic bias,

$$\partial p(\beta^N_0, \lambda_n)/\partial\beta = 0 \quad \text{for some } n > n^*,$$

cannot hold for LASSO since $\partial(\lambda_n\beta)/\partial\beta = \lambda_n \neq 0$ for all n. In contrast, SCAD and hard thresholding satisfy this condition because their penalties flatten out for large β, leaving large coefficients unpenalised and recovering the oracle property. The trade-off is therefore:

- **LASSO**: correctly zeros out irrelevant coefficients, but introduces shrinkage bias in the nonzero estimates.
- **SCAD / Hard threshold**: correctly zeros out irrelevant coefficients *and* leaves nonzero coefficients asymptotically unbiased, achieving the oracle property.

Appendix A

Theorems

A.1 Theorem Reference

The following theorems are invoked throughout the analysis. All convergence results hold as $T \to \infty$ unless otherwise stated.

Theorem A.1 Ergodic Law of Large Numbers

Let $\{x_t\}_{t=1}^{T}$ be a strictly stationary and ergodic sequence with $\mathbb{E}[|x_t|] < \infty$. Then

$$\frac{1}{T}\sum_{t=1}^{T} x_t \xrightarrow{p} \mathbb{E}[x_t].$$

Theorem A.2 Central Limit Theorem (i.i.d.)

Let $\{\varepsilon_t\}_{t=1}^{T}$ be an i.i.d. sequence with $\mathbb{E}[\varepsilon_t] = 0$ and $\mathbb{E}[\varepsilon_t^2] = \sigma^2 < \infty$. Then

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T} \varepsilon_t \xrightarrow{d} N\left(0,\, \sigma^2\right).$$

Theorem A.3 Chebyshev's Inequality

For any random variable X with $\mathbb{E}[X^2] < \infty$ and any $\delta > 0$,

$$\mathbb{P}(|X| > \delta) \;\leq\; \frac{\mathbb{E}[X^2]}{\delta^2}.$$

Bootstrap version. Conditional on the data, for any bootstrap random variable X^* with $\mathbb{E}^*[(X^*)^2] < \infty$,

$$\mathbb{P}^*(|X^*| > \delta) \;\leq\; \frac{\mathbb{E}^*[(X^*)^2]}{\delta^2}.$$

Theorem A.4 Martingale Difference CLT (Brown, 1971)

Let $\{m_t, \mathcal{F}_t\}_{t=1}^{T}$ be a martingale difference sequence, so $\mathbb{E}[m_t \mid \mathcal{F}_{t-1}] = 0$. Suppose:

(i) **Conditional variance condition:** $\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\big[m_t^2 \mid \mathcal{F}_{t-1}\big] \xrightarrow{p} \sigma_m^2 > 0.$

(ii) **Lindeberg condition:** $\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\Big[m_t^2\, \mathbf{1}\Big(|m_t| > \sqrt{T}\,\delta\Big) \mid \mathcal{F}_{t-1}\Big] \xrightarrow{p} 0 \quad$ for all $\delta > 0$.

Then

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T} m_t \xrightarrow{d} N\big(0, \sigma_m^2\big).$$

Theorem A.5 Donsker's Invariance Principle

Let $\{\varepsilon_t\}$ be i.i.d. with $\mathbb{E}[\varepsilon_t] = 0$ and $\mathbb{E}[\varepsilon_t^2] = \sigma^2 < \infty$. Define the partial-sum process $S_T(u) = T^{-1/2}\sum_{s=1}^{\lfloor Tu \rfloor} \varepsilon_s$ for $u \in [0,1]$. Then

$$S_T(\cdot) \; d \; B(\cdot) \quad \text{in } C[0,1],$$

where $B(\cdot)$ is a Brownian motion with variance parameter σ^2.

Theorem A.6 Continuous Mapping Theorem (CMT)

Let $\mathcal{S}$ be a metric space, X_T, $X : \mathcal{S}$-valued random elements, and $f : \mathcal{S} \to \mathcal{S}'$ continuous. If $X_T \xrightarrow{d} X$, then

$$f(X_T) \xrightarrow{d} f(X).$$

The result extends to the function space $C[0,1]$ under the sup-norm metric, permitting the application of continuous functionals (integration, sup, etc.) to weak limits of partial-sum processes.

Theorem A.7 Slutsky's Theorem

Let $X_T \xrightarrow{d} X$ and $Y_T \xrightarrow{p} c$ for some constant c. Then

$$X_T + Y_T \xrightarrow{d} X + c \qquad \text{and} \qquad X_T \cdot Y_T \xrightarrow{d} c\,X.$$

Theorem A.8 Riemann Approximation / Functional Limit

Let $x_t = \sum_{s=1}^{t} \varepsilon_s$ be a random walk with ε_t i.i.d. $(0, \sigma^2)$. For any continuous functional $g : \mathbb{R} \to \mathbb{R}$,

$$\frac{1}{T} \sum_{t=1}^{T} g\Big(T^{-1/2}\, x_{t-1}\Big) \xrightarrow{d} \int_0^1 g(B(u))\, \mathrm{d}u,$$

by Donsker's theorem combined with the CMT applied to the Riemann integral functional. In particular,

$$\frac{1}{T^2} \sum_{t=1}^{T} x_{t-1}^2 \xrightarrow{d} \int_0^1 B(u)^2\, \mathrm{d}u, \qquad \frac{1}{T^{3/2}} \sum_{t=1}^{T} x_{t-1} \xrightarrow{d} \int_0^1 B(u)\, \mathrm{d}u.$$

Theorem A.9 Stochastic Integral Limit

Under the same conditions as Theorem A.8,

$$\frac{1}{T} \sum_{t=1}^{T} \varepsilon_t\, x_{t-1} \xrightarrow{d} \int_0^1 B(u)\, \mathrm{d}B(u) \;=\; \tfrac{1}{2}\big[B(1)^2 - \sigma^2\big],$$

where the last equality is the discrete Itô formula.

Theorem A.10 Standard Exponential Limit

For any constant $k \in \mathbb{R}$,

$$\lim_{T \to \infty} \left(1 - \frac{k}{T}\right)^T = e^{-k}.$$

This identity is used in characteristic function arguments to pass from finite-product approximations to the Gaussian moment generating function.

Theorem A.11 Chi-Squared Characterisation

If $X \sim N(0,1)$, then $X^2 \sim \chi^2_1$. More generally, if $\mathbf{X} \sim N_k(\mathbf{0},\, I_k)$, then $\mathbf{X}'\mathbf{X} \sim \chi^2_k$.

Theorem A.12 Sylvester's Criterion

Let A be a real symmetric matrix of order n. Denote by Δ_k the k-th *leading principal minor*, i.e. the determinant of the upper-left $(k \times k)$ submatrix of A, for $k = 1, \ldots, n$. Then

$$A \succ 0 \quad \Longleftrightarrow \quad \Delta_k > 0 \quad \text{for all } k = 1, \ldots, n.$$

Theorem A.13 Dominated Convergence Theorem

Let $\{X_t\}_{t\geq 1}$ be a sequence of random variables defined on a common probability space, and suppose:

(i) $X_t \xrightarrow{a.s.} X$ as $t \to \infty$;

(ii) there exists an integrable dominating function $Y \geq 0$ with $E[Y] < \infty$ such that $|X_t| \leq Y$ a.s. for all $t \geq 1$.

Then $E[|X_t|] < \infty$ for all t, $E[|X|] < \infty$, and

$$\lim_{t\to\infty} E[X_t] \;=\; E\Big[\lim_{t\to\infty} X_t\Big] \;=\; E[X].$$

That is, almost sure convergence and expectation may be freely interchanged under uniform integrability guaranteed by the dominating function Y.

Appendix B

Further Arguments

B.1 Characteristic Function

Given $X \sim N(0, \sigma^2)$ it holds that $\phi_x(s)$ is the characteristic function that uniquely characterize the distribution of X.

1. For X random variable $\phi_x(t) = E[e^{itx}], t \in R$, which is the Fourier transform of the probability distribution function associated with this random variable. Unlike the moment generating function, the characteristic function always exists, encoding all the existing moments.

2. We move from to the frequency domain since it is more easy to work the proof out.

   Let $X \sim \mathcal{N}(0, \sigma^2)$. The characteristic function of X is

   $$\phi_X(t) = \mathbb{E}\left[e^{itX}\right], \qquad t \in \mathbb{R}.$$

   Since X has density

   $$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right),$$

we compute

$$\begin{aligned}\phi_X(t) &= \int_{-\infty}^{+\infty} e^{itx} f_X(x)\, dx \\ &= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right) \exp(itx)\, dx.\end{aligned}$$

Combine the exponentials:

$$-\frac{x^2}{2\sigma^2} + itx = -\frac{1}{2\sigma^2}\left(x^2 - 2i\sigma^2 tx\right).$$

Complete the square:

$$x^2 - 2i\sigma^2 tx = (x - i\sigma^2 t)^2 - (i\sigma^2 t)^2.$$

Hence,

$$-\frac{1}{2\sigma^2}\left(x^2 - 2i\sigma^2 tx\right) = -\frac{(x - i\sigma^2 t)^2}{2\sigma^2} + \frac{(i\sigma^2 t)^2}{2\sigma^2}.$$

Since $(i\sigma^2 t)^2 = -\sigma^4 t^2$, we obtain

$$\frac{(i\sigma^2 t)^2}{2\sigma^2} = -\frac{\sigma^2 t^2}{2}.$$

Therefore,

$$\begin{aligned}\phi_X(t) &= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - i\sigma^2 t)^2}{2\sigma^2}\right) \exp\left(-\frac{\sigma^2 t^2}{2}\right) dx \\ &= \exp\left(-\frac{\sigma^2 t^2}{2}\right) \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - i\sigma^2 t)^2}{2\sigma^2}\right) dx.\end{aligned}$$

The remaining integral equals 1 (it is the integral of a Gaussian density, shifted in the complex plane but with the same variance), hence

$$\boxed{\phi_X(t) = \exp\left(-\frac{\sigma^2 t^2}{2}\right).}$$

B.2 Projection Identity

B.2.1 Orthogonal Projection Identity

We now introduce $\beta \in \mathrm{R}^{\mathrm{p}\times\mathrm{r}}$ and $\beta_\perp \in \mathrm{R}^{\mathrm{p}\times(\mathrm{p}-\mathrm{r})}$ as basis for R^p where:

$$I_p = \beta(\beta'\beta)^{-1}\beta' + \beta_\perp(\beta'_\perp\beta_\perp)^{-1}\beta'_\perp.$$

For $X \in \mathbb{R}^p$, we have

$$X = \beta(\beta'\beta)^{-1}\beta'X + \beta_\perp(\beta'_\perp\beta_\perp)^{-1}\beta'_\perp X = \beta v_1 + \beta_\perp v_2.$$

Basically we project our vector into the space span by β and $\beta_\perp$.

B.2.2 Projection Identity

Consider $\alpha, \beta \in \mathbb{R}^{p\times r}$ such that $\beta'\alpha$ has full rank r. Then the pair $(\beta_\perp, \alpha)$ forms a basis of $\mathbb{R}^p$ and the identity matrix can be written as

$$I_p = \beta_\perp(\alpha'_\perp\beta_\perp)^{-1}\alpha'_\perp + \alpha(\beta'\alpha)^{-1}\beta'.$$

For any vector $X \in \mathbb{R}^p$, we obtain the decomposition

$$X = \beta_\perp(\alpha'_\perp\beta_\perp)^{-1}\alpha'_\perp X + \alpha(\beta'\alpha)^{-1}\beta'X = \beta_\perp v_1 + \alpha v_2.$$

Unlike the decomposition $(\beta, \beta_\perp)$, the pair $(\beta_\perp, \alpha)$ is generally not orthogonal. Nevertheless, if $\beta'\alpha$ has full rank, the two matrices span complementary subspaces of $\mathbb{R}^p$ and therefore form a basis.

This representation is particularly useful in the context of cointegrated VAR models. The matrix β spans the cointegration space, while α describes the adjustment of the system toward the long-run equilibrium. Using $(\beta_\perp, \alpha)$ therefore allows us to decompose any vector into a component associated with the common stochastic trends (spanned by $\beta_\perp$) and a component associated with the equilibrium adjustment dynamics (spanned by α).

B.3 Matrix Derivatives for the Multivariate GARCH Likelihood

1. Proof: $d\Omega^{-1} = -\Omega^{-1}(d\Omega)\,\Omega^{-1}$

Start from the definition $\Omega\,\Omega^{-1} = I$ and differentiate both sides:

$$d\big(\Omega\,\Omega^{-1}\big) = dI = 0.$$

By the product rule for differentials:

$$(d\Omega)\,\Omega^{-1} + \Omega\,d(\Omega^{-1}) = 0.$$

Solving for $d(\Omega^{-1})$ by pre-multiplying by Ω^{-1}:

$$\Omega^{-1}\Omega\,d(\Omega^{-1}) = -\Omega^{-1}(d\Omega)\,\Omega^{-1},$$

$$\boxed{d\Omega^{-1} = -\Omega^{-1}(d\Omega)\,\Omega^{-1}.}$$

This is the exact matrix analogue of $dx^{-1} = -x^{-2}dx$.

2. Proof: $d\log|\Omega| = \operatorname{tr}\{\Omega^{-1}\,d\Omega\}$

Step 1: Differential of $|\Omega|$ via eigendecomposition. Since Ω is symmetric positive definite, write $\Omega = P\Lambda P'$ where $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Then $|\Omega| = \prod_i \lambda_i$, so:

$$d\log|\Omega| = d\log\prod_i \lambda_i = \sum_i d\log\lambda_i = \sum_i \frac{d\lambda_i}{\lambda_i}.$$

Step 2: Linking eigenvalue differentials to the trace. It can be shown (via Jacobi's formula) that the differential of the determinant satisfies:

$$d|\Omega| = |\Omega|\operatorname{tr}\big(\Omega^{-1}\,d\Omega\big)\,.$$

Step 3: Chain rule on the logarithm. Applying $d\log|\Omega| = \frac{1}{|\Omega|}d|\Omega|$:

$$d\log|\Omega| = \frac{1}{|\Omega|}\cdot|\Omega|\operatorname{tr}\big(\Omega^{-1}\,d\Omega\big)\,,$$

$$\boxed{d\log|\Omega| = \operatorname{tr}\big\{\Omega^{-1}\,d\Omega\big\}\,.}$$

This is the exact matrix analogue of $d\log(x) = x^{-1}dx$.

(ABC)^-1 = C^-1B^-1A^-1

|ABC| = |A||B||C|